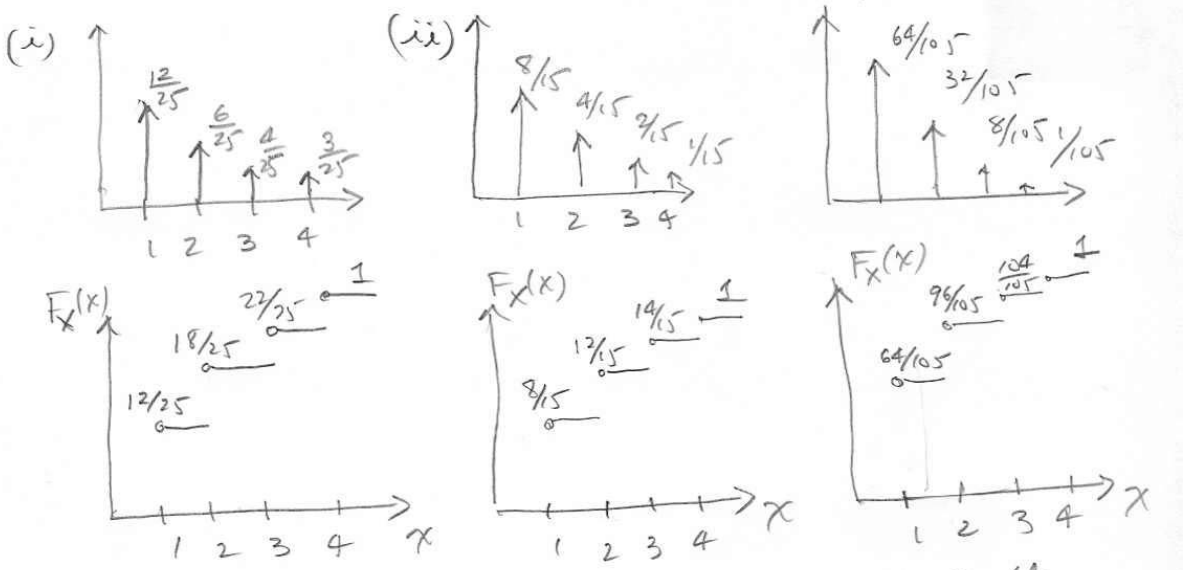
Chapter 4: One Random Variable

4.1 The Cumulative Distribution Function

4.1 (i) $1 = p_1\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right) = \frac{25}{12}p_1 \Rightarrow p_1 = \frac{12}{25}$ $\quad p = \left(\frac{12}{25}, \frac{6}{25}, \frac{4}{25}, \frac{3}{25}\right)$

(ii) $1 = p_1\left(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right) = \frac{15}{8}p_1 \Rightarrow p_1 = \frac{8}{15}$ $\quad p = \left(\frac{8}{15}, \frac{4}{15}, \frac{2}{15}, \frac{1}{15}\right)$

(iii) $1 = p_1\left(1+\frac{1}{2}+\frac{1}{8}+\frac{1}{64}\right) = \frac{105}{64}p_1 \Rightarrow p_1 = \frac{64}{105}$ $\quad p = \left(\frac{64}{105}, \frac{32}{105}, \frac{8}{105}, \frac{1}{105}\right)$



$P[X \le 1] = \frac{12}{25}$ $\quad P[X \le 1] = \frac{8}{15}$ $\quad P[X \le 1] = \frac{64}{105}$

$P[X < 2.5] = \frac{18}{25}$ $\quad P[X \le 2.5] = \frac{12}{15}$ $\quad P[X \le 2.5] = \frac{96}{105}$

$P[0.5 < X \le 2] = \frac{6}{25}$ $\quad P[0.5 < X \le 2] = \frac{4}{15}$ $\quad P[0.5 < X \le 2] = \frac{32}{105}$

$P[1 < X < 4] = \frac{10}{25}$ $\quad P[1 < X < 4] = \frac{6}{15}$ $\quad P[1 < X < 4] = \frac{40}{105}$

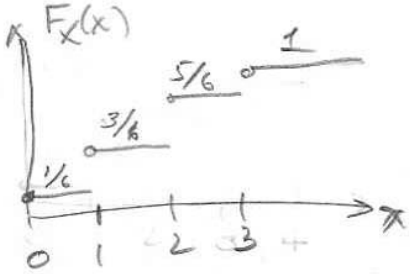
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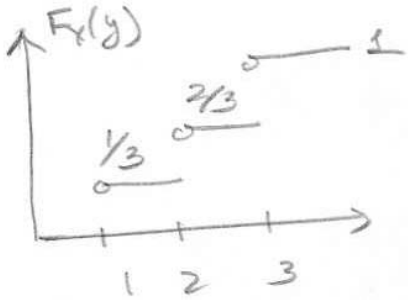
4.2

ξ	1	2	3	4	5	6
$X(\xi)$	0	1	1	2	2	3
$Y(\xi)$	1	1	2	2	3	3

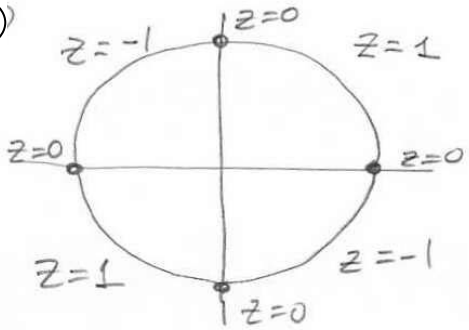
$S_X = \{0, 1, 2, 3\}$ $\quad p = (\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6})$

$S_Y = \{1, 2, 3\}$ $\quad p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$





4.3



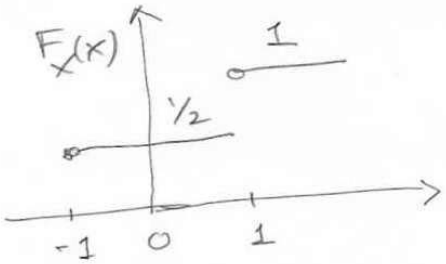
$S_X = \{-1, 0, 1\}$

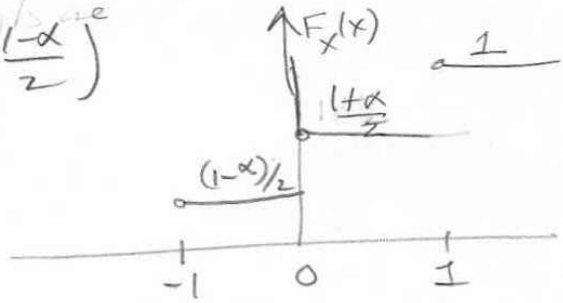
$p = (\frac{1}{4} + \frac{1}{4}, 0, \frac{1}{4} + \frac{1}{4})$

$= (\frac{1}{2}, 0, \frac{1}{2})$

If clock has property to stop at 3/6/9/12 α of the three then

$p = (\frac{1-\alpha}{2}, \alpha, \frac{1-\alpha}{2})$





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4.4

8 × $1 2 × $5

without replacement

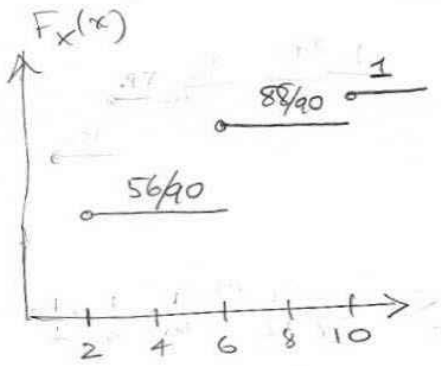
1+1
1+5
5+5

$S_X = \{2, 6, 10\}$

with replacement

1+1
1+5
5+5

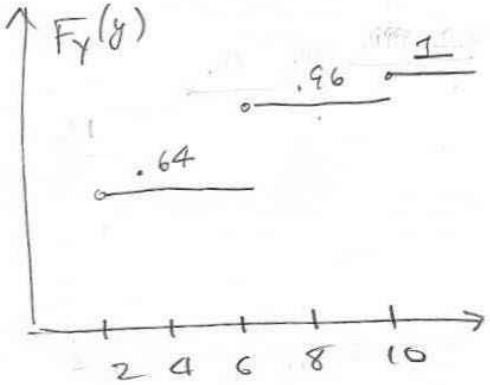
$S_Y = \{2, 6, 10\}$



$P[2] = \frac{8}{10} \cdot \frac{7}{9} = \frac{56}{90}$

$P[6] = 2 \cdot \frac{8}{10} \cdot \frac{2}{9} = \frac{32}{90}$

$P[10] = \frac{2}{10} \cdot \frac{1}{9} = \frac{2}{90}$



$P[2] = .8^2 = .64$

$P[6] = 2(.8)(.2) = .32$

$P[10] = (.2)^2 = .04$

$P[X=2] = \frac{56}{90}$ $P[Y=2] = \frac{64}{100}$

$P[X<7] = \frac{88}{90}$ $P[Y<7] = \frac{96}{100}$

$P[X \geq 6] = 1 - \frac{56}{90} = \frac{34}{90}$ $P[Y \geq 6] = 1 - .64 = \frac{.36}{100}$

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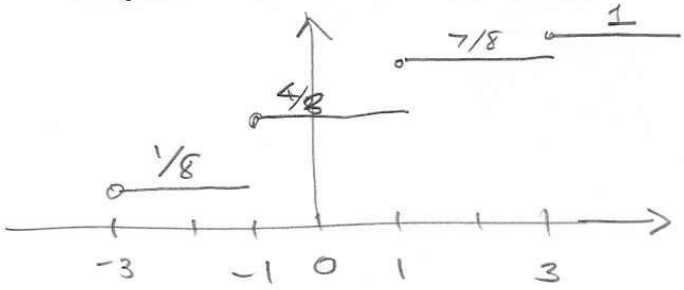
4.5

$Y = N_H - N_T = N_H - (n - N_H) = 2N_H - n$

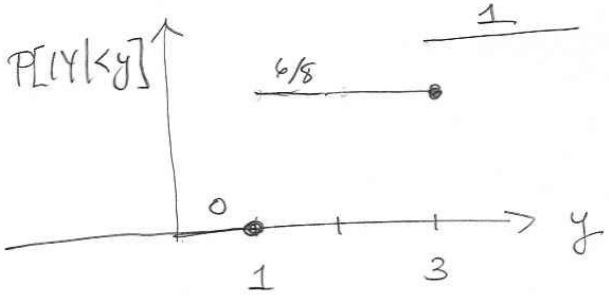
$n = 3$

n_H	0	1	2	3
$y(n_H)$	−3	−1	1	3
$P[y]$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$P[N_H = k] = \binom{3}{k}\left(\frac{1}{2}\right)^3$



$P[|Y| < y] = P[-y < Y < y] = F_Y(y^-) - F_Y(-y)$

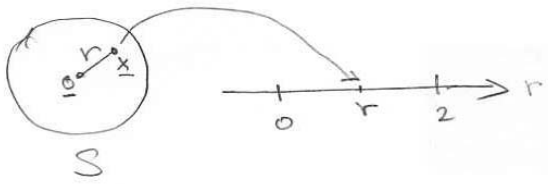


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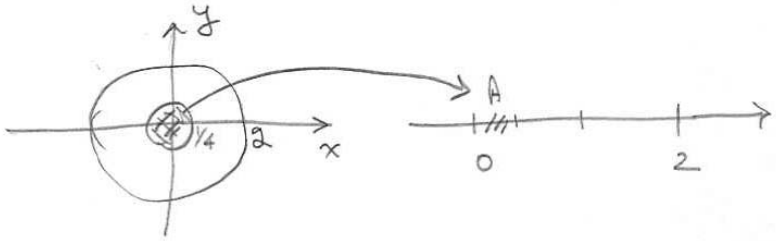
4.6

$S = \{(x,y): x^2+y^2 \le 4\}$ $S_R = \{r: 0 \le r \le 2\}$

$R = \sqrt{X^2+Y^2}$

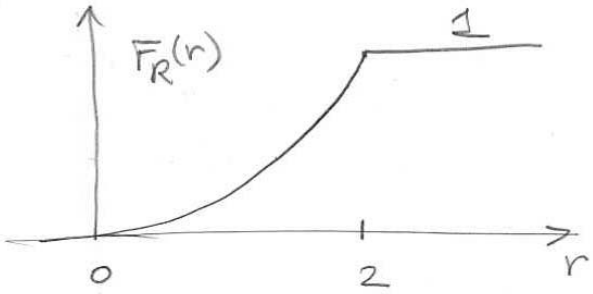


$$P[A] = P\left[R \le \tfrac{1}{4}\right] = \frac{\pi(\frac{1}{4})^2}{\pi(2)^2} = \frac{1}{64}$$



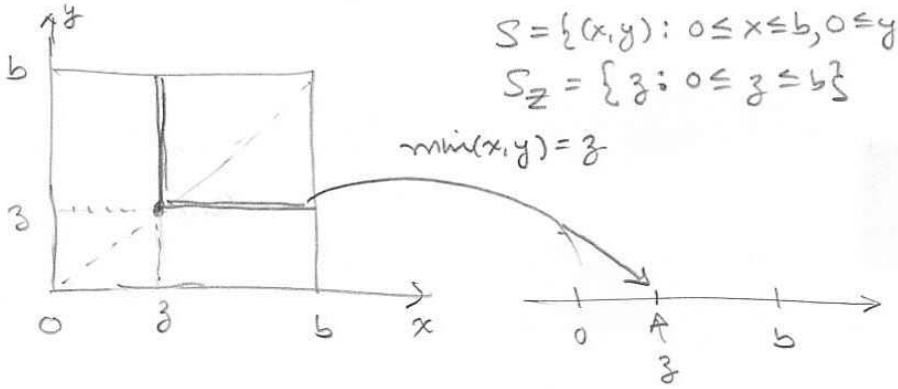
for $0 \le r \le 2$

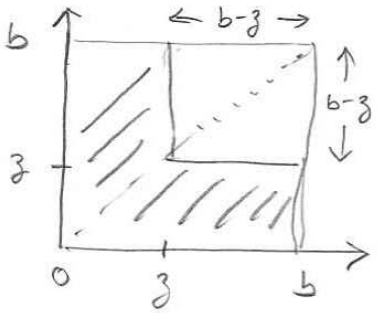
$$F_R(r) = P[R \le r] = \frac{\pi r^2}{\pi 2^2} = \left(\frac{r}{2}\right)^2$$



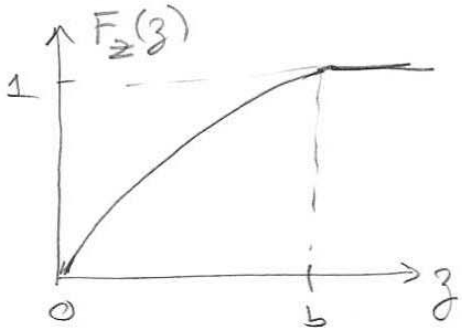
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4.7





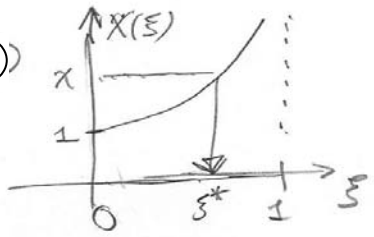
$$P[Z\le z] = 1-\left(\frac{b-z}{b}\right)^2$$
$$= 1-\left(1-2\frac{z}{b}+\frac{z^2}{b^2}\right)$$
$$= \frac{2z}{b}-\frac{z^2}{b^2}$$
$$= \frac{z}{b}\left(2-\frac{z}{b}\right)$$



$$P[Z>0] = 1$$
$$P[Z>b] = 1$$
$$P[Z\le \tfrac{b}{2}] = F_Z(\tfrac{b}{2})$$
$$= \tfrac{1}{2}(2-\tfrac{1}{2}) = \tfrac{3}{4}$$
$$P[Z>\tfrac{b}{4}] = 1-F_Z(\tfrac{b}{4})$$
$$= 1-\tfrac{1}{4}(2-\tfrac{1}{4})$$
$$= \tfrac{9}{16}$$

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4.8

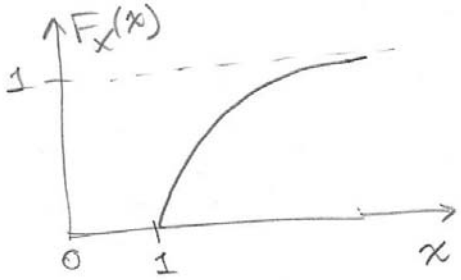


$$X(\xi) = \frac{1}{\sqrt{1-\xi^2}}$$

$$S_X = \{x: 1 \le x < \infty\}$$

$$P[X(\xi) \le x] = P\left[\frac{1}{\sqrt{1-\xi^2}} \le x\right] = P\left[\frac{1}{1-\xi^2} \le x^2\right]$$

$$= P\left[\frac{1}{x^2} \le 1-\xi\right] = P\left[\xi \le 1-\frac{1}{x^2}\right] = 1-\frac{1}{x^2}$$



$$P[X>1] = 1 - F_X(1)$$

$$= 1-\left(1-\frac{1}{1^2}\right) = 0$$

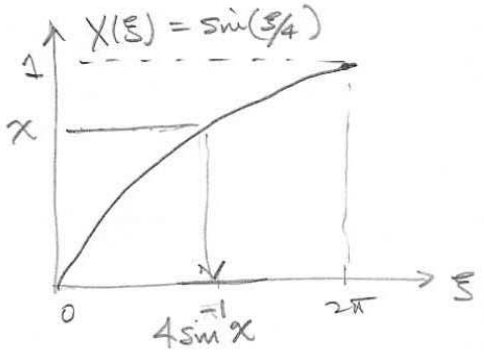
$$P[5<X<7] = F_X(7) - F_X(5) = \left(1-\frac{1}{49}\right) - \left(1-\frac{1}{25}\right)$$

$$= \frac{1}{25} - \frac{1}{49} = 0.01959$$

$$P[X \le 20] = 1 - \frac{1}{400} = 0.9975$$

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4.9

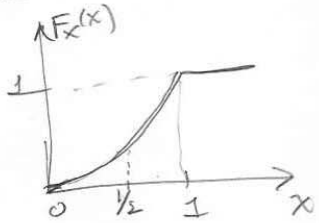


$S = \{\xi : 0 \le \xi < 2\pi\}$

$S_X = \{x : 0 \le x < 1\}$

$$P[X \le x] = P[\sin(\tfrac{\xi}{4}) \le x] = P[\xi \le 4\sin^{-1} x]$$

$$= \frac{4\sin^{-1} x}{2\pi} \qquad 0 \le x \le 1$$

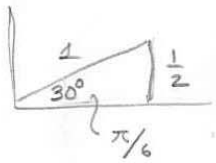


$$P[X > 1] = 1 - F_X(1) = 1 - 1 = 0$$

$$P[-\tfrac{1}{2} < X < \tfrac{1}{2}] = P[0 < X < \tfrac{1}{2}]$$

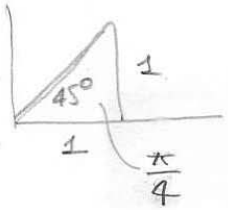
$$= F_X(\tfrac{1}{2})$$

$$= \frac{4\sin^{-1}\frac{1}{2}}{2\pi}$$



$$= \frac{4}{2\pi}\frac{\pi}{6} = \frac{1}{3}$$

$$P[X \le \tfrac{1}{\sqrt{2}}] = \frac{4\sin^{-1}\frac{1}{\sqrt{2}}}{2\pi}$$



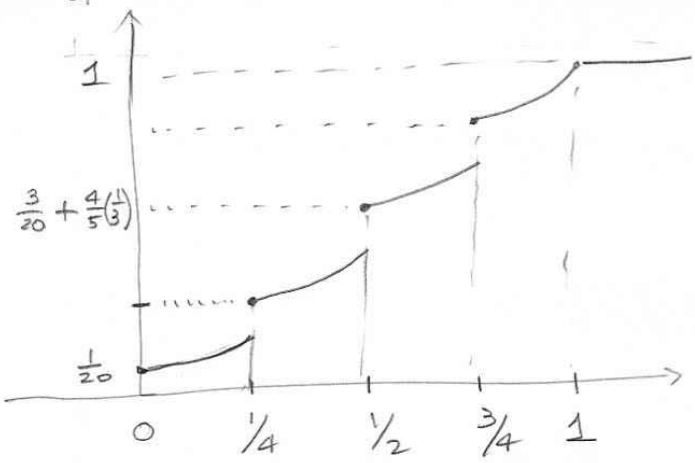
$$= \frac{4}{2\pi}\frac{\pi}{4} = \frac{1}{2}$$

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4.10

The probability law over $S = \{\xi : 0 \le x < 2\pi\}$ places probability $\frac{0.2}{4}$ on the points $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ and probability 0.8 uniformly in the interval $[0, 2\pi)$.

$X(\xi)$ has probability mass at the points $0, 1/4, 1/2, 3/4$



$$F_X(x) = \frac{1}{20}u(x) + \frac{1}{20}u\left(x - \frac{1}{4}\right) + \frac{1}{20}u\left(x - \frac{1}{2}\right) + \frac{1}{20}u\left(x - \frac{3}{4}\right) + \frac{4}{5}\,\frac{4\sin^{-1}x}{2\pi}$$

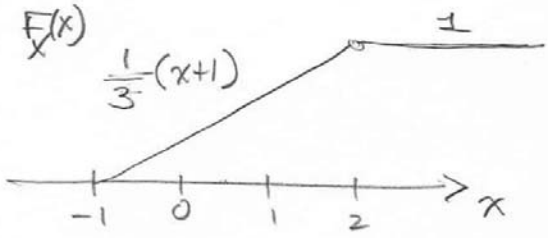
$$P[X > 1] = 1 - F_X(1) = 0$$

$$P\left[-\tfrac{1}{2} < X < \tfrac{1}{2}\right] = P\left[0 < X < \tfrac{1}{2}\right] = \frac{2}{20} + \frac{4}{5}\,\frac{4\sin^{-1}\left(\frac{1}{2}\right)}{2\pi} = \frac{2}{20} + \frac{4}{5}\left(\frac{1}{3}\right) = \frac{11}{30}$$

$$P\left[X \le \frac{1}{\sqrt{2}}\right] = \frac{3}{20} + \frac{4}{5}\,\frac{\sin^{-1}\frac{1}{\sqrt{2}}}{2\pi} = \frac{3}{20} + \frac{4}{5}\left(\frac{1}{2}\right) = \frac{11}{20}$$

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4.11



$$P[X<0] = F_X(0) = \frac{1}{3}$$

$$P[|X-\tfrac{1}{2}|<1] = P[-1<X-\tfrac{1}{2}<1] = P[-\tfrac{1}{2}<X<\tfrac{3}{2}]$$

$$= \frac{1}{3}\left(\frac{3}{2}+1\right) - \frac{1}{3}\left(-\frac{1}{2}+1\right)$$

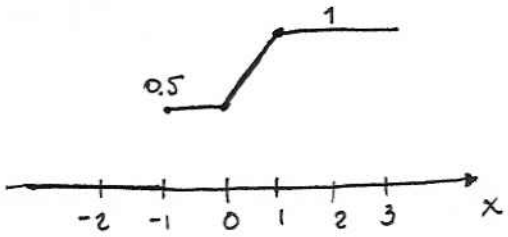
$$= \frac{1}{3}\left(\frac{3}{2}+1+\frac{1}{2}-1\right) = \frac{2}{3}$$

$$P[X>-\tfrac{1}{2}] = 1 - P[X\leq -\tfrac{1}{2}] = 1-\frac{1}{3}\left(-\frac{1}{2}+1\right) = \frac{5}{6}$$

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4.12

a)



Mixed type random variable

b) $P[X \leq -1] = 0.5$

$P[X = -1] = 0.5$

$P[X < 0.5] = P[X \leq 0.5] - P[X = 0.5]$

$= \frac{1+0.5}{2} - 0$

$= 0.75$

$P[-0.5 < X < 0.5] = P[X \leq 0.5] - P[X = 0.5] - P[X \leq -0.5]$

$= \frac{1+0.5}{2} - 0 - 0.5$

$= 0.75 - 0.5$

$= 0.25$

$P[X > -1] = 1 - P[X \leq -1]$

$= 1 - 0.5$

$= 0.5$

$P[X \leq 2] = 1$

$P[X > 3] = 1 - P[X \leq 3]$

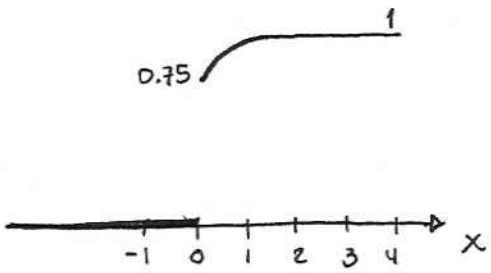
$= 1 - 1$

$= 0$

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4.13

a)



Mixed type random variable

b) $P[X \leq 2] = 1 - \frac{1}{4}e^{-2(2)}$

$= 0.9954$

$P[X = 0] = 1 - \frac{1}{4}e^{-2(0)}$

$= 0.75$

$P[X < 0] = 0$

$P[2 < X < 6] = P[X \leq 6] - P[X \leq 2]$

$= 1 - \frac{1}{4}e^{-2(6)} - 1 + \frac{1}{4}e^{-2(2)}$

$= 0.0046$

$P[X > 10] = 1 - P[X \leq 10]$

$= 1 - \left(1 - \frac{1}{4}e^{-2(10)}\right)$

$= 5.15 \times 10^{-10}$

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4.14

a) Mixed Type

b) $P[X<-1]=0$

$P[X\leq -1]=\frac{2}{10}$

$P[-1<X<-0.75]=P[X\leq -0.75]-P[X\leq -1]$

$=\frac{2}{10}-\frac{2}{10}$

$=0$

$P[-0.5\leq X\leq 0.5]=P[X\leq 0.5]-P[X\leq -0.5]$

$=\frac{8}{10}-\frac{2}{10}$

$=\frac{6}{10}$

$P[|X-0.5|<0.5]=P[\{X<1\}\cup\{X>0\}]=P[\{0<X<1\}]$

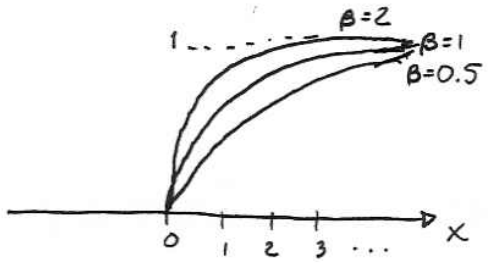
$=1-\frac{6}{10}$

$=\frac{4}{10}$

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4.15

a)



b) $P[j\lambda < X < (j+1)\lambda] = P[X \le (j+1)\lambda] - P[X \le j\lambda]$

$= 1 - e^{-\left((j+1)\frac{\lambda}{\lambda}\right)^\beta} - 1 + e^{-\left(j\frac{\lambda}{\lambda}\right)^\beta}$ $\quad j \ge 0$

$= e^{-j^\beta} - e^{-(j+1)^\beta}$ $\quad j \ge 0$

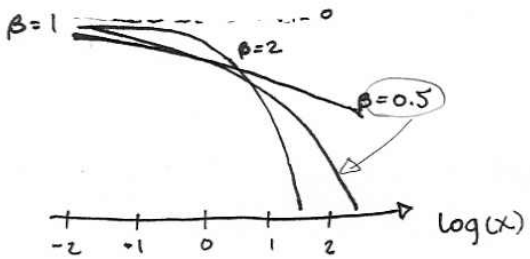
$$= \begin{cases} 0 & j < -1 \\ 1 - e^{-(j+1)^\beta} & -1 \le j < 0 \\ e^{-j^\beta} - e^{-(j+1)^\beta} & j \ge -1 \end{cases}$$

$P[X > j\lambda] = 1 - P[X \le j\lambda] = 1 - \left(1 - e^{-\left(j\frac{\lambda}{\lambda}\right)^\beta}\right)$ $\quad j \ge 0$

$$= \begin{cases} 0 & j < 0 \\ e^{-j^\beta} & j \ge 0 \end{cases}$$

c) $\log P[X > x] = \begin{cases} \log(1) & x < 0 \\ \log\left(e^{-\left(\frac{x}{\lambda}\right)^\beta}\right) & x \ge 0 \end{cases}$

$$= \begin{cases} 0 & x < 0 \\ -\left(\frac{x}{\lambda}\right)^\beta & x \ge 0 \end{cases}$$

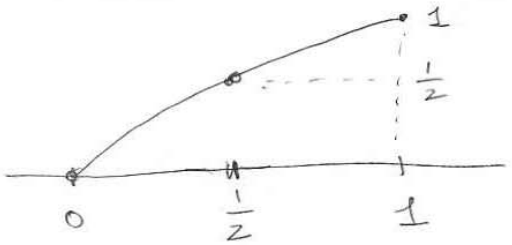


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4.16

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} + c\sin^2\frac{\pi}{2}x & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

$\sin^2\frac{\pi}{2}x$



$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

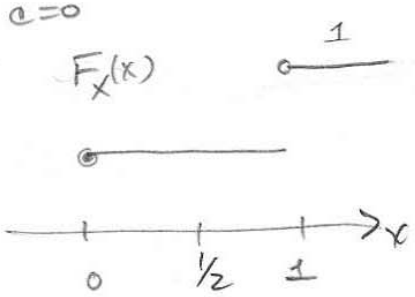
$\sin^2\frac{\pi}{4} = \frac{1}{2}$



The constant c should be selected so properties of the cdf are satisfied: $c > 0$ so cdf is increasing

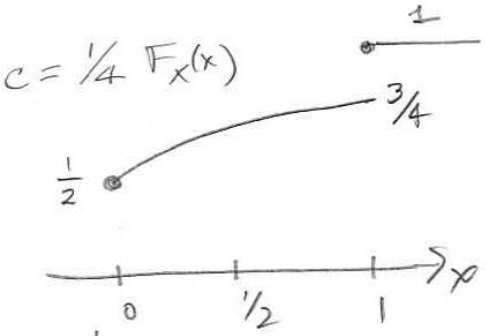
$c < \frac{1}{2}$ so cdf ≤ 1

$c = 0$



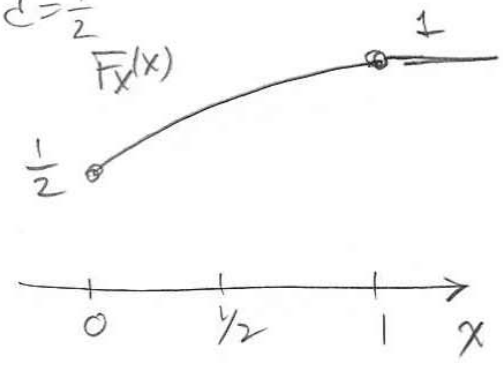
$P[X > 0] = 1 - P[X \le 0] = \frac{1}{2}$

$c = 1/4$ $F_X(x)$



$P[X > 0] = \frac{1}{2}$

$c = \frac{1}{2}$



$P[X > 0] = \frac{1}{2}$

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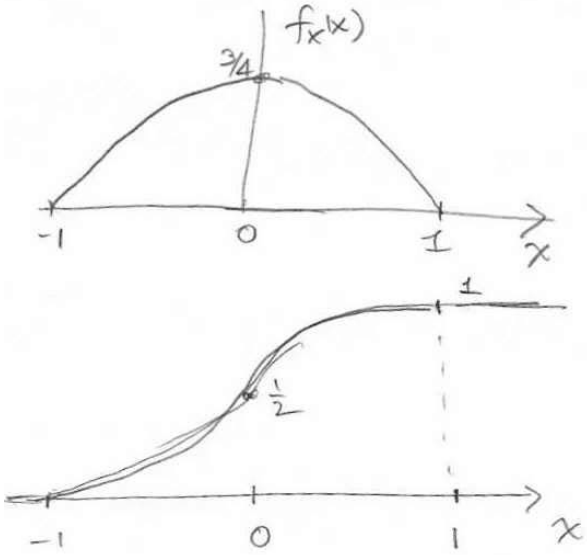
4.2 The Probability Density Function

4.17

$$1 = c\int_{-1}^{1}(1-x^2)\,dx = c\left[x\Big|_{-1}^{1} - \frac{x^3}{3}\Big|_{-1}^{1}\right] = c\left[2 - \tfrac{1}{3}2\right] = \frac{4}{3}c$$

$$\Rightarrow c = \frac{3}{4}$$

$$f_X(x) = \frac{3}{4}(1-x^2) \quad -1 \le x \le 1$$



$$F_X(x) = \frac{3}{4}\int_{-1}^{x}(1-y^2)\,dy = \frac{3}{4}\left[y\Big|_{-1}^{x} - \frac{y^3}{3}\Big|_{-1}^{x}\right]$$

$$= \frac{3}{4}\left[(x+1) - \tfrac{1}{3}(x^3+1)\right]$$

$$P[X=0] = F_X(0) = 0$$

$$P[0 < X < 0.5] =$$

$$= \frac{3}{4}\left[\left(\tfrac{1}{2}+1\right) - \tfrac{1}{3}\left(\tfrac{1}{8}+1\right)\right] - \frac{3}{4}\left[1 - \tfrac{1}{3}\right]$$

$$= \frac{11}{32}$$

$$P\left[\,|X - \tfrac{1}{2}| < \tfrac{1}{4}\,\right] = P\left[\,\tfrac{1}{4} < X < \tfrac{3}{4}\,\right]$$

$$= \frac{3}{4}\left[\left(\tfrac{3}{4}+1\right) - \tfrac{1}{3}\left(\left(\tfrac{3}{4}\right)^3+1\right)\right] - \frac{3}{4}\left[\left(\tfrac{1}{4}+1\right) - \tfrac{1}{3}\left(\left(\tfrac{1}{4}\right)^3+1\right)\right]$$

$$= 0.2734$$

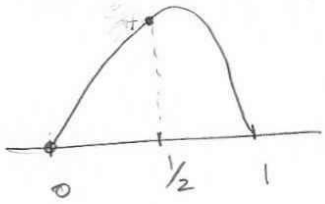
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4.18

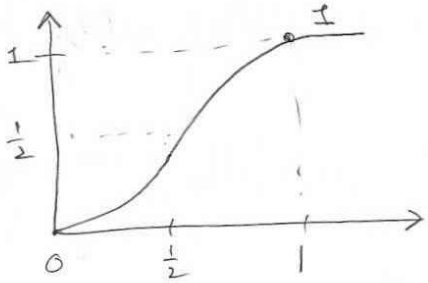
$$1 = c\int_0^1 x(1-x^2)dx = c\left[\frac{x^2}{2}\Big|_0^1 - \frac{x^4}{4}\Big|_0^1\right] = c\left[\frac{1}{2}-\frac{1}{4}\right] = c\left[\frac{1}{4}\right]$$

$$\Rightarrow c = 4$$

$$f_X(x) = 4x(1-x^2) \qquad 0 \le x \le 1$$



$$F_X(x) = 4\int_0^x (y-y^3)dy = 4\left[\frac{x^2}{2} - \frac{x^4}{4}\right]$$



$$P[0 < X < \tfrac{1}{2}] = F_X(\tfrac{1}{2}) = 7/16 = .4375$$

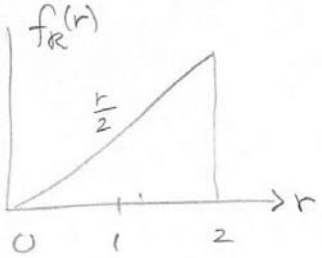
$$P[X = 1] = 0$$

$$P[\tfrac{1}{4} < X < \tfrac{1}{2}] = F_X(\tfrac{1}{2}) - F_X(\tfrac{1}{4}) = \frac{7}{16} - 4\left[\frac{1}{32} - \frac{1}{1024}\right] = .3164$$

4.19 from 4.6

$$F_R(r) = \left(\frac{r}{2}\right)^2 \qquad 0 \le r \le 2$$

(a) $$f_R(r) = \frac{d}{dx}F_R(r) = 2\left(\frac{r}{2}\right)\left(\frac{1}{2}\right) = \frac{r}{2} \qquad 0 \le r \le 2$$



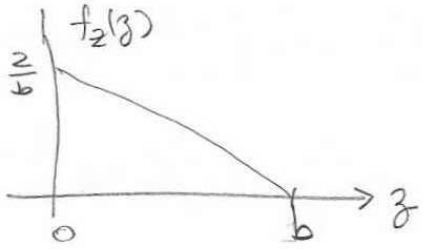
(b) $$P[R > \tfrac{1}{4}] = \int_{1/4}^{2} \frac{r}{2}dr = \frac{1}{2}\frac{r^2}{2}\Big|_{1/4}^{2} = \frac{1}{4}\left[4 - \frac{1}{16}\right] = \frac{63}{64}$$

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4.20

$F_Z(z) = 2\frac{z}{b} - \frac{z^2}{b^2}$ from prob. 4.7

(a) $f_Z(z) = \frac{d}{dz}F_Z(z) = \frac{2}{b} - \frac{2z}{b^2} = \frac{2}{b}\left(1 - \frac{z}{b}\right)$



(b) $P[Z > \frac{b}{3}] = \int_{b/3}^{b} \frac{2}{b}\left(1 - \frac{z}{b}\right)dz = \frac{2}{b}\left[z\Big|_{b/3}^{b} - \frac{1}{b}\frac{z^2}{2}\Big|_{b/3}^{b}\right]$

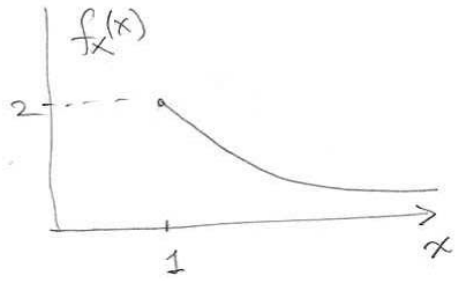
$= \frac{2}{b}\left(b - \frac{b}{3}\right) - \frac{2}{2b^2}\left[b^2 - \frac{b^2}{9}\right]$

$= \frac{4}{3} - \frac{8}{9} = \frac{4}{9}$

4.21

from 4.8 $F_X(x) = 1 - \frac{1}{x^2}$ $x \geq 1$.

$f_X(x) = \frac{d}{dx}F_X(x) = 2\frac{1}{x^3}$ $x \geq 1$



$a > 1$

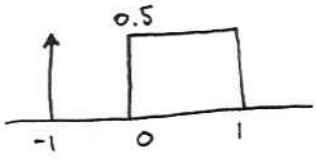
$P[X > a] = 2\int_a^{\infty} \frac{1}{x^3}dx = \frac{1}{x^2}\Big|_a^{\infty} = \frac{1}{a^2}$

$P[X > 2a] = \frac{1}{x^2}\Big|_{2a}^{\infty} = \frac{1}{4a^2}$

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4.22

a) $$f_X(x) = \begin{cases} 0.5\delta(x+1) & x \le 0 \\ 0.5 & 0 \le x \le 1 \\ 0 & x \ge 1 \end{cases}$$



b) $$P[-1 \le X < 0.25] = \int_{-1^-}^{0.25^-} f_X(x)\,dx$$

$$= \int_{-1^-}^{0} 0.5\,\delta(x+1)\,dx + \int_{0}^{0.25} 0.5\,dx$$

$$= 0.5 \times \int_{0}^{0.25} + 0.5$$

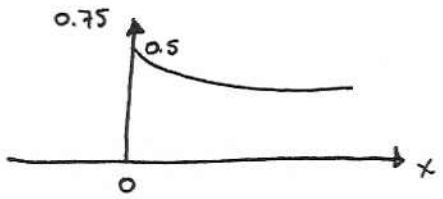
$$= 0.125 + 0.5$$

$$= 0.625$$

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4.23

a) $$f_X(x) = \begin{cases} 0 & x<0 \\ 0.75\delta(x) & x=0 \\ 0.5e^{-2x} & x>0 \end{cases}$$



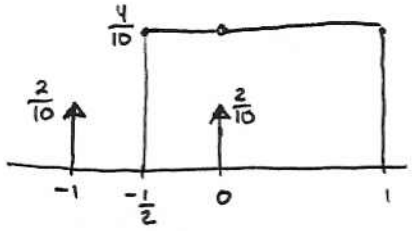
b) $P[X=0] = 0.75$

$$P[X>8] = \int_8^\infty 0.5e^{-2x}\,dx$$
$$= -0.25e^{-2x}\Big|_8^\infty$$
$$= 0 + 0.25e^{-2(8)}$$
$$= 0.25e^{-16}$$

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4.24

a)



b) $P[X<-1]=0$

$P[X\le -1]=\int_{-\infty}^{-1}\frac{2}{10}\delta(x+1)dx=\frac{2}{10}$

$P[-1<X<-0.75]=\int_{-1^+}^{-0.75^-}0\,dx=0$

$P[-0.5\le X<0]=\int_{-0.5}^{0^-}\frac{4}{10}dx=\frac{4}{10}\left(\frac{1}{2}\right)=\frac{2}{10}$

$P[-0.5\le X\le 0.5]=\int_{-0.5}^{0.5}f_X(x)dx=\int_{-0.5}^{0^-}\frac{4}{10}dx+\int_{0^-}^{0^+}\frac{2}{10}\delta(x)\,dx$

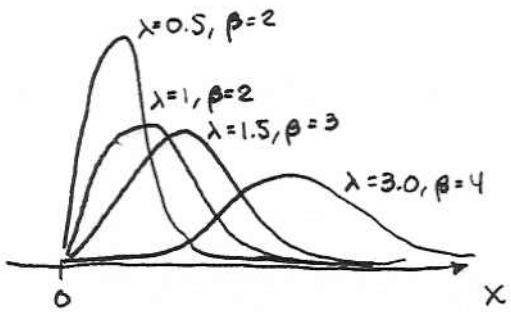
$+\int_{0^+}^{0.5}\frac{4}{10}dx=\frac{4}{10}(0+0.5)+\frac{2}{10}+\frac{4}{10}(0.5-0)$

$=\frac{6}{10}$

$P[|X-0.5|<0.5]=P[\{X<1\}\cup\{X>0\}]=\int_{0^+}^{1}\frac{4}{10}dx=\frac{4}{10}$

4.25

a) $$f_X(x)=\begin{cases}0 & x<0\\ \frac{\beta}{\lambda}\left(\frac{x}{\lambda}\right)^{\beta-1}e^{-\left(\frac{x}{\lambda}\right)^{\beta}} & x\ge 0\end{cases}$$



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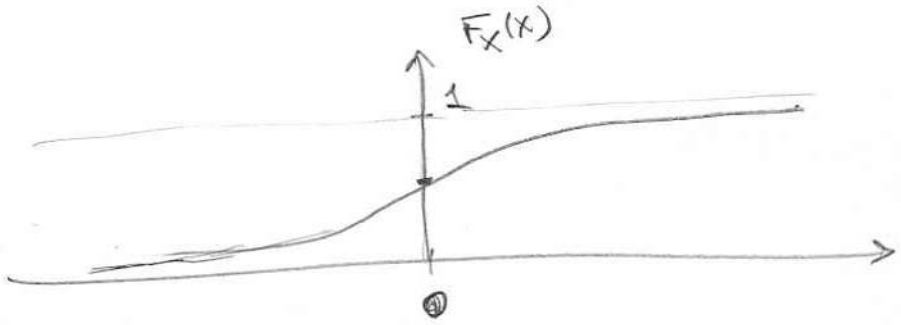
4.26

$$f_X(x) = \frac{\alpha/\pi}{x^2+\alpha^2}$$

$$F_X(x) = \int_{-\infty}^{x} \frac{\alpha/\pi}{t^2+\alpha^2}\,dt = \int_{-\infty}^{x} \frac{1/\pi}{1+\left(\frac{t}{\alpha}\right)^2} d\left(\frac{t}{\alpha}\right)$$

$$= \frac{1}{\pi}\tan^{-1}\left(\frac{t}{\alpha}\right)\Bigg|_{-\infty}^{x}$$

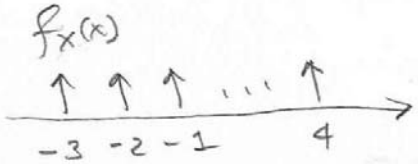
$$= \frac{1}{\pi}\left[\tan^{-1}\left(\frac{x}{\alpha}\right) + \frac{\pi}{2}\right] \qquad -\infty < x < \infty$$



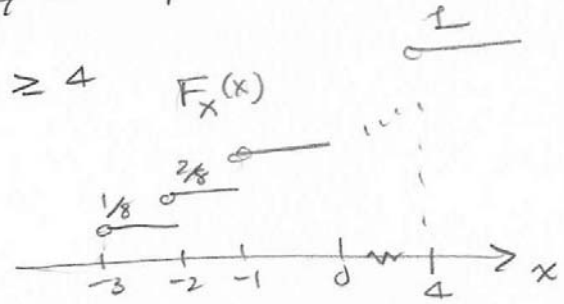
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4.27

(a) $f_X(x) = \sum_{j=-3}^{4} \frac{1}{8}\delta(x-j)$



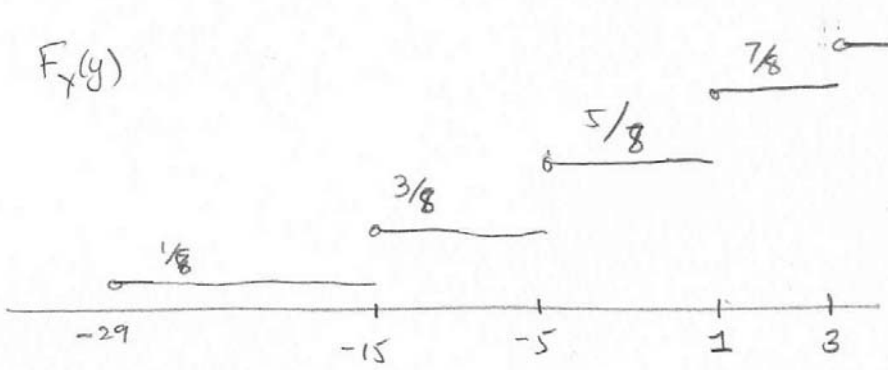
$$F_X(x) = \begin{cases} 0 & x < -3 \\ \frac{j+3}{8} & \frac{j+3}{7} \le x < \frac{j+4}{7} \quad j = -3, \cdots, 3 \\ 1 & x \ge 4 \end{cases}$$



(b)

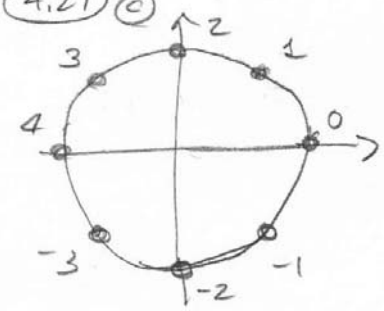
x	-3	-2	-1	0	1	2	3	4
$Y = -2X^2 + 3$	-15	-5	1	3	1	-5	-15	-29

$$f_Y(y) = \frac{1}{8}\delta(y+29) + \frac{2}{8}\delta(y+15) + \frac{2}{8}\delta(y+5) + \frac{2}{8}\delta(y-1) + \frac{1}{8}\delta(y-3)$$



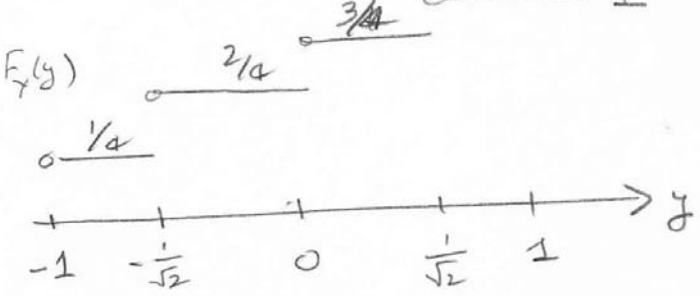
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x	-3	-2	-1	0	1	2	3	4
$\cos\frac{\pi x}{8}$	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1

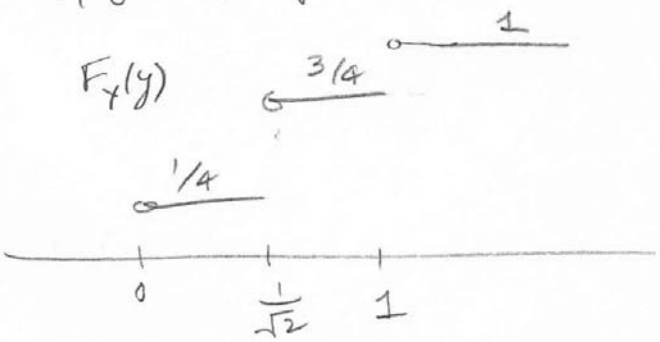
$$f_Y(y) = \frac{2}{8}\delta(y+1) + \frac{2}{8}\delta\left(y+\frac{1}{\sqrt{2}}\right) + \frac{2}{8}\delta\left(y-\frac{1}{\sqrt{2}}\right) + \frac{2}{8}\delta(y-1)$$



4.27 d

$$Y = \cos^2\frac{\pi X}{8} = \begin{cases} \frac{1}{2} & x = -3, -1, 1, 3 \\ 1 & x = 0, 4 \\ 0 & x = 2, -2 \end{cases}$$

$$f_Y(y) = \frac{1}{4}\delta(y) + \frac{2}{4}\delta\left(y-\frac{1}{\sqrt{2}}\right) + \frac{1}{4}\delta(y-1)$$

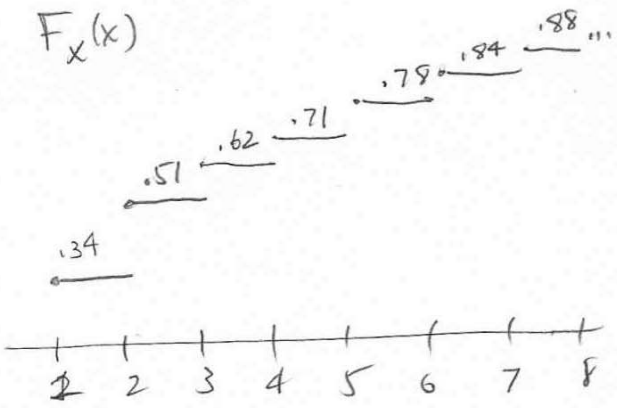


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4.28

$$p_X(i) = \frac{1}{c_{10}} \frac{1}{i} \qquad i = 1, 2, \ldots, 10$$

$$c_{10} = \sum_{i=1}^{10} \frac{1}{i} = 2.9289$$



	$P[X=i]$
1	0.341
2	0.171
3	0.114
4	0.0853
5	0.0683
6	0.0569
7	0.0488
8	0.0427
9	0.0379
10	0.0341
	1

Note slow decay rate

$$f_X(x) = \sum_{i=1}^{10} p_X(i)\, \delta(x-i)$$

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4.29

3.26 $F_X(x|A) = \dfrac{P[\{X \le x\} \cap A]}{P[A]}$ if $P[A] > 0$

i) $P[A] > 0, \quad P[\{x \le x\} \cap A] \ge 0 \Rightarrow F_x(x|A) \ge 0$
$\{\{X \le x\} \cap A\} \subset A \Rightarrow P[\{X \le x\} \cap A] \le P[A]$

Therefore

$$F_X(x|A) \le 1$$

ii) $\lim_{x\to\infty} F_X(x|A) = \lim_{x\to\infty} \dfrac{P[\{X \le x\} \cap A]}{P[A]} = \dfrac{P[A]}{P[A]} = 1$

iii) $\lim_{x\to-\infty} P[\{X \le x\} \cap A] = P[\Phi] = 0 \Rightarrow \lim_{x\to\infty} F_X(x|A) = 0$

iv)
$$\begin{aligned} a < b &\Rightarrow \{\{X \le a\} \cap A\} \subset \{\{X \le b\} \cap A\} \\ &\Rightarrow P[\{X \le a\} \cap A] \le P[\{x \le b\} \cap A] \\ &\Rightarrow F_x(a|A) \le F_x(b|A) \end{aligned}$$

v)
$$\begin{aligned} P[\{X \le b\} \cap A] &= \lim_{h\to 0} P[\{X \le b+h\} \cap A] \\ F_X(b|A) &= \lim_{h\to 0} F_X(b+h|A) \end{aligned}$$

vi) $\{\{X \le a\} \cap A\} \cup \{\{a < X \le b\} \cap A\} = \{\{X \le b\} \cap A\}\}$
The two event on LHS are mutually exclusive. Therefore,

$$P[\{X \le a\} \cap A] + P[\{a < X \le b\} \cap A] = P[\{X \le b\} \cap A]$$

$$P[a < X < b|A] = F_X(b|A) - F_X(a|A)$$

vii)
$$\begin{aligned} P[\{X = b\} \cap A] &= P[\{X \le b\} \cap A] - P[\{X < b^-\} \cap A] \\ P[\{X = b\}|A] &= F_X(b|A) - F_X(b^-|A) \end{aligned}$$

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4.30

a) $$F_X(x|C) = \frac{P[\{X \le x\} \cap \{X > 0\}]}{P[X>0]} = \frac{P[0 < X \le x]}{P[X>0]} \qquad x \ge 0$$

$$= \begin{cases} 0 & x \le 0 \\ \dfrac{F_X(x) - F_X(0)}{1 - F_X(0)} & x \ge 0 \end{cases}$$

$$= \begin{cases} 0 & x \le 0 \\ \dfrac{-\frac{1}{4}e^{-2x} + \frac{1}{4}}{1-(1-\frac{1}{4})} = 1 - e^{-2x} & x \ge 0 \end{cases}$$

b) $$F_X(x|C) = \frac{P[\{X \le x\} \cap \{X = 0\}]}{P[X=0]}$$

$$= \begin{cases} 0 & x < 0 \\ 1 & x \ge 0 \end{cases}$$

4.31

$$B = \left\{0, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}\right\}^c \qquad P[B] = 1 - 0.2 = 4/5$$

(a)

$$F_X(x|B) = \frac{P[X \le x, B]}{P[B]} = \frac{\frac{4}{5}\frac{4\sin^{-1}x}{2\pi}}{4/5} = \frac{4\sin^{-1}x}{2\pi}$$

(b) $$B^c = \left\{0, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}\right\}$$

$$F_X(x|B^c) = \frac{P[X \le x, B^c]}{P[B^c]} \qquad P[B^c] = 1/5$$

$$= \frac{1}{1/5}\left[\frac{1}{20}u(x) + \frac{1}{20}u\left(x-\tfrac{1}{4}\right) + \frac{1}{20}u\left(x-\tfrac{1}{2}\right) + \frac{1}{20}u\left(x-\tfrac{3}{4}\right)\right]$$

$$= \frac{1}{4}u(x) + \frac{1}{4}u\left(x-\tfrac{1}{4}\right) + \frac{1}{4}u\left(x-\tfrac{1}{2}\right) + \frac{1}{4}u\left(x-\tfrac{3}{4}\right)$$

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4.32

$$F_X(x|B) = \frac{P[\{X \le x\} \cap \{X > 0.25\}]}{P[X > 0.25]} = \frac{P[0.25 < X \le x]}{P[X > 0.25]}$$

$$= \begin{cases} 0 & x < 0.25 \\ \dfrac{F_X(x) - F_X(0.25)}{1 - F_X(0.25)} & x \ge 0.25 \end{cases}$$

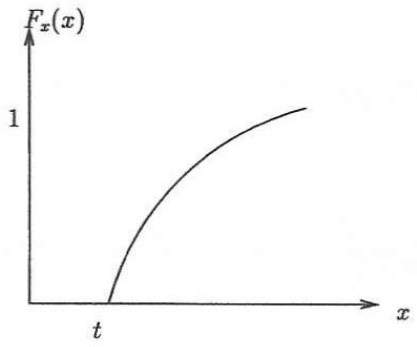
$$= \begin{cases} 0 & x \le 0.25 \\ \dfrac{-\frac{1}{4}e^{-2x} + \frac{1}{4}e^{-2(\frac{1}{4})}}{1-\left(1-\frac{1}{4}e^{-2(\frac{1}{4})}\right)} = \dfrac{e^{-\frac{1}{2}} - e^{-2x}}{e^{-\frac{1}{2}}} = 1 - e^{-(2x-\frac{1}{2})} & x \ge 0.25 \end{cases}$$

$$f_X(x|B) = \begin{cases} \dfrac{f_X(x)}{1 - F_X(0.25)} & x \ge 0.25 \\ 0 & x < 0.25 \end{cases}$$

$$= \begin{cases} \dfrac{\frac{1}{2}e^{-2x}}{\frac{1}{4}e^{-2(\frac{1}{4})}} = 2e^{-(2x-\frac{1}{2})} & x \ge 0.25 \\ 0 & x \le 0.25 \end{cases}$$

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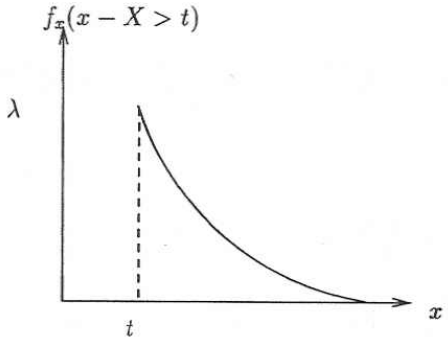
4.33 a)



$$\begin{aligned} F_X(x|X>T) &= \frac{P[\{X\le x\}\cap\{X>t\}]}{P[X>t]} \\ &= \begin{cases} 0 & x<t \\ \frac{F_X(x)-F_X(t)}{1-F_X(t)} & x\ge t \end{cases} \\ &= \begin{cases} 0 & x<t \\ \frac{(1-e^{-\lambda x})-(1-e^{-\lambda t})}{1-(1-e^{-\lambda t})} & x\ge t \end{cases} \\ &= \begin{cases} 0 & x<t \\ \frac{e^{-\lambda x}-e^{-\lambda t}}{e^{-\lambda t}} & x\ge t \end{cases} \end{aligned}$$

$F_x(x|x>t)$ is delayed version of $F_x(x)$

b) $f_x(x|x>t)=\dfrac{f_x(x)}{1-F_x(t)}=\dfrac{\lambda e^{-\lambda x}}{e^{-\lambda t}}=\lambda e^{-\lambda(x-t)}, \quad x\ge t$



c)

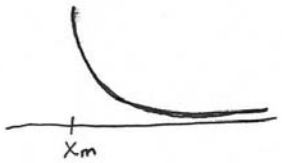
$$\begin{aligned} P &= [X>t+x|X>t] \qquad x\ge 0 \\ &= \frac{P[\{X>t+x\}\cap\{X>t\}]}{P[X>t]} \\ &= \frac{1-F_X(t+x)}{1-F_X(t)} \\ &= \frac{1-(1-e^{-\lambda(t+x)})}{1-(1-e^{-\lambda t})} \\ &= e^{-\lambda x} \\ &= P[X>x] \end{aligned}$$

Additional waiting time does not depend on time already spent waiting ⇒ "memoryless"

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4.34

a) $$f_X(x) = \begin{cases} 0 & x < x_m \\ \dfrac{\alpha x_m^\alpha}{x^{\alpha+1}} & x \geq x_m \end{cases}$$



b) $$F_X(x|X>t) = \frac{P[\{X \leq x\} \cap \{X > t\}]}{P[X>t]} = \frac{P[t < X \leq x]}{P[X>t]}$$

$$= \begin{cases} 0 & x \leq t \\ \dfrac{F_X(x) - F_X(t)}{1 - F_X(t)} & x \geq t \end{cases}$$

if $t \geq x_m$ $$F_X(x|X>t) = \frac{1 - \frac{x_m^\alpha}{x^\alpha} - 1 + \frac{x_m^\alpha}{t^\alpha}}{1 - \left(1 - \frac{x_m^\alpha}{t^\alpha}\right)} = t^\alpha\left(\frac{1}{t^\alpha} - \frac{1}{x^\alpha}\right) = 1 - \left(\frac{t}{x}\right)^\alpha \quad x \geq t$$

if $t < x_m$ $$F_X(x|X>t) = 1 - \left(\frac{x_m}{x}\right)^\alpha \quad x \geq x_m$$

$$f_X(x|X>t) = \frac{f_X(x)}{1 - F_X(t)} \quad x \geq t$$

if $t \geq x_m$ $$f_X(x|X>t) = \frac{\frac{\alpha x_m^\alpha}{x^{\alpha+1}}}{\frac{x_m^\alpha}{t^\alpha}} = \alpha t \left(\frac{t}{x}\right)^{\alpha+1} \quad x \geq t$$

if $t < x_m$ $$f_X(x|X>t) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \quad x \geq x_m$$

c) $$\frac{P[\{X > t+x\} \cap \{X > t\}]}{P[X>t]} \xrightarrow[t \to \infty]{} 1$$

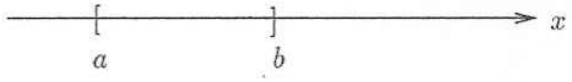
The longer you wait the longer you are likely to wait more!

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4.35

3.28 a) From the definition of conditional probability we have:

$$F_X(x|a \le X \le b) = \frac{P[\{X \le x\} \cap \{a \le X \le b\}]}{P[a \le X \le b]}$$



From the above figure we see that

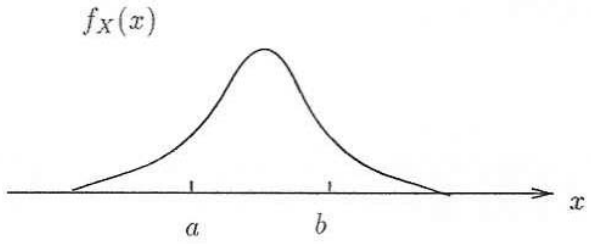
$$\{X \le x\} \cap \{a \le X \le b\} = \begin{cases} \emptyset & \text{for } x < a \\ \{a \le X \le x\} & \text{for } a \le x \le b \\ \{a \le X \le b\} & \text{for } x > b \end{cases}$$

Therefore

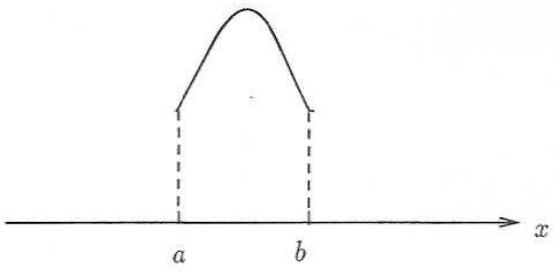
$$F_X(x|a \le X \le b) = \begin{cases} \frac{P[\emptyset]}{P[a \le X \le b]} = 0 & x < a \\ \frac{P[a \le X \le x]}{P[a \le X \le b]} = \frac{F_X(x) - F_X(a^-)}{F_X(b) - F_X(a^-)} & a \le x \le b \\ \frac{P[a \le X \le b]}{P[a \le X \le b]} = 1 & x > b \end{cases}$$

b)
$$\begin{aligned} f_X(x|a \le X \le b) &= \frac{d}{dx} F_X(x|a \le X \le b) \\ &= \begin{cases} 0 & x < a \\ \frac{f_X(x)}{F_X(b) - F_X(a)} & a \le x \le b \\ 0 & x > b \end{cases} \end{aligned}$$

Thus if X has pdf:



then $f_X(x|a \le X \le b)$ is



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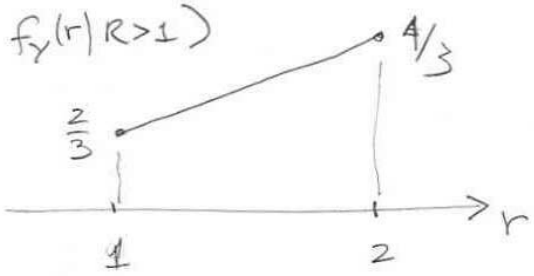
4.36

$$F_R(r|R>1) = \frac{P[R\le r, R>1]}{P[R>1]} \qquad 1 \le r \le 2$$

$$= \frac{P[1\le R\le r]}{P[R>1]} = \frac{\left(\frac{r}{2}\right)^2 - \left(\frac{1}{2}\right)^2}{1-\left(\frac{1}{2}\right)^2}$$

$$= \frac{r^2-1}{4-1} = \frac{1}{3}(r^2-1)$$

$$f_X(r|R>1) = \frac{d}{dr}\frac{1}{3}(r^2-1) = \frac{2}{3}r \qquad 1\le r\le 2$$



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4.37

(a) $$F_Z(z|\tfrac{b}{4}\le Z\le\tfrac{b}{2}) = \frac{P[Z\le z,\ \frac{b}{4}\le Z\le\frac{b}{2}]}{P[\frac{b}{4}\le Z\le\frac{b}{2}]}$$

$$= \frac{P[\frac{b}{4}\le Z\le z]}{P[\frac{b}{4}\le Z\le\frac{b}{2}]} \quad \text{for } \frac{b}{4}\le z\le\frac{b}{2}$$

$$= \frac{F_Z(z)-F_Z(\frac{b}{4})}{F_Z(\frac{b}{2})-F_Z(\frac{b}{4})} = \frac{\frac{z}{b}(2-\frac{z}{b})-\frac{1}{4}(2-\frac{1}{4})}{\frac{1}{2}(2-\frac{1}{2})-\frac{1}{4}(2-\frac{1}{4})}$$

$$= \frac{\frac{z}{b}(2-\frac{z}{b})-\frac{7}{16}}{5/16}$$

$$f_Z(z|\tfrac{b}{4}\le Z\le\tfrac{b}{2}) = \frac{16}{5}\left[\frac{2}{b}-\frac{2z}{b^2}\right] \quad \frac{b}{4}\le z\le\frac{b}{2}$$

$$= \frac{32}{5b}\left[1-\frac{z}{b}\right]$$

(b) $$F_Z(z|X>\tfrac{b}{2}) = \frac{P[\frac{b}{2}<X<z]}{P[X>\frac{b}{2}]} \quad \frac{b}{2}<z<b$$

$$= \frac{\frac{z}{b}(2-\frac{z}{b})-\frac{3}{4}}{1-\frac{3}{4}}$$

$$f_Z(z|X>\tfrac{b}{2}) = \frac{\frac{2}{b}\left[1-\frac{z}{b}\right]}{1/4} \quad \frac{b}{2}<z<b$$

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4.38

a) $$F_Y(x) = F_Y(x|B_0)P[B_0] + F_Y(x|B_1)P[B_1]$$
$$= P[Y \le x \mid X=-1](1-p) + P[Y \le x|X=1]p$$
$$= P[X+N \le x|X=-1](1-p) + P[X+N \le x|X=1]p$$
$$= P[N \le x+1](1-p) + P[N \le x-1]p$$
$$= F_N(x+1)(1-p) + F_N(x-1)p$$

$$f_Y(x) = \frac{d}{dx}F_Y(x)$$
$$= (1-p)f_N(x+1) + pf_N(x-1)$$
$$f_Y(x|B_0) = f_N(x+1) = \frac{\alpha}{2}e^{-\alpha|x+1|}$$
$$f_Y(x|B_1) = f_N(x-1) = \frac{\alpha}{2}e^{-\alpha|x-1|}$$
$$f_Y(x) = \frac{1}{2}\left[\frac{\alpha}{2}e^{-\alpha|x+1|} + \frac{\alpha}{2}e^{-\alpha|x-1|}\right] = \frac{1}{4}\alpha\left[e^{-\alpha|x+1|} + e^{-\alpha|x-1|}\right]$$

b) $$P[Y<0|B_1] = P[X+N<0|X=1] = P[N<-1]$$
$$= \frac{\alpha}{2}e^{-\alpha|-1|} = \frac{\alpha}{2}e^{-\alpha}$$
$$P[Y \ge 0|B_0] = P[X+N \ge 0|X=-1] = P[N \ge 1]$$
$$= \frac{\alpha}{2}e^{-\alpha}$$

c) $$P_E = P[Y<0|B_1]P[B_1] + P[Y \ge 0|B_0]P[B_0]$$
$$= 0.5\frac{\alpha}{2}e^{-\alpha} + 0.5\frac{\alpha}{2}e^{-\alpha} = \frac{\alpha}{2}e^{-\alpha}$$

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4.3 The Expected Value of *X*

4.39

$$f_X(x) = \frac{3}{4}(1-x^2) \qquad -1 \le x \le 1$$

$$E[X] = \frac{3}{4}\int_{-1}^{1} x(1-x^2)\,dx = \frac{3}{4}\left[\frac{x^2}{2}\Big|_{-1}^{1} - \frac{x^4}{4}\Big|_{-1}^{1}\right] = 0$$

$$E[X^2] = \frac{3}{4}\int_{-1}^{1} x^2(1-x^2)\,dx = \frac{3}{4}\left[\frac{x^3}{3}\Big|_{-1}^{1} - \frac{x^5}{5}\Big|_{-1}^{1}\right]$$

$$= \frac{3}{4}\left[\frac{2}{3} - \frac{2}{5}\right] = \frac{1}{5}$$

$$VAR[X] = E[X^2] - E[X]^2 = \frac{1}{5}$$

4.40

$$f_X(x) = 4x(1-x^2) \qquad 0 \le x \le 1$$

$$E[X] = 4\int_0^1 x\left(x(1-x^2)\right)dx = 4\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = 4\left[\frac{1}{3} - \frac{1}{5}\right]$$

$$= 8/15$$

$$E[X^2] = 4\int_0^1 x^2\left(x(1-x^2)\right)dx = 4\left[\frac{x^4}{4} - \frac{x^6}{6}\right]_0^1 = \frac{1}{3}$$

$$VAR[X] = E[X^2] - E[X]^2 = \frac{1}{3} - \frac{64}{225} = \frac{11}{225}$$

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4.41

$$f_R(r) = \frac{r}{2} \qquad 0 \le r \le 2$$

$$E[R] = \int_0^r \left(\frac{r}{2}\right) dr = \frac{1}{2}\int_0^2 r^2\, dr = \frac{1}{2}\frac{r^3}{3}\Big|_0^2 = \frac{8}{6} = \frac{4}{3}$$

$$E[R^2] = \int_0^2 r^2 \frac{r}{2} dr = \frac{1}{2}\left[\frac{r^4}{4}\right]_0^2 = \frac{16}{8} = 2$$

$$VAR[R] = E[R^2] - E[R]^2 = 2 - \frac{16}{9} = \frac{2}{9}$$

4.42

$$f_Z(z) = \frac{2}{b}\left(1 - \frac{z}{b}\right) \qquad 0 \le z \le b$$

$$E[Z] = \frac{2}{b}\int_0^b z\left(1 - \frac{z}{b}\right) dz = \frac{2}{b}\left[\frac{z^2}{2} - \frac{1}{b}\frac{z^3}{3}\right]_0^b$$

$$= \frac{2}{b}\left[\frac{b^2}{2} - \frac{1}{b}\frac{b^3}{3}\right] = b\left[1 - \frac{2}{3}\right] = \frac{b}{3}$$

$$E[Z^2] = \frac{2}{b}\int_0^b z^2\left(1 - \frac{z}{b}\right) dz = \frac{2}{b}\left[\frac{z^3}{3} - \frac{1}{b}\frac{z^4}{4}\right]_0^b$$

$$= 2b^2\left[\frac{1}{3} - \frac{1}{4}\right] = \frac{b^2}{6}$$

$$VAR[Z] = E[Z^2] - E[Z]^2 = \frac{b^2}{6} - \frac{b^2}{9} = \frac{b^2}{18}$$

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4.43

$$f_X(x) = \frac{2}{x^3} \qquad x \geq 1$$

$$E[X] = \int_1^\infty x\,\frac{2}{x^3}\,dx = \int_1^\infty \frac{2}{x^2}\,dx = -\frac{2}{x}\Big|_1^\infty = 2$$

$$E[X^2] = \int_1^\infty x^2\,\frac{2}{x^3}\,dx = \int_1^\infty \frac{2}{x}\,dx = 2\ln x\Big|_1^\infty = \infty$$

Second moment does not exist

$\Rightarrow$ variance does not exist.

Using Eqn. 4.28

$$E[X] = \int_0^\infty (1 - F_X(t))\,dt = \int_0^1 1\,dt + \int_1^\infty \frac{1}{t^2}\,dt$$

$$= 1 + -\frac{1}{t}\Big|_1^\infty = 1 + 1 = 2 \quad \checkmark$$

4.44

$$E[X] = \frac{1}{2}\int_{-1^-}^{-1^+} (-1)\,\delta(x+1)\,dx + \int_0^1 x\,\frac{1}{2}\,dx$$

$$= -1\left(\frac{1}{2}\right) + \frac{1}{2}\frac{x^2}{2}\Big|_0^1 = \frac{3}{4}$$

$$E[X^2] = \frac{1}{2}\int_{-1^-}^{-1^+} (-1)^2\,\delta(x+1)\,dx + \int_0^1 \frac{1}{2}x^2\,dx$$

$$= \frac{1}{2} + \frac{1}{2}\frac{x^3}{3}\Big|_0^1 = \frac{1}{2} + \frac{1}{6} = \frac{4}{6}$$

$$VAR[X] = E[X^2] - E[X]^2$$

$$= \frac{4}{6} - \frac{9}{16} = \frac{32-27}{48} = \frac{5}{48}$$

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4.45

$$f_X(x) = \frac{3}{4}\delta(x) + \frac{1}{4}\,2e^{-2x} \quad x > 0.$$

$$E[X] = \int_0^\infty x f_X(x)\,dx = 0 + \frac{1}{4}\underbrace{\int_0^\infty 2x e^{-2x}\,dx}_{\text{mean of exponential RV}}$$

$$= \frac{1}{4}\,\frac{1}{2} = \frac{1}{8}$$

$$E[X^2] = \int_0^\infty x^2 f_X(x)\,dx = 0 + \frac{1}{4}\underbrace{\int_0^\infty 2x^2 e^{-2x}\,dx}_{\substack{\text{2nd moment of} \\ \text{exponential} \\ \text{RV} \\ \frac{2}{2^2}}}$$

$$= \frac{1}{4}\,\frac{1}{2}$$

$$= \frac{1}{8}$$

see Problem 4.48 for solution of this integral

$$VAR[X] = E[X^2] - E[X]^2$$

$$= \frac{1}{8} - \left(\frac{1}{8}\right)^2$$

$$= \frac{7}{64}$$

4.46

$$\begin{aligned}
\mathcal{E}[X] &= \int_{-\infty}^{\infty} x\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-m)^2/2\sigma^2} = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}(y+m)e^{-y^2/2\sigma^2}dy \\
&= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} ye^{-y^2/2\sigma^2}dy + \frac{m}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-y^2/2\sigma^2}dy \\
&= \frac{-\sigma^2}{\sqrt{2\pi}\sigma}\left[e^{-y^2/2\sigma^2}\right]_{-\infty}^{\infty} + m = m \\
\sigma_X^2 &= \int_{-\infty}^{\infty}(x-m)^2\frac{e^{-(x-m)^2/2\sigma^2}}{\sqrt{2\pi}\sigma}dx = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} y^2e^{-y^2/2\sigma^2}dy \\
&= \frac{1}{\sqrt{2\pi}\sigma}\left\{\left[-\sigma^2 ye^{-y^2/2\sigma^2}\right]_{-\infty}^{\infty} + \sigma^2\int_{-\infty}^{\infty} e^{-y^2/2\sigma^2}dy\right\} \\
&= \sigma^2 \quad \text{where we used integration by parts with} \\
&\qquad u = y \quad dv = e^{-y^2/2\sigma^2}
\end{aligned}$$

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4.47

$$\begin{aligned}\int_0^x t f_X(t)dt &= tF_X(t)|_0^x - \int_0^x F_X(t)dt \text{ where we let } u = t,\ dv = f_X(t) \\ &= xF_X(x) - \int_0^x F_X(t)dt \\ &= x(F_X(x)-1) + \int_0^x (1-F_X(t))dt\end{aligned}$$

If $\mathcal{E}[X] < \infty$ then the first term on the right-hand side approaches zero as $x \to \infty$, so

$$\mathcal{E}[X] = \int_0^t t f_X(t)dt = \int_0^\infty (1-F_X(t))dt$$

For the discrete case, we have

$$\begin{aligned}\sum_{k=0}^{\infty} P[X>k] &= \sum_{k=0}^{\infty}\left(\sum_{j=k+1}^{\infty} P[X=j]\right) \\ &= (P[X=1]+P[X=2]+\ldots)+(P[X=2]+\ldots)+(P[X=3]+\ldots)+\ldots \\ &= P[X=1]+2P[X=2]+3P[X=3]+\ldots \\ &= \sum_{k=0}^{\infty} kP[X=k] \triangleq \mathcal{E}[X]\end{aligned}$$

4.48

$$E[X]^2 = \left(\frac{1}{\lambda}\right)^2$$

$$E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$$

$u = x^2 \qquad dv = \lambda e^{-\lambda x} dx$

$du = 2x\,dx \qquad v = -e^{-\lambda x}$

$$= -x^2 e^{-\lambda x}\Big|_0^\infty + \int_0^\infty e^{-\lambda x}(2x)\,dx$$

$u = 2x \qquad dv = e^{-\lambda x} dx$

$du = 2dx \qquad v = -\frac{1}{\lambda} e^{-\lambda x}$

$$= -x^2 e^{-\lambda x}\Big|_0^\infty + \left(-\frac{2x}{\lambda} e^{-\lambda x}\right)\Big|_0^\infty + \int_0^\infty \frac{1}{\lambda} e^{-\lambda x}(2)\,dx$$

$$= -\frac{2}{\lambda^2} e^{-\lambda x}\Big|_0^\infty = \frac{2}{\lambda^2}$$

$$VAR[X] = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

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4.49

a) $$E[X] = \int_0^\infty x\frac{\beta}{\lambda}\left(\frac{x}{\lambda}\right)^{\beta-1} e^{-\left(\frac{x}{\lambda}\right)^\beta} dx = \int_0^\infty \beta\left(\frac{x}{\lambda}\right)^\beta e^{-\left(\frac{x}{\lambda}\right)^\beta} dx$$

$$y = \left(\frac{x}{\lambda}\right)^\beta \qquad dy = \frac{\beta}{\lambda}\left(\frac{x}{\lambda}\right)^{\beta-1} dx$$

$$x = y^{1/\beta}\lambda \qquad dy = \frac{\beta}{\lambda} y^{1-\frac{1}{\beta}} dx$$

$$= \int_0^\infty \beta\left(y^{1/\beta}\frac{\lambda}{\lambda}\right)^\beta e^{-y} \frac{\lambda}{\beta} y^{-(1-\frac{1}{\beta})} dy$$

$$= \int_0^\infty \lambda y y^{-1+\frac{1}{\beta}} e^{-y} dy$$

$$= \int_0^\infty \lambda y^{1+\frac{1}{\beta}-1} e^{-y} dy$$

$$= \lambda\,\Gamma\left(1+\frac{1}{\beta}\right)$$

b) $$E[X^2] = \int_0^\infty x^2 \frac{\beta}{\lambda}\left(\frac{x}{\lambda}\right)^{\beta-1} e^{-\left(\frac{x}{\lambda}\right)^\beta} dx = \int_0^\infty (y^{1/\beta}\lambda)\,\beta\left(\frac{y^{1/\beta}\lambda}{\lambda}\right)^\beta e^{-y} \frac{\lambda}{\beta} y^{\frac{1}{\beta}-1} dy$$

$$= \int_0^\infty \lambda^2 y^{\frac{1}{\beta}} y\, y^{\frac{1}{\beta}-1} e^{-y} dy$$

$$= \int_0^\infty \lambda^2 y^{1+\frac{2}{\beta}-1} e^{-y} dy$$

$$= \lambda^2\,\Gamma\left(\frac{2}{\beta}+1\right)$$

$$VAR[X] = E[X^2] - E[X]^2$$

$$= \lambda^2\,\Gamma\left(1+\frac{2}{\beta}\right) - \lambda^2\,\Gamma\left(1+\frac{1}{\beta}\right)^2$$

4.50

3.72 $$\mathcal{E}[X] = \int_{-\infty}^0 x\frac{1}{\pi(1+x^2)}dx + \int_0^\infty \frac{x}{\pi(1+x^2)}dx$$

Consider the latter term:

$$\int_0^y \frac{x}{\pi(1+x^2)}dx = \frac{1}{2\pi}\ln(1+x^2)\Big|_0^y == \frac{\ln(1+y)}{2\pi} \to \infty$$

Thus the integrals do not exist $\Rightarrow \mathcal{E}[X]$ does not exist.

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4.51

$$E[X] = \int_0^\infty (1 - F_X(t))\,dt = \int_0^{x_m} dt + \int_{x_m}^\infty \left(\frac{x_m}{x}\right)^\alpha dx$$

$$= x_m - \frac{x_m^\alpha}{(\alpha-1)} \frac{1}{x^{\alpha-1}}\Big|_{x_m}^\infty$$

$$= x_m + \frac{x_m^\alpha}{\alpha-1} \frac{1}{x_m^{\alpha-1}}$$

$$= x_m + \frac{x_m}{\alpha-1}$$

$$= \frac{\alpha x_m}{\alpha-1} \quad \rightarrow E[X] \text{ DNE } \alpha = 1$$

Alternatively, consider

$$E[X] = \int_{x_m}^\infty x \cdot \alpha \frac{x_m^\alpha}{x^{\alpha+1}}\,dx$$

$$= \alpha x_m^\alpha \int_{x_m}^\infty \frac{1}{x^\alpha}\,dx \quad \text{when } \alpha = 1 \quad \int_{x_m}^\infty \frac{1}{x}\,dx$$

does not exist.

4.52

$$\mathcal{E}[c] = \int_{-\infty}^{\infty} c f_X(x)dx = c\int_{-\infty}^{\infty} f_X(x)dx = c$$

$$\mathcal{E}[c^2] = \int_{-\infty}^{\infty} c^2 f_X(x)dx = c^2$$

$$VAR[c] = \mathcal{E}[c^2] - \mathcal{E}[c]^2 = c^2 - c^2 = 0 \qquad 4.36\ (3.68)$$

$$VAR[X+c] = \mathcal{E}[((X+c) - \mathcal{E}[X+c])^2]$$

$$= \mathcal{E}[(X + C - \mathcal{E}(X) - C)^2]$$

$$= \mathcal{E}[(X - \mathcal{E}(X))^2] = VAR[X] \qquad 4.37\ (3.69)$$

$$VAR[cX] = \mathcal{E}[(cX - \mathcal{E}[cX])^2]$$

$$= \mathcal{E}[c^2(X - \mathcal{E}[X])^2]$$

$$= c^2 VAR[X] \qquad 4.38\ (3.70)$$

4.53

$$Y = A\cos\omega t + c$$

$$E[Y] = E[A]\cos\omega t + c = m\cos\omega t + c$$

$$VAR[Y] = \cos^2\omega t\, VAR[A] = \sigma^2\cos^2\omega t$$

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4.54

3.71

(a)

$$\begin{aligned}
\mathcal{E}[Y] &= \int_{-\infty}^{\infty} g(x)f_X(x)dx \quad \text{write integral into three parts} \\
&= -a\int_{-\infty}^{-a} f_X(x)dx + \int_{-a}^{a} xf_X(x)dx + a\int_{a}^{\infty} f_X(x)dx \\
&= -aF_X(-a) + \int_{-a}^{a} xf_X(x)dx + a(1-F_X(a^-)) \\
\mathcal{E}[Y^2] &= a^2F_X(-a) + \int_{-a}^{a} x^2f_X(x)dx + a^2(1-F_X(a^-)) \\
VAR[Y] &= \mathcal{E}[Y^2] - \mathcal{E}[Y]^2
\end{aligned}$$

4.54 (b)

$$E[Y] = \underbrace{(-1)P[Y\le -1]}_{\frac{1}{2}e^{-1}} + \underbrace{(1)P[Y\ge 1]}_{\frac{1}{2}e^{-1}} + \underbrace{\int_{-1}^{1} x\,\tfrac{1}{2}e^{-|x|}dx}_{\text{odd}} = \underbrace{0}_{=0}$$

$$VAR[Y] = E[Y^2] = (-1)^2P[Y\le -1] + (1)^2P[Y\ge 1] + \int_{-1}^{1} x^2\,\tfrac{1}{2}e^{-|x|}dx$$

$$= e^{-1} + 2\cdot\tfrac{1}{2}\int_{0}^{1} x^2e^{-x}dx$$

$$\int_0^1 x^2e^{-x}dx = e^{-x}(x^2+2x+2)\Big|_0^1 \quad \text{from appendix}$$

$$= e^{-1} + 5e^{-1} - 2 = 6e^{-1} - 2$$

(c)

$$E[Y] = (-\tfrac{1}{2})\underbrace{P[X\le -\tfrac{1}{2}]}_{5/32} + (\tfrac{1}{2})\underbrace{P[X\ge \tfrac{1}{2}]}_{5/32} + \int_{-\frac{1}{2}}^{\frac{1}{2}} x\cdot\tfrac{3}{4}(1-x^2)dx = 0$$

(odd)

$$VAR[Y] = E[Y^2] = (\tfrac{1}{2})^2P[X\le -\tfrac{1}{2}] + (\tfrac{1}{2})^2P[X\ge \tfrac{1}{2}] + \tfrac{3}{4}\int_{-\frac{1}{2}}^{\frac{1}{2}} x^2(1-x^2)dx$$

$$= 2\cdot\tfrac{1}{4}\cdot\tfrac{5}{32} + \tfrac{3}{4}\cdot 2\int_0^{\frac{1}{2}}(x^2-x^4)dx$$

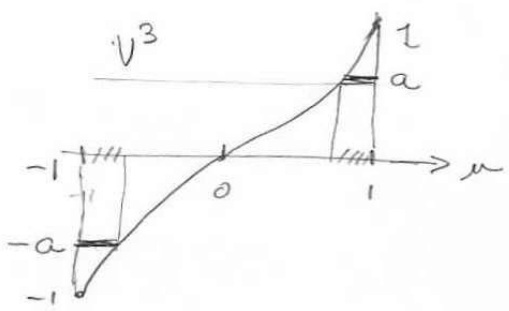
$$\int_0^{1/2}(x^2-x^4)dx = \frac{x^3}{3} - \frac{x^5}{5}\Big|_0^{1/2} = \frac{1}{24} - \frac{1}{160}$$

$$= \frac{5}{64} + \frac{3}{2}\left(\frac{17}{480}\right)$$

$$= 0.13125$$

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(4.54) (d)



$$\mu_+^3 = \frac{1}{2}$$
$$\mu_+ = \left(\frac{1}{2}\right)^{1/3} = .7937$$
$$\mu_-^3 = -\frac{1}{2}$$
$$\mu_- = \left(-\frac{1}{2}\right)^{1/2} = -.7937$$

$$Y = \begin{cases} \frac{1}{2} & \mu^3 > \frac{1}{2} \\ -\frac{1}{2} & \mu^3 < -1/2 \\ \mu^3 & -\frac{1}{2} < \mu^3 < \frac{1}{2} \end{cases}$$

$$E[Y] = \underbrace{\frac{1}{2}P\left[Y \geq \frac{1}{2}\right]}_{\frac{1-(\frac{1}{2})^{1/3}}{2}} - \underbrace{\frac{1}{2}P\left[Y \leq \frac{1}{2}\right]}_{\frac{(-\frac{1}{2})^{1/3}-(-1)}{2}} + \underbrace{\int_{-1/2}^{1/2} \mu^3 \frac{du}{2}}_{0}$$

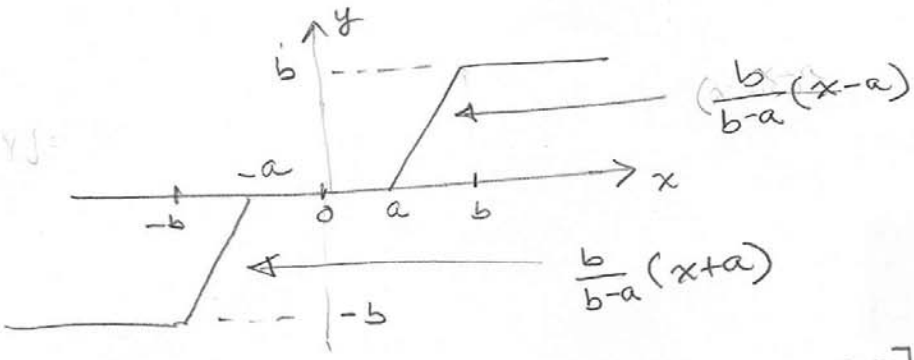
$$= 0$$

$$VAR[Y] = E[Y^2]$$
$$= \left(\frac{1}{2}\right)^2 P\left[Y \geq \frac{1}{2}\right] + \left(-\frac{1}{2}\right)^2 P\left[Y \leq \frac{1}{2}\right] + \int_{-1/2}^{1/2} \mu^6 \frac{du}{2}$$
$$= 2\left(\frac{1}{4}\right)\frac{\left(1-\left(\frac{1}{2}\right)^{1/3}\right)}{2} + \frac{2}{7}\left(\frac{1}{2}\right)^7$$
$$= \frac{1}{4}\left(1-\left(\frac{1}{2}\right)^{1/3}\right) + \frac{2}{7}\left(\frac{1}{2}\right)^7$$

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4.55

(a)



$$E[Y] = -b\,P[X \le -b] + b\,P[X \ge b] + 0 \cdot P[-a \le X \le a]$$
$$+ \int_{-b}^{a} \frac{b}{b-a}(x+a) f_X(x)\,dx + \int_{a}^{b} \frac{b}{b-a}(x-a) f_X(x)\,dx$$

$$E[Y^2] = b^2\,P[X \le -b] + b^2\,P[X \ge b]$$
$$+ \int_{-b}^{a} \frac{b^2}{(b-a)^2}(x+a)^2 f_X(x)\,dx + \int_{a}^{b} \frac{b^2}{(b-a)^2}(x-a)^2 f_X(x)\,dx$$

$$VAR[Y] = E[Y^2] - E[Y]^2$$

(b) $f_X(x) = \frac{1}{2} e^{-|x|}\,dx \quad -\infty < x < \infty, \quad a = 1, \; b = 2$

$$E[Y] = -2\underbrace{P[X \le -2]}_{\frac{1}{2}e^{-2}} + 2\underbrace{P[X \ge 2]}_{\frac{1}{2}e^{-2}}$$
$$+ \int_{-2}^{1} 2(x+1) e^{x}\,dx + \underbrace{\int_{1}^{2} 2(x-1) e^{-x}\,dx}_{=0}$$

$$E[Y^2] = 4 \cdot \tfrac{1}{2} e^{-2} + 4 \cdot \tfrac{1}{2} e^{-2} + 4\int_{-2}^{1} (x+1)^2 e^{x}\,dx + 4\int_{1}^{2} (x-1)^2 e^{-x}\,dx$$
$$= 4e^{-2} + 8\int_{1}^{2} (x-1)^2 e^{-x}\,dx$$

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4.55 (c) $\int_1^2 (x-1)^2 e^{-x}\,dx = \int_0^1 y^2 e^{-y-1}\,dy = e^{-1}\left[e^{-y}(x^2+2x+2)\right]_0^1$

$= e^{-1}\left[5e^{-1} - 2\right] = 5e^{-2} - 2e^{-1}$

$\therefore \text{VAR}[Y] = 4e^{-2} + 40e^{-2} - 16e^{-1} = 44e^{-2} - 16e^{-1}$

(c) $a = \frac{1}{2}$ $b = \frac{3}{2}$ $P[X \le -\frac{3}{2}] = 0 = P[X \ge \frac{3}{2}]$

$f_X(x)$: $\frac{1}{2}\delta(x+1)$ at -1; $\frac{1}{2}$ on $(0,1)$; x

$E[Y] = \int_{-3/2}^{-1/2} \frac{3}{2}\left(x+\frac{1}{2}\right)\frac{1}{2}\delta(x+1)\,dx + \int_{1/2}^{1} \frac{3}{2}\left(x-\frac{1}{2}\right)\cdot\frac{dx}{2}$

$= \frac{3}{4}\left(-1+\frac{1}{2}\right) + \frac{3}{4}\int_0^{1/2} x'\,dx' = -\frac{3}{8} + \frac{3}{4}\frac{1}{2}\left(\frac{1}{2}\right)^2 = -\frac{9}{32}$

$E[Y^2] = \int_{-3/2}^{-1/2} \frac{9}{4}\left(x+\frac{1}{2}\right)^2\frac{1}{2}\delta(x+1)\,dx + \frac{9}{4}\int_{1/2}^{1}\left(x-\frac{1}{2}\right)^2\frac{dx}{2}$

$= \frac{9}{4}\left(-\frac{1}{2}\right)^2\left(\frac{1}{2}\right) + \frac{9}{8}\int_0^{1/2} x'^2\,dx' = \frac{21}{64}$

$\text{VAR}(Y) = \frac{21}{64} - \left(\frac{9}{32}\right)^2 = \frac{255}{1024}$

(d)

$$Y = \begin{cases} \left(\frac{1}{2}-1\right) & u^3 > \frac{1}{2} \\ 2\left(u^3 - \frac{1}{4}\right) & \frac{1}{4} < u^3 < \frac{1}{2} \\ 0 & -\frac{1}{4} < u^3 < \frac{1}{4} \\ 2\left(u^3 + \frac{1}{4}\right) & -\frac{1}{2} < u^3 < -\frac{1}{4} \\ -\frac{1}{2} & u^3 < -\frac{1}{2} \end{cases}$$

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4.55d – continued –

$$E[Y] = \underbrace{\tfrac{1}{2}P[U^3 > \tfrac{1}{2}] - \tfrac{1}{2}P[U^3 < -\tfrac{1}{2}]}_{=0} + \int_{-1/2}^{-1/4} 2(u^3+\tfrac{1}{4})\frac{du}{2} + \int_{1/4}^{1/2} 2(u^3-\tfrac{1}{4})\frac{du}{2}$$

let $u' = -u$

$$= 0$$

$$E[Y] = 0$$

$$VAR[Y] = E[Y^2] = (\tfrac{1}{2})^2 P[U^3 > \tfrac{1}{2}] + (-\tfrac{1}{2})^2 P[U^3 < -\tfrac{1}{2}] + \int_{-1/2}^{-1/4} 4(u^3+\tfrac{1}{4})^2\frac{du}{2} + \int_{1/4}^{1/2} 4(u^3-\tfrac{1}{4})^2\frac{du}{2}$$

$$= 2(\tfrac{1}{2})^2\frac{1-(\tfrac{1}{2})^{1/3}}{2} + 2\cdot 4\int_{1/4}^{1/2}\underbrace{(u^3-\tfrac{1}{4})^2}_{u^6-\frac{1}{2}u^3+\frac{1}{16}}\frac{du}{2}$$

$$= \frac{1-(\tfrac{1}{2})^{1/3}}{1/4} + 4\int_{1/4}^{1/2} u^6\,du - 2\int_{1/4}^{1/2} u^3\,du + \frac{1}{4}\int_{1/4}^{1/2} du$$

$$\frac{u^7}{7}\Big|_{1/4}^{1/2} \qquad \frac{u^4}{4}\Big|_{1/4}^{1/2} \qquad \Big|_{1/4}^{1/2}$$

$$= \frac{1-(\tfrac{1}{2})^{1/3}}{1/4} + \frac{4}{7}\left[(\tfrac{1}{2})^7 - (\tfrac{1}{4})^7\right] - \frac{1}{2}\left[(\tfrac{1}{2})^4 - (\tfrac{1}{4})^4\right] + \frac{1}{16}$$

$$= 4(.7937) + \frac{4}{7}(\tfrac{1}{2})^7\left[1-(\tfrac{1}{2})^7\right] - \frac{1}{2}(\tfrac{1}{2})^4\left[1-(\tfrac{1}{2})^4\right] + \frac{1}{16}$$

$$= 3.212$$

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4.56

a) $E[Y] = 3E[X]+2$

$VAR[Y] = VAR[3X+2] = VAR[3X] = 9\,VAR[X]$

b) Laplacian R.V. $E[X]=0$

$VAR[X] = \frac{2}{\alpha^2}$

$E[Y] = 2$

$VAR[Y] = 9\left(\frac{2}{\alpha^2}\right) = \frac{18}{\alpha^2}$

c) Gaussian R.V. $E[X]=m$

$VAR[X]=\sigma^2$

$E[Y] = 3m+2$

$VAR[Y] = 9\sigma^2$

d) $E[X] = b\int_0^1 \cos(2\pi u)\,du = -b\sin(2\pi u)\Big|_0^1 = 0$

$VAR[X] = b^2\int_0^1 \cos^2(2\pi u)\,du$

$= b^2\int_0^1 \frac{1}{2}du + \frac{b^2}{2}\int_0^1 \cos 4\pi u\,du$

$= b^2\frac{1}{2} + b^2\left(\frac{1}{4\pi}\right)(-\sin 4\pi u)\Big|_0^1$

$= \frac{b^2}{2}$

$E[Y]=2$

$VAR[Y] = \frac{9b^2}{2}$

4.57

$$\mathcal{E}[X^n] = \int_0^1 x^n dx = \frac{x^{n+1}}{n+1}\Big|_0^1 = \frac{1}{n+1}$$

$$\mathcal{E}[Y^n] = \frac{1}{b-a}\int_a^b y^n dy = \frac{1}{b-a}\left[\frac{b^{n+1}-a^{n+1}}{n+1}\right]$$

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4.58

a) $$F_X(x|d\le x\le 2d)=\begin{cases}0 & x\le d\\ \dfrac{F_X(x)-F_X(d)}{F_X(2d)-F_X(d)} & d\le x\le 2d\\ 1 & x\ge 2d\end{cases}$$

$$f_X(x|d\le x\le 2d)=\frac{f_X(x)}{F_X(2d)-F_X(d)} \qquad d\le x\le 2d$$

$$f_X(x)=\frac{1}{2x_{max}} \qquad -x_{max}\le x\le x_{max}$$

$$F_X(x)=\frac{x+x_{max}}{2x_{max}} \qquad -x_{max}\le x\le x_{max}$$

$$f_X(x|d\le x\le 2d)=\frac{1}{2d+x_{max}-d-x_{max}}=\frac{1}{d}$$

b) $$E[X|d\le X\le 2d]=\int_{-\infty}^{\infty} x f_X(x|d\le x\le 2d)\,dx=\int_d^{2d}\frac{x}{d}dx$$
$$=\frac{1}{d}\frac{x^2}{2}\Big|_d^{2d}=\frac{1}{d}\left(\frac{4d^2}{2}-\frac{d^2}{2}\right)=\frac{3d}{2}$$

$$VAR[X|d\le X\le 2d]=\int_{-\infty}^{\infty} x^2 f_X(x|d\le x\le 2d)dx-E^2[X|d\le x\le 2d]$$
$$=\int_d^{2d}\frac{x^2}{d}dx-\left(\frac{3d}{2}\right)^2$$
$$=\frac{1}{d}\frac{x^3}{3}\Big|_d^{2d}-\frac{9d^2}{4}=\frac{7d^2}{3}-\frac{9d^2}{4}=\frac{d^2}{12}$$

c) $$E[(X-c)^2|d<X<2d]=\int_d^{2d}\frac{x^2}{d}dx-2c\int_d^{2d}\frac{x}{d}dx+c^2\int_d^{2d}\frac{1}{d}dx$$
$$=\frac{7d^2}{3}-2c\left(\frac{3d}{2}\right)+c^2\left(\frac{1}{d}\right)d=c^2-3cd+\frac{7}{3}d^2$$

d) $2e-3d+0=0$

$c=\dfrac{3d}{2}$ it is the midpoint of the interval $(d, 2d)$

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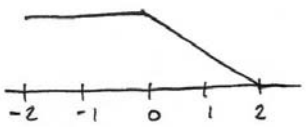
4,4 Important Continuous Random Variables

4.59

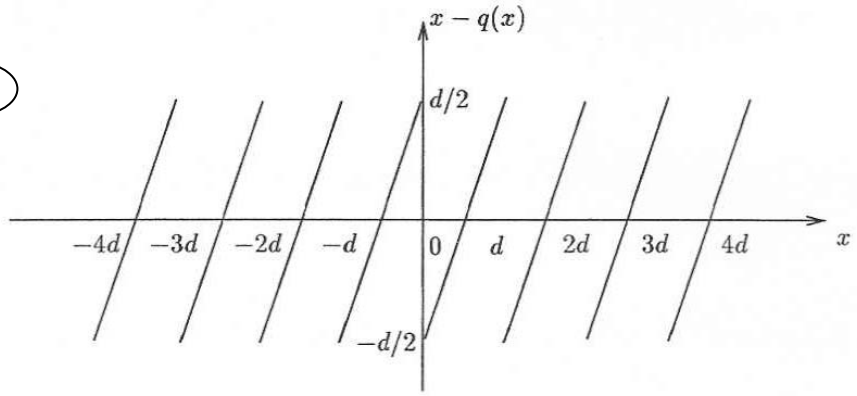
$$P[|X|>x] = P[\{X>x\}\cup\{X<-x\}]$$
$$= P[X>x] + P[X<-x]$$
$$= 1 - F_X(x) + F_X(-x)$$

$$F_X(x) = \begin{cases} 0 & x\le -2 \\ \frac{x+2}{4} & -2\le x\le 2 \\ 1 & x\ge 2 \end{cases}$$

$$P[|X|>x] = \begin{cases} 1 & x\le 0 \\ 1-\left(\frac{x+2}{4}\right)+\left(\frac{-x+2}{4}\right) = 1-\frac{x}{2} & 0\le x\le 2 \\ 0 & x\ge 2 \end{cases}$$



4.60



for $-\frac{d}{2} < y < \frac{d}{2}$ the equation $y = x - q(x)$ has 8 roots, thus from Eqn. 3.55:

$$f_Y(y) = \sum_{k=1}^{8} \frac{f_X(x_k)}{\frac{dy}{dx}|_{x=x_k}}$$

Since $x-q(x)$ consists of piecewise linear unit-slope segments, we have that $\left.\frac{dy}{dx}\right|_{x=x_k} =$ 1 all x_k.

Thus

$$f_Y(y) = \sum_{k=1}^{9} f_X(x_k) = \sum_{k=1}^{8} \frac{1}{8d} = \frac{1}{d}$$

for $-\frac{d}{2} < y < \frac{d}{2}$ ✓

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4.61

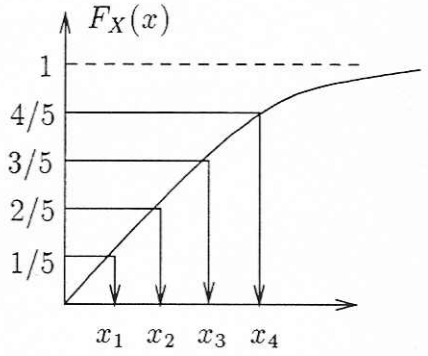
~~3.18~~ a) $P[X \leq d] = F_X(d) = 1 - e^{-\lambda d} \qquad d > 0$

$$\begin{aligned} P[kd \leq X \leq (k+1)d] &= F_X((k+1)d) - F_X(kd) \\ &= e^{-\lambda kd} - e^{-\lambda(k+1)d} = e^{-\lambda kd}(1 - e^{-\lambda d}) \end{aligned}$$

$$P[X > kd] = 1 - P[X \leq kd] = 1 - F_X(kd) = e^{-\lambda kd}$$

b) Find x_k, $k = 1, 2, 3, 4$ such that

$$\begin{aligned} F_X(x_k) &= \frac{k}{5} \\ &= 1 - e^{-\lambda x_k} \end{aligned}$$



$$x_1 = \frac{\ln \frac{5}{4}}{\lambda} \qquad x_2 = \frac{\ln \frac{5}{3}}{\lambda} \qquad x_3 = \frac{\ln \frac{5}{2}}{\lambda} \qquad x_4 = \frac{\ln 5}{\lambda}$$

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4.62 a)

$$P[X \leq x] = 1 - e^{-\lambda x} = \frac{r}{100}$$

$$1 - \frac{r}{100} = e^{-\lambda x}$$

$$\Rightarrow \pi(r) = x = -\frac{1}{\lambda}\ln\left(1 - \frac{r}{100}\right) = \frac{1}{\lambda}\ln\left(\frac{100}{100 - r}\right)$$

$$\pi(90) \cong \frac{23}{\lambda} \quad \pi(95) \approx \frac{3}{\lambda} \quad \pi(99) \approx \frac{4.6}{\lambda}$$

b) $P[X \leq x] = 1 - Q\left(\frac{x}{\sigma}\right)$ Using Tables 4.2 and 4.3:

$$1 - Q\left(\frac{x}{\sigma}\right) = 0.90 \Rightarrow \tfrac{x}{\sigma} = 1.28 \Rightarrow \pi(90) = 1.28\sigma$$

$$1 - Q\left(\frac{x}{\sigma}\right) = 0.95 \Rightarrow \tfrac{x}{\sigma} \approx 1.5 \Rightarrow \pi(95) \approx 1.5\sigma$$

$$1 - Q\left(\frac{x}{\sigma}\right) = 0.99 \Rightarrow \tfrac{x}{\sigma} \approx 2.33 \Rightarrow \pi(99) \approx 2.33\sigma$$

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4.63

a) $P[X>4] = 1-F_X(4) = 1-\Phi\left(\frac{4-5}{4}\right) = 1-\Phi\left(-\frac{1}{4}\right) = \Phi\left(\frac{1}{4}\right) = 0.598$

$P[X\geq 7] = 1-F_X(7) = 1-\Phi\left(\frac{7-5}{4}\right) = 1-\Phi\left(\frac{1}{2}\right) = 0.308$

$P[6.72<X<10.16] = \Phi\left(\frac{10.16-5}{4}\right)-\Phi\left(\frac{6.72-5}{4}\right) = \Phi(1.29)-\Phi(0.43) = 0.235$

$P[2<X<7] = \Phi\left(\frac{7-5}{4}\right)-\Phi\left(\frac{2-5}{4}\right) = \Phi\left(\frac{1}{2}\right)-\Phi\left(-\frac{3}{4}\right) = 0.465$

$P[6\leq X\leq 8] = \Phi\left(\frac{8-5}{4}\right)-\Phi\left(\frac{6-5}{4}\right) = \Phi\left(\frac{3}{4}\right)-\Phi\left(\frac{1}{4}\right) = 0.175$

b) $P[X<a] = 0.8869$

$\Phi\left(\frac{a-5}{4}\right) = 0.8869 = 1-Q(x)$

$Q(x) = 0.1131 \rightarrow x = 1.2 = \frac{a-5}{4} \rightarrow a = 9.8$

c) $P[X>b] = 1-\Phi\left(\frac{b-5}{4}\right) = 0.11131$

$Q(x) = 0.11131 \rightarrow x = 1.2 = \frac{b-5}{4} \rightarrow b = 9.8$

d) $P[13<X\leq c] = 0.0123$

$\Phi\left(\frac{c-5}{4}\right)-\Phi\left(\frac{13-5}{4}\right) = \Phi\left(\frac{c-5}{4}\right)-\Phi(2) = 0.0123$

$\Phi\left(\frac{c-5}{4}\right) = 0.0123+0.9772 = 0.9895$

$Q\left(\frac{c-5}{4}\right) = 0.0105 \rightarrow x = 2.3 = \frac{c-5}{4} \rightarrow c = 14.2$

4.64

$$\begin{aligned} Q(-x) &= \frac{1}{\sqrt{2\pi}}\int_{-x}^{\infty} e^{t^2/2}dt = 1-\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{-x} e^{-t^2/2}dt \\ &= 1-\frac{1}{\sqrt{2\pi}}\int_{\infty}^{x} e^{-t'^2/2}(-dt') \text{ where } t' = -t \\ &= 1-\frac{1}{\sqrt{2\pi}}\int_{x}^{\infty} e^{-t'^2/2}dt' = 1-Q(x) \end{aligned}$$

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4.65

To generate Q values

```
> x = [0:0.1:10];
> format shorte
> 1-normal_cdf(x)
```

To generate $Q(x_k) = 10^{-k}$

$$10^{-k} = Q(x_k) = 1 - Q(-x_k) = F_X(-x_k)$$

$$x_k = -F_X^{-1}(10^{-k})$$

```
> k = [1:1:10];
> p = ones(1,10)/10;
> p2 = p.^k;
> -normal_inv(p2)
```

4.66

$$P[X < m] = P[X \leq m] = \Phi\left(\frac{n-m}{\sigma}\right) = \Phi(0) = \frac{1}{2}$$

$$\begin{aligned} P[|X-m| > k\sigma] &= 1 - P[-k\sigma + m \leq X \leq m + k\sigma] \\ &= 1 - \left(\Phi\left(\frac{m+k\sigma-m}{\sigma}\right) - \Phi\left(\frac{m-k\sigma-m}{\sigma}\right)\right) \\ &= \underbrace{1-\Phi(k)}_{Q(k)} + \underbrace{\Phi(-k)}_{Q(k)} \\ &= Q(k) + Q(k) = 2Q(k) \end{aligned}$$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	from 4.2 Table ~~3.3~~
$2Q(k)$	0.318	$4.56(10^{-2})$	$4.05(10^{-3})$	$6.34(10^{-5})$	$5.74(10^{-7})$	

$$P[X > m + k\sigma] = Q\left(\frac{m+k\sigma-m}{\sigma}\right) = Q(k)$$

	$k=1.28$	$k=3.09$	$k=4.26$	$k=5.20$	from 4.3 Table ~~3.4~~
$Q(k)$	$\approx 10^{-1}$	$\approx 10^{-3}$	$\approx 10^{-5}$	$\approx 10^{-7}$	

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4.67

a) $F_Y(X+N \le y \mid X=+1) = F_N(y-1)$

$F_Y(X+N \le y \mid X=-1) = F_N(y+1)$

$f_Y(y \mid X=+1) = f_N(y-1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(y-1)^2/2\sigma^2}$

$f_Y(y \mid X=-1) = f_N(y+1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(y+1)^2/2\sigma^2}$

b) $f_Y(y \mid X=-1)\,P[X=-1] > f_Y(y \mid X=+1)\,P[X=+1]$ decide "0"

$\frac{e^{-(y+1)^2/2\sigma^2}}{\sigma\sqrt{2\pi}}(3p_1) > \frac{e^{-(y-1)^2/2\sigma^2}}{\sigma\sqrt{2\pi}} p_1$

$3e^{-(y+1)^2/2\sigma^2} e^{(y-1)^2/2\sigma^2} > 1$

$3e^{-y^2-2y-1+y^2-2y+1/2\sigma^2} > 1$

$3e^{-4y/2\sigma^2} > 1$

$-\frac{4y}{2\sigma^2} > \ln\left(\frac{1}{3}\right)$

$y < -\frac{\sigma^2}{2}\ln\left(\frac{1}{3}\right)$ decide "0" $T = -\frac{\sigma^2}{2}\ln\left(\frac{1}{3}\right)$

c) $P[X+N<T \mid X=+1] = P[N<T-1] = \Phi\left(\frac{T-1}{\sigma^2}\right)$

$P[X+N \ge T \mid X=-1] = P[N \ge T+1] = \left(1-\Phi\left(\frac{T+1}{\sigma^2}\right)\right)$

d) $P[X+N<T \mid X=+1]\,P[X=+1] + P[X+N \ge T \mid X=-1]\,P[X=-1] =$

$= p_1\,\Phi\left(\frac{T-1}{\sigma^2}\right) + \left(1-\Phi\left(\frac{T+1}{\sigma^2}\right)\right)3p_1$

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4.68

$$P[X_1 > x] = Q\left(\frac{x-20}{5}\right)$$
$$P[X_2 > x] = Q\left(\frac{x-22}{1}\right)$$
$$P[X_1 > 20] = Q\left(\frac{20-20}{5}\right) = Q(0) = \frac{1}{2}$$
$$P[X_2 > 20] = Q\left(\frac{20-22}{1}\right) = Q(-2) = 1 - Q(2) = 0.9722$$

} pick #2

$$P[X_1 > 24] = Q\left(\frac{24-20}{5}\right) = Q(1) = 0.159 \quad (0.8413)$$
$$P[X_2 > 24] = Q\left(\frac{24-22}{1}\right) = Q(2) = 0.023$$

} pick #1

Note that in the second case, the chip with the smaller mean (but larger variance) is selected.

4.69

$$P[X \geq 10] = 1 - P[X \leq 10]$$
$$= \sum_{k=0}^{7-1} \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$
$$= \sum_{k=0}^{6} \frac{10^k}{k!} e^{-10}$$
$$= 0.1301$$

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4.70) Gamma RV

(a) $$E[X] = \int_0^\infty x \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} dx = \int_0^\infty \frac{(\lambda x)^{\alpha} e^{-\lambda x}}{\lambda \Gamma(\alpha)} dx$$

$$= \int_0^\infty \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \frac{\lambda(\lambda x)^{\alpha}}{\Gamma(\alpha+1)} e^{-\lambda x} dx = \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \underbrace{\int_0^\infty \frac{\lambda(\lambda x)^{\alpha}}{\Gamma(\alpha+1)} e^{-\lambda x} dx}_{1}$$

$$= \frac{\alpha}{\lambda}$$

(b) $$E[X^2] = \int_0^\infty x^2 \frac{\lambda(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} e^{-\lambda x} dx = \frac{\Gamma(\alpha+2)}{\Gamma(\alpha)} \frac{1}{\lambda^2} \underbrace{\int_0^\infty \frac{\lambda(\lambda x)^{\alpha+1}}{\Gamma(\alpha+2)} e^{-\lambda x} dx}_{1}$$

$$= \frac{(\alpha+1)(\alpha)}{\lambda^2}$$

$$\Gamma(\alpha+2) = (\alpha+1)\Gamma(\alpha+1) = (\alpha+1)\alpha\Gamma(\alpha)$$

$$VAR[X] = \frac{(\alpha+1)\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2}$$

$$= \frac{\alpha}{\lambda^2}$$

(c) m-Erlang $\alpha = m$

$$E[X] = \frac{m}{\lambda} \quad VAR[X] = \frac{m}{\lambda^2}$$

(d) chi-square $\alpha = k/2$ $\lambda = \frac{1}{2}$

$$E[X] = \frac{k}{2}\frac{1}{1/2} = k$$

$$VAR[X] = \frac{k}{2}\frac{1}{1/4} = 2k$$

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4.71 Gamma RV

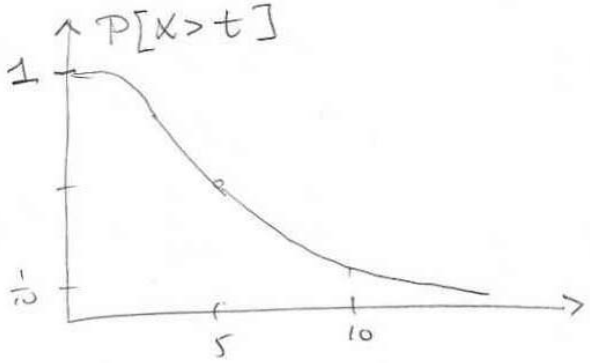
$$m = 4 = \frac{\alpha}{\lambda} \qquad \sigma^2 = 8 = \frac{\alpha}{\lambda^2}$$

$$\frac{\sigma^2}{m} = 2 = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{2}$$

$$\alpha = m\lambda = 4 \cdot \frac{1}{2} = 2$$

Note: Gamma pdf frequently stated in terms of α and $\beta = 1/\lambda$.
Octave uses this latter notation

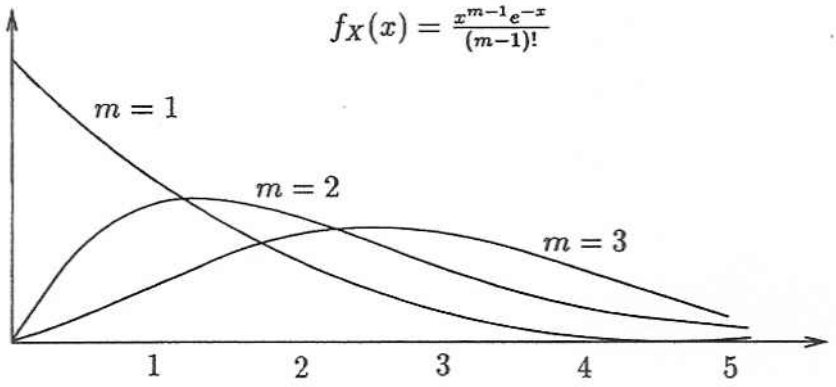
```
> x=[0: 0.1: 5];
> plot (1-gamma_cdf(x, 2, 2))
```



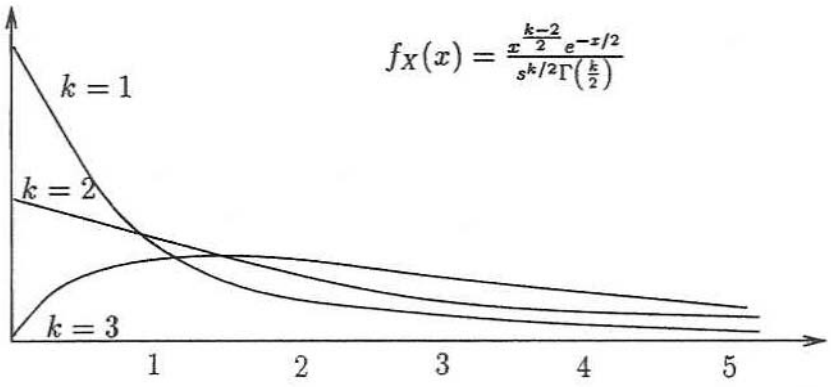
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4.72 Use gamma_cdf with appropriate adjustments for m-Erlang and chi-square RV's.

(a)



(b)



4.73

$$P[X \geq 15] = \sum_{k=0}^{4-1} \frac{\left(\frac{1}{3}15\right)^k}{k!} e^{-\frac{15}{3}}$$

$$= \sum_{k=0}^{3} \frac{5^k}{k!} e^{-5}$$

$$= 0.2650$$

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4.74

3.50 a) The cdf is given by the integral

$$F_X(x) = \frac{\lambda^{m-1}}{(m-1)!}\int_0^x y^{m-1}\lambda e^{-\lambda y}dy,$$

since $\Gamma(m)(m-1)!$. Integrate by parts using $u = y^{m-1}$ and $dv = \lambda e^{-\lambda y}dy$ so that $du = (m-1)y^{m-2}dy$ and $v = -e^{-\lambda y}$:

$$\begin{aligned} F_X(x) &= \frac{\lambda^{m-1}}{(m-1)!}\left\{-y^{m-1}e^{-\lambda y}\Big|_0^x - \int_0^x -(m-1)y^{m-2}e^{-\lambda y}dy\right\} \\ &= -\frac{(\lambda x)^{m-1}}{(m-1)!}e^{-\lambda x} + \frac{\lambda^{m-2}}{(m-2)!}\int_0^x y^{m-2}\lambda e^{-\lambda y}dy. \end{aligned}$$

The integral on the righ-hand side is identical to the equation for $F_X(x)$ with $m-1$ replaced by $m-2$. We can therefore repeatedly perform integration by parts to obtain the cdf of X:

$$\begin{aligned} F_X(x) &= -\frac{(\lambda x)^{m-1}}{(m-1)!}e^{-\lambda x} - \frac{(\lambda x)^{m-2}}{(m-2)!}e^{-\lambda x} - \ldots + \int_0^x e^{-\lambda y}dy \\ &= -\frac{(\lambda x)^{m-1}}{(m-1)!}e^{-\lambda x} - \frac{(\lambda x)^{m-2}}{(m-2)!}e^{-\lambda x}\ldots - e^{-\lambda x} + 1 \\ &= 1 - \sum_{k=0}^{m-1}\frac{(\lambda x)^k}{k!}e^{-\lambda x}. \end{aligned}$$

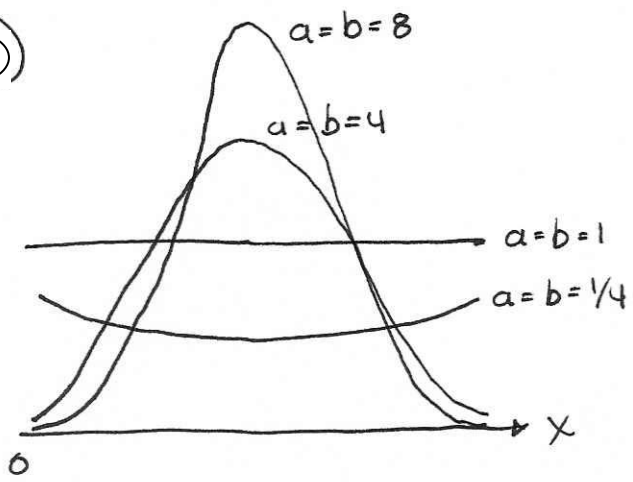
b) From part a)

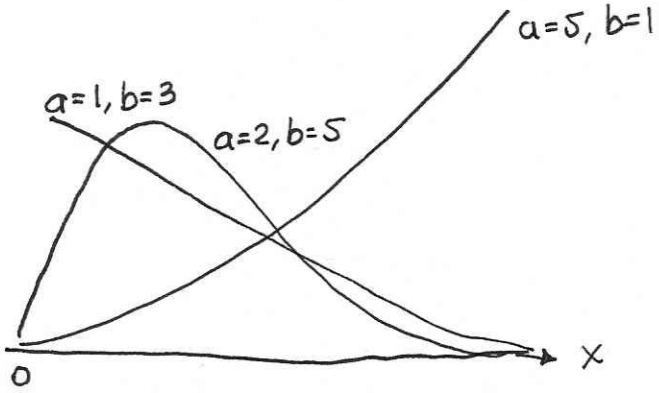
$$F_X(x) = 1 - \sum_{k=0}^{m-1}\frac{(\lambda x)^k}{k!}e^{-\lambda x} = \frac{\lambda^{m-1}}{(m-1)!}\int_0^x y^{m-1}\lambda e^{-\lambda y}dy$$

$$f_X(x) = \frac{df_X(x)}{dx} = \frac{\lambda(\lambda y)^{m-1}e^{-\lambda y}}{(m-1)!}$$

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4.75



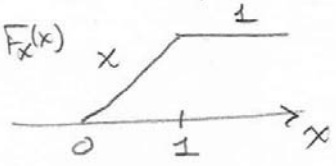


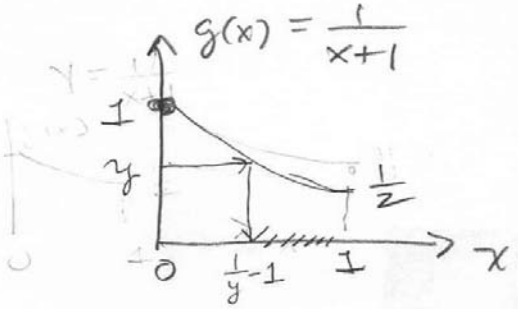
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4.5 Functions of a Random Variable

4.76

X uniform on [0,1].





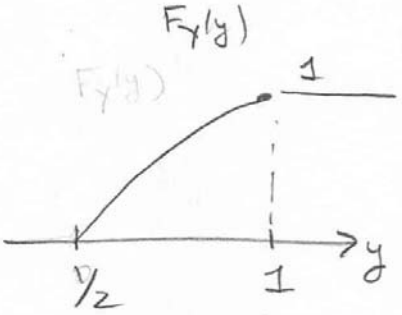
$$P[Y \le y] = P\left[\frac{1}{X+1} \le y\right] \qquad \frac{1}{2} < y < 1$$

$$= P\left[\frac{1}{y} \le X+1\right]$$

$$= P\left[X \ge \frac{1}{y} - 1\right]$$

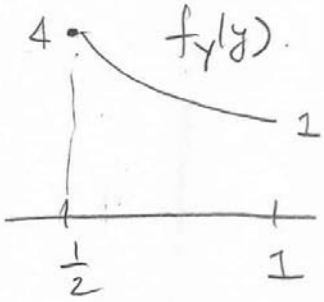
$$= 1 - \left(\frac{1}{y} - 1\right)$$

$$= 2 - \frac{1}{y}$$



$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

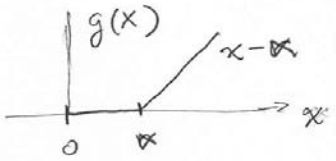
$$= +\frac{1}{y^2} \qquad \frac{1}{2} < y < 1$$



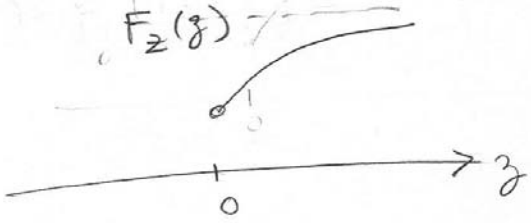
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4.77

(a) $Z = (X-\alpha)^+ = \begin{cases} 0 & X \le \alpha \\ X-\alpha & X > \alpha \end{cases}$



$$P[Z \le z] = \begin{cases} 0 & z < 0 \\ P[X \le r] = 1 - e^{-\alpha^2/2\alpha^2} = 1 - e^{-1/2} & z = 0 \\ P[X - \alpha \le z] = P[X \le z+\alpha] & z > 0 \\ \quad = 1 - e^{-(z+\alpha)^2/2\alpha^2} & \end{cases}$$



$$f_Z(z) = \frac{d}{dz} F_Z(z) = (1-e^{-1/2})\,\delta(z) + \frac{d}{dz} F_X(z+\alpha)$$

$$= (1-e^{-1/2})\,\delta(z) + f_X(z+\alpha)$$

$$= (1-e^{-1/2})\,\delta(z) + \frac{z+\alpha}{\alpha^2} e^{-(z+\alpha)^2/2\alpha^2}$$

(b) $P[Z \le z] = P[X^2 \le z] = P[X \le \sqrt{z}]$ since X is nonnegative

$$f_Z(z) = \frac{d}{dx} F_X(\sqrt{z}) = f_X(\sqrt{z})\,\tfrac{1}{2} z^{-1/2}$$

$$= \frac{f_X(\sqrt{z})}{2\sqrt{z}} = \frac{\sqrt{z}}{2\sqrt{z}} \frac{1}{\alpha^2} e^{-\sqrt{z}^2/2\alpha^2} = \frac{1}{2\alpha^2} e^{-z/2\alpha^2}$$

exponential RV.

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4.78

$$P[N=0] = F_X(\pi)$$

$$P[N=1] = F_X(2\pi) - F_X(\pi)$$

$$\vdots$$

$$P[N=n] = F_X((n+1)\pi) - F_X(n\pi)$$

$$= 1 - e^{-\lambda(n+1)\pi} - (1 - e^{-\lambda n\pi})$$

$$= (e^{-\lambda n\pi} - e^{-\lambda(n+1)\pi})$$

$$= e^{-\lambda n\pi}(1 - e^{-\lambda\pi}) \qquad \lambda = \frac{1}{5\pi}$$

$$= e^{-\frac{n\pi}{5\pi}}\left(1 - e^{-\frac{\pi}{5\pi}}\right)$$

$$= (1 - e^{-1/5})(e^{-1/5})^n \qquad n = 0, 1, 2, \ldots$$

geometric RV

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4.79

a) $P[\{X<-4d\} \cup \{X>4d\}] = 0.01$

$$P[X<-4d] + P[X>4d] = \Phi(-4d) + 1 - \Phi(4d)$$
$$= 1 - (1 - Q(4d)) + Q(4d)$$
$$= 2Q(4d)$$

$$Q(4d) = 0.005 \rightarrow 4d = 2.57 \rightarrow d = 0.6425$$

b) $P[0<X<d] = F_X(d) - F_X(0) = Q(0) - Q(0.64) = 0.5 - 0.258$
$= 0.242$

$P[d<X<2d] = F_X(2d) - F_X(d) = Q(0.64) - Q(1.28) = 0.258 - 0.0994$
$= 0.1586$

$P[2d<X<3d] = F_X(3d) - F_X(2d) = Q(1.28) - Q(1.92) = 0.0994 - 0.0273$
$= 0.0721$

$P[3d<X<4d] = Q(3d) - Q(4d) = Q(1.92) - Q(2.57) = 0.0273 - 0.005$
$= 0.0223$

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4.80

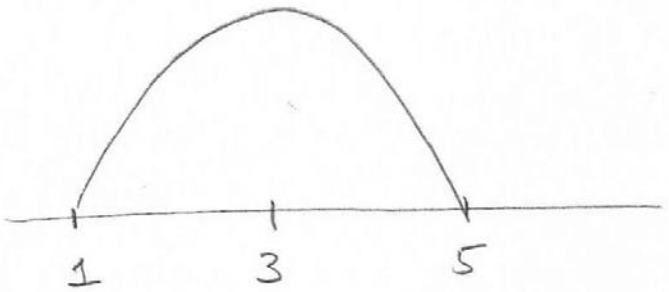
$Y = 2X+3$

$$F_Y(y) = P[2X+3 \le y] = P[X \le \tfrac{y-3}{2}]$$
$$= F_X(\tfrac{y-3}{2})$$

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(\tfrac{y-3}{2})$$
$$= \tfrac{1}{2} f_X(\tfrac{y-3}{2})$$

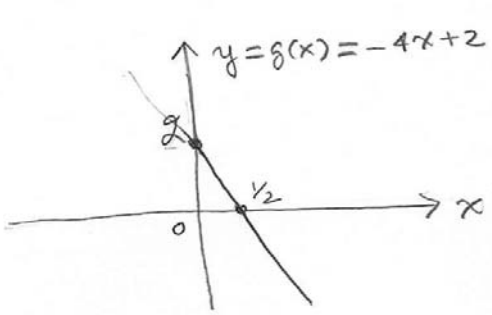
$$f_X(x) = \tfrac{3}{4}(1-x^2) \quad -1 < x < 1$$

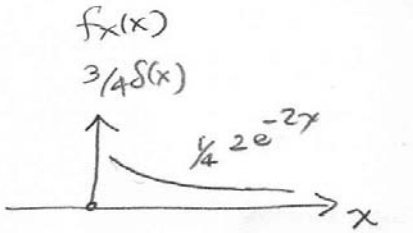
$$\Rightarrow f_Y(y) = \tfrac{3}{8}\left(1-\left(\tfrac{y-3}{2}\right)^2\right) \quad 1 < y < 5$$



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4.81



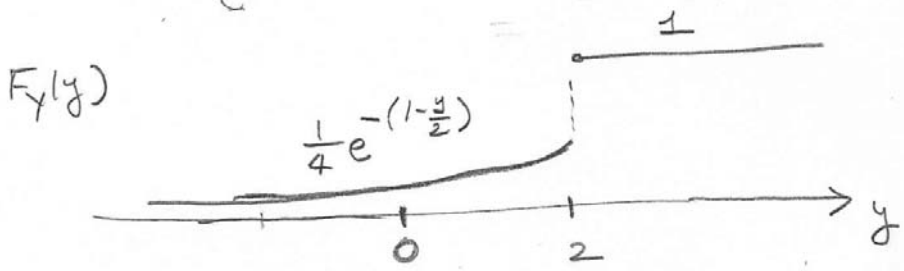


$F_X(x) = \frac{3}{4} + \frac{1}{4}(1 - e^{-2x})$

$$F_Y(y) = P[-4X + 2 \le y] = P[-4X \le y - 2]$$

$$= P\left[X \ge \frac{2-y}{4}\right] \qquad -\infty < y < 2$$

$$= \begin{cases} P[X \ge 0] = 1 & y = 2 \\ P\left[X \ge \frac{2-y}{4}\right] = \frac{1}{4} e^{-2\left(\frac{2-y}{4}\right)} & y < 2 \end{cases}$$



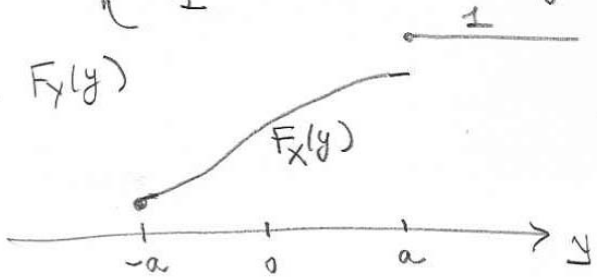
$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{4} e^{-\left(1-\frac{y}{2}\right)}\left(\frac{1}{2}\right) + \frac{3}{4}\delta(y-2)$$

$$= \frac{1}{8} e^{-(1-y/2)} + \frac{3}{4}\delta(y-2)$$

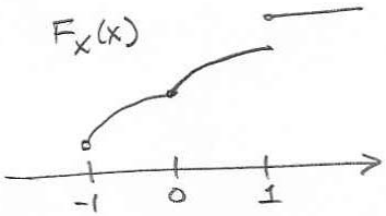
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4.82

$$F_Y(y) = \begin{cases} 0 & y < -a \\ F_X(-a) = P[X \le -a] & y = -a \\ F_X(y) = P[X \le y] & -a < y < a \\ 1 & y = +a \end{cases}$$



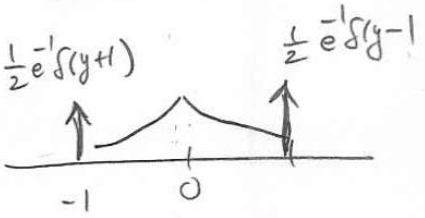
(b) X Laplacian $\lambda = a = 1$



$$F_Y(y) = \begin{cases} 0 & y < -1 \\ \frac{1}{2}e^{-1} & y = -1 \\ \frac{1}{2}e^{x} & -1 \le y \le 0 \\ \frac{1}{2} + \frac{1}{2}(1 - e^{-x}) & 0 \le y \le 1 \\ 1 & y \ge 1 \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-1}\delta(y+1) & y = -1 \\ \frac{1}{2}e^{-|x|} & -1 < y < 1 \\ \frac{1}{2}e^{-1}\delta(y-1) & y = +1 \end{cases}$$

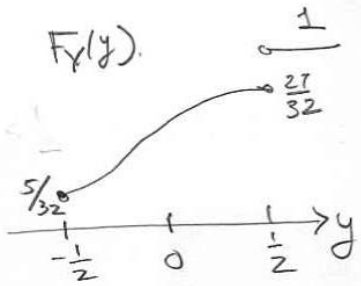
$f_X(x)$.



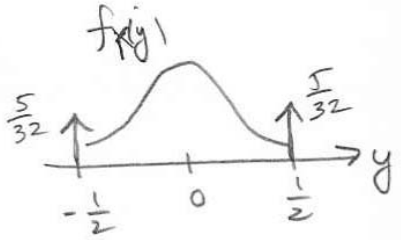
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(c)

$$F_Y(y) = \begin{cases} 0 & y < -1/2 \\ \frac{5}{32} & y = -\frac{1}{2} \\ \frac{3}{4}\left[(x+1) - \frac{1}{3}(x^3+1)\right] & -\frac{1}{2} < y < \frac{1}{2} \\ 1 & y = 1 \end{cases}$$

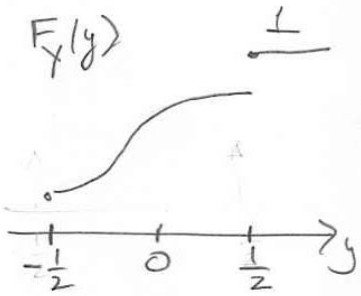


$$f_Y(y) = \begin{cases} \frac{5}{32}\delta(y+\frac{1}{2}) & y = -\frac{1}{2} \\ \frac{3}{4}(1-y^2) & -\frac{1}{2} < y < \frac{1}{2} \\ \frac{5}{32}\delta(y-\frac{1}{2}) & y = \frac{1}{2} \end{cases}$$

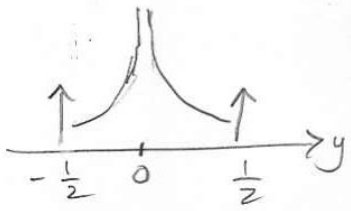


(d)

$$F_Y(y) = \begin{cases} 0 & y < -\frac{1}{2} \\ (1-(\frac{1}{2})^{1/3})/2 & y = -\frac{1}{2} \\ P[U^3 \le y] = \frac{y^{1/3}-(-1)}{2} & -\frac{1}{2} < y < \frac{1}{2} \\ 1 & y = \frac{1}{2} \end{cases}$$



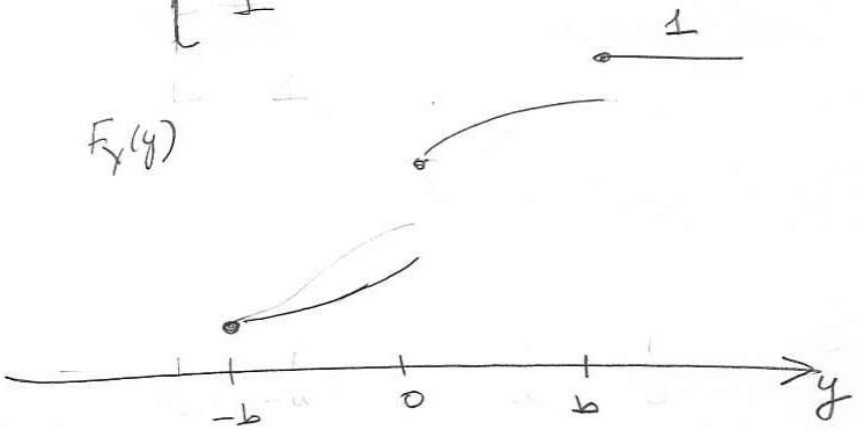
$$f_Y(y) = \begin{cases} (1-(\frac{1}{2})^{1/3})/2\,\delta(y+\frac{1}{2}) & y = -\frac{1}{2} \\ \frac{1}{6y^{2/3}} & -\frac{1}{2} < y < \frac{1}{2} \\ (1-(\frac{1}{2})^{1/3})/2\,\delta(y-\frac{1}{2}) & y = \frac{1}{2} \end{cases}$$

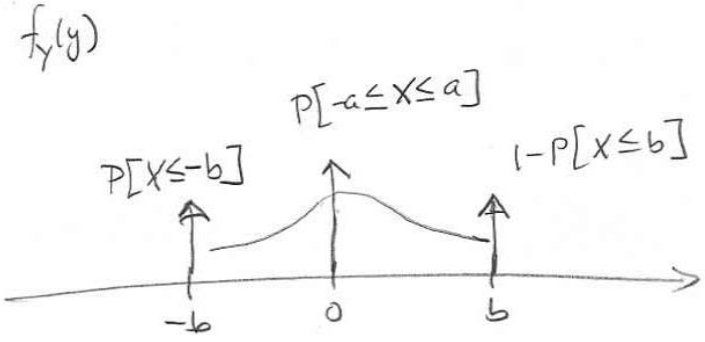


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4.83a

$$F_Y(y) = \begin{cases} 0 & y < -b \\ P[X \le -b] & y = -b \\ P\left[\frac{b}{b-a}(X+a) \le y\right] = F_X\left(\frac{b-a}{b}y - a\right) & -b < y < 0 \\ P[X \le a] & y = 0 \\ P\left[\frac{b}{b-a}(X-a) \le y\right] = F_X\left(\frac{b-a}{b}y + a\right) & 0 < y < b \\ 1 & y = b \end{cases}$$





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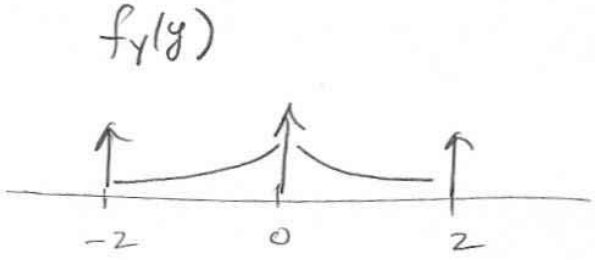
4.83b) $a=1, b=2$

$$P[X \le -2] = \tfrac{1}{2}e^{-2} = P[X \ge 2]$$

$$P[-1 \le X \le 1] = 2P[0 \le X \le 1] = 2 \cdot \tfrac{1}{2}(1-e^{-1})$$

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{1}{2}e^{-2} & y = -2 \\ \frac{1}{2}e^{\frac{y}{2}-1} & -2 < y < 0 \\ \frac{1}{2} + \frac{1}{2}(1-e^{-1}) & y = 0 \\ \frac{1}{2} + \frac{1}{2}(1-e^{-(\frac{y}{2}+1)}) & 0 < y < 2 \\ 1 & y = 2 \end{cases}$$

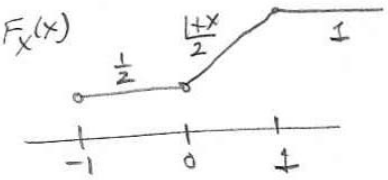
$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-2}\delta(y+2) & y = 2 \\ \frac{1}{4}e^{\frac{y}{2}-1} & -2 < y < 0 \\ (1-e^{-1})\delta(y) & y = 0 \\ \frac{1}{4}e^{-(\frac{y}{2}+1)} & 0 < y < 2 \\ \frac{1}{2}e^{-2}\delta(y-2) & y = 2 \end{cases}$$

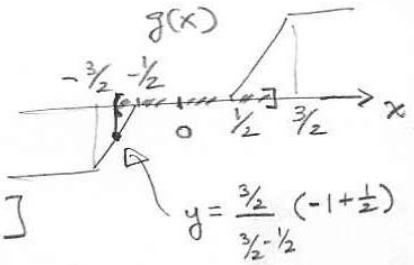


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4.83c

Recall

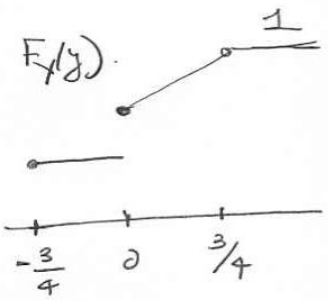




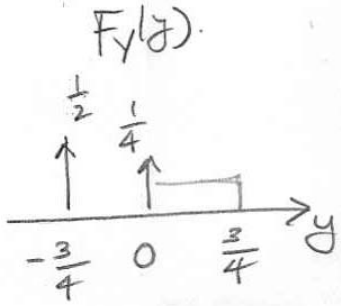
$P[X \le -5] = P[X \le -\frac{3}{2}] = 0 = P[X \ge \frac{3}{2}]$

$P[-\frac{1}{2} \le X \le \frac{1}{2}] = P[0 \le X \le \frac{1}{2}] = \frac{1}{4}$

$$F_Y(y) = \begin{cases} 0 & y < -\frac{3}{4} \\ \frac{1}{2} & y = -\frac{3}{4} \\ \frac{1}{2} & -\frac{3}{4} \le y < 0 \\ \frac{3}{4} & y = 0 \\ F_X\left(\frac{2}{3}y+\frac{1}{2}\right) = \frac{1+\frac{2}{3}y+\frac{1}{2}}{2} = \frac{2}{3}y+\frac{3}{4} & 0 < y \le \frac{3}{4} \end{cases}$$



$$f_Y(y) = \begin{cases} \frac{1}{2}\delta(y+\frac{3}{4}) & y = -\frac{3}{4} \\ \frac{1}{4}\delta(y) & y = 0 \\ \frac{1}{3} & 0 < y \le \frac{3}{4} \\ 0 & \text{elsewhere} \end{cases}$$



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4.84 $F_Y(y) = F_X\left(\frac{y-2}{3}\right)$

$f_Y(y) = \frac{1}{3} f_X\left(\frac{y-2}{3}\right)$

- X is laplacian

$$F_Y(y) = \begin{cases} \frac{1}{2} e^{\alpha\left(\frac{y-2}{3}\right)} & y \le 2 \\ 1 - \frac{1}{2} e^{-\alpha\left(\frac{y-2}{3}\right)} & y \ge 2 \end{cases} \qquad f_Y(y) = \frac{1}{3}\frac{\alpha}{2} e^{-\alpha\left|\frac{y-2}{3}\right|}$$

- X is Gaussian

$$F_Y(y) = \Phi\left(\frac{\frac{y-2}{3} - m}{\sigma}\right) = \Phi\left(\frac{y-(2+3m)}{3\sigma}\right)$$

$$f_Y(y) = \frac{1}{3\sigma\sqrt{2\pi}} e^{-\left(\frac{y-2}{3}-m\right)^2/2\sigma^2} = \frac{1}{3\sigma\sqrt{2\pi}} e^{-(y-(2+3m))^2/2(3\sigma)^2}$$

- $X = b\cos(2\pi U)$

$$F_Y(y) = \begin{cases} 0 & y < -3b+2 \\ \frac{1}{\pi}\sin^{-1}\left(\frac{y-2}{3b}\right) + \frac{1}{\pi}\sin^{-1}\left(-\frac{1}{b}\right) & 2-3b \le y \le 3b+2 \\ 1 & y > 3b+2 \end{cases}$$

$$f_Y(y) = \frac{1}{3}\frac{1}{\pi b\sqrt{1-\left(\frac{y-2}{3b}\right)^2}} \qquad 2-3b \le y < 3b+2$$

4.85 X : Gaussian, $Y = aX + b$, a linear combination of X.
Y is also Gaussian

$$E[Y] = aE[X] + b = am + b = m'$$

$$Var[Y] = a^2 Var[X] = a^2\alpha^2 = \alpha'^2$$

$$a = \alpha'/\alpha, \quad b = m' - am = m' - m\alpha'/\alpha$$

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4.86

$$f_X(x) = \sum_k f_U(u) \left| \frac{1}{n} x^{\frac{1}{n}-1} \right| \Big|_{x=x_k}$$

$$f_X(x) = f_U(\sqrt[n]{x}) \left(\frac{1}{n} x^{\frac{1}{n}-1} \right)$$

$$= \frac{1}{n} x^{\frac{1}{n}-1} \qquad 0 \le x \le 1$$

$$F_X(x) = \int_0^x \frac{1}{n} x^{\frac{1}{n}-1} \, dx = x^{\frac{1}{n}} \Big|_0^x = x^{\frac{1}{n}} \qquad 0 \le x \le 1$$

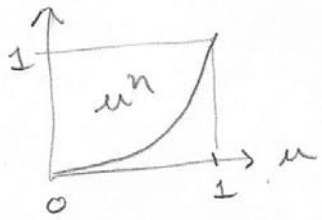
$$= \begin{cases} 0 & x < 0 \\ x^{1/n} & x \ge 0 \end{cases}$$

Alternatively we could start with the cdf:

$$F_X(x) = P[U^n \le x]$$

$$= P[U \le x^{1/n}]$$

$$= x^{1/n}$$



$$0 \le x \le 1$$

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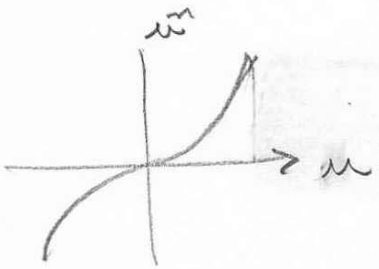
4.87

$X = U^n$ n odd

$$F_X(x) = P[U^n \le x] = P[U \le x^{1/n}]$$

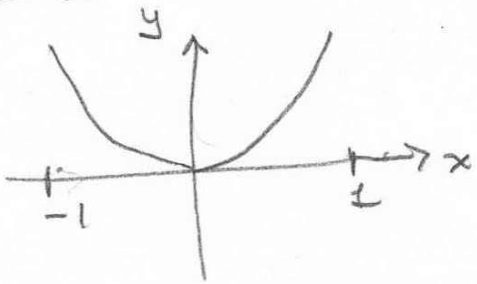
$$= \frac{1}{2}(x^{1/n} - (-1))$$

$$= \frac{1}{2}(x^{1/n} + 1)$$



$$f_X(x) = \frac{1}{2}\frac{1}{n}x^{\frac{1}{n}-1} \quad -1 \le x \le 1$$

$X = U^n$ n even



$$F_X(x) = P[U^n \le x]$$

$$= P[-x^{1/n} \le U \le x^{1/n}]$$

$$= 2\left(\frac{1}{2}x^{1/n}\right) \qquad 0 \le x \le 1$$

$$= x^{1/n}$$

$$f_X(x) = \frac{1}{n}x^{1/n-1}$$

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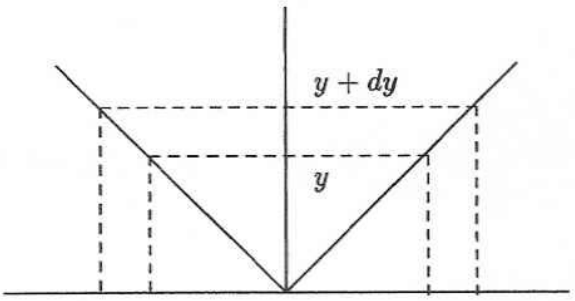
4.88 a) The equivalent event for $\{Y \leq y\}$ is $\{|X| \leq y\}$, therefore:

$$\begin{aligned} F_Y(y) &= P[|X| \leq y] = P[-y \leq X \leq y] \\ &= \begin{cases} 0 & y < 0 \\ F_X(y) - F_X(y^-) & y \geq 0 \end{cases} \end{aligned}$$

Assuming X is a continuous random variable,

$$f_Y(y) = F'_Y(y) = f_X(y) + f_X(-y) \quad \text{for } y > 0 .$$

b) The equivalent event for $\{dy < Y \leq y + dy\}$ is shown below:



Therefore

$$\begin{aligned} P[y < Y \leq y + dy] &= P[y < X \leq y + dy] \\ &\quad + P[-y - dy < X \leq -y] \end{aligned}$$

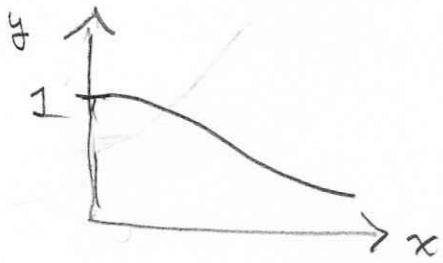
$$\begin{aligned} \Rightarrow f_Y(y)dy &= f_X(y)dy + f_X(-y)|dy| \\ \Rightarrow f_Y(y) &= f_X(y) + f_X(-y) \qquad \text{for } y > 0 . \end{aligned}$$

c) If $f_X(x)$ is an even factor of x, then $f_X(x) = f_X(-x)$ and thus $f_Y(y) = 2f_X(y)$.

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4.89

$$Y = \frac{1}{(X+1)^2}$$

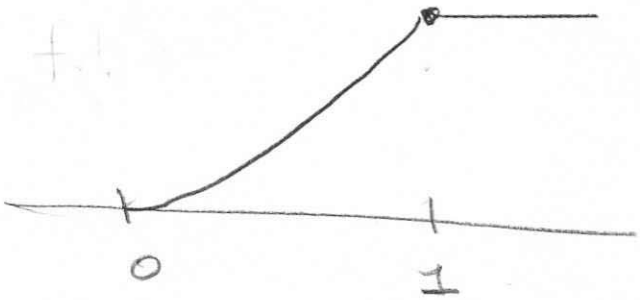


$$P[Y \le y] = P\left[\frac{1}{(X+1)^2} \le y\right]$$

$$= P\left[\frac{1}{y} \le (X+1)^2\right]$$

$$= P\left[\sqrt{\frac{1}{y}} - 1 \le X\right]$$

$$= e^{-\sqrt{1/y - 1}} \qquad 0 < y < 1$$



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4.90

$$X = \pm\sqrt{\frac{P}{R}} \qquad \frac{dx}{dp} = \pm\frac{1}{2}\frac{p^{\frac{1}{2}-1}}{\sqrt{R}} = \pm\frac{1}{2\sqrt{RP}}$$

$$f_P(p) = [f_X(x) + f_X(-x)]\left|\frac{dx}{dP}\right|$$

$$= \left[f_X\left(\sqrt{\frac{P}{R}}\right) + f_X\left(-\sqrt{\frac{P}{R}}\right)\right]\frac{1}{2\sqrt{RP}}$$

$$= \left[\frac{1}{2\sqrt{2\pi}}\left(e^{-\left(\sqrt{\frac{P}{R}}-1\right)^2/2(2)} + e^{-\left(-\sqrt{\frac{P}{R}}-1\right)^2/2(2)}\right)\right]\frac{1}{2\sqrt{RP}}$$

$$= \frac{1}{2\sqrt{2\pi}}\frac{1}{2\sqrt{RP}}\left(e^{-(\sqrt{P}-\sqrt{R})^2/2(2R)} + e^{-(\sqrt{P}-(-\sqrt{R}))^2/2(2R)}\right)$$

4.91

a) For $y \le 0$ $\quad P[Y \le y] = 0$

For $y > 0$ $\quad P[Y \le y] = P[e^X \le y] = P[X \le \ln y] = F_X(\ln y)$

$$\therefore \quad F_Y(y) = \begin{cases} 0 & y \le 0 \\ F_X(\ln y) & y > 0 \end{cases}$$

For $y > 0$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = F_X'(\ln y)\frac{d}{dy}\ln y$$

$$= \frac{1}{y}f_X(\ln y)$$

b) If X is a Gaussian random variable, then

$$f_Y(y) = \begin{cases} 0 & y \le 0 \\ \dfrac{e^{-(\ln y - m)^2/2\sigma^2}}{y\sqrt{2\pi}\sigma} & y > 0 \end{cases}$$

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4.92 $f_X(x) = 4x(1-x^2) \qquad 0 \le x \le 1$

(a) $Y = \pi X^2 \qquad \frac{dy}{dx} = 2\pi x_1 \qquad x_1 = \sqrt{\frac{y}{\pi}}$

only a positive root since x ≥ 0 non-negative

$$f_Y(y) = \frac{f_X(x_1)}{|2\pi x_1|} = \frac{4\sqrt{\frac{y}{\pi}}\left(1-\frac{y}{\pi}\right)}{|2\pi\sqrt{\frac{y}{\pi}}|} = \frac{2}{\pi}\left(1-\frac{y}{\pi}\right) \qquad 0 < y < \pi$$

(b) $Y = \frac{4}{3}\pi X^3$

$$x_1 = \left(\frac{3}{4\pi}y\right)^{1/3} \qquad \frac{dy}{dx} = 4\pi x^2$$

$$f_Y(y) = \frac{f_X(x_1)}{|4\pi x_1^2|} = \frac{4x_1(1-x_1^2)}{4\pi x_1^2} = \frac{1-x_1^2}{\pi x_1}$$

$$= \frac{1-\left(\frac{3}{4\pi}y\right)^{2/3}}{\pi\left(\frac{3}{4\pi}y\right)^{1/3}} \qquad 0 < y < \frac{4}{3}\pi$$

(c) $Y = cX^n$

$$x_1 = \left(\frac{y}{c}\right)^{1/n} \qquad \frac{dy}{dx} = ncx^{n-1}$$

$$f_Y(y) = \frac{f_X(x_1)}{|ncx^{n-1}|} = \frac{4x_1(1-x_1^2)}{nc\,x_1^{n-1}} = \frac{4(1-x_1^2)}{nc\,x_1^{n-2}}$$

$$= \frac{4}{nc}\,\frac{\left(1-\left(\frac{y}{c}\right)^{2/n}\right)}{(y/c)^{\frac{n-2}{n}}}$$

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4.93) Q(z) defined on intervals

$(-\infty, -3d], (-3d, -2d], (-2d, -d], (-d, 0]$

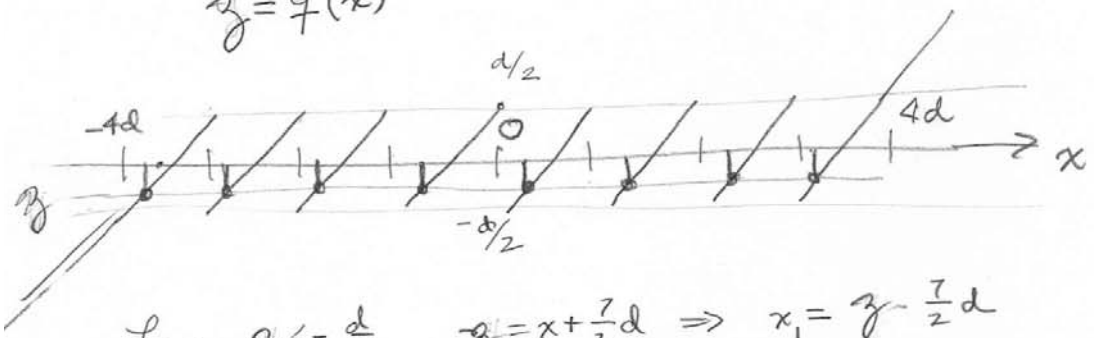
$(0, d], (d, 2d], (2d, 3d], (3d, \infty)$

for $(kd, (k+1)d)$ $\quad Z = X - (k+\frac{1}{2})d \quad k = -3, \ldots, 3$

$(-\infty, -3d)$ $\quad Z = X + \frac{7}{2}d$

$(3d, \infty)$ $\quad Z = X - \frac{7}{2}d$

$z = g(x)$



for $z < -\frac{d}{2}$ $\quad z = x + \frac{7}{2}d \Rightarrow x_1 = z - \frac{7}{2}d$

$z > \frac{d}{2}$ $\quad z = x - \frac{7}{2}d \Rightarrow x_1 = z + \frac{7}{2}d$.

for $-\frac{d}{2} < z < \frac{d}{2}$ there are 8 solutions to

$z = g(x_i)$

$z = x_1 + \frac{7}{2}d \Rightarrow x_1 = z - \frac{7}{2}d$ $\quad x_5 = z + \frac{1}{2}d$

$x_2 = z - \frac{5}{2}d$ $\quad x_6 = z + \frac{3}{2}d$

$x_3 = z - \frac{3}{2}d$ $\quad x_7 = z - \frac{5}{2}d$

$x_4 = z - \frac{1}{2}d$ $\quad x_7 = z + \frac{7}{2}d$

also $\frac{dz}{dx} = 1$.

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(4.93) continued —

$$f_X(x) = \frac{1}{2}\frac{2}{d}e^{-2|x|/d} \qquad \frac{1}{\lambda} = \frac{d}{2}$$

for $-\frac{d}{2} < z < \frac{d}{2}$ the pdf of Z is

$$f_Z(z) = \sum_{i=1}^{8} \frac{f_X(x_i)}{|dz/dx|} = \underbrace{\sum_{i=1}^{4} f_X(x_i)}_{\text{negative roots}} + \underbrace{\sum_{i=5}^{8} f_X(x_i)}_{\text{positive roots}}$$

$$= \frac{1}{d}\left[e^{+\frac{2}{d}(z-\frac{7}{2}d)} + e^{+\frac{2}{d}(z-\frac{5}{2}d)} + e^{+\frac{2}{d}(z-\frac{3}{2}d)} + e^{+\frac{2}{d}(z-\frac{d}{2})} \right]$$

$$+ \frac{1}{d}\left[e^{-\frac{2}{d}(z+\frac{1}{2}d)} + e^{-\frac{2}{d}(z+\frac{3}{2}d)} + e^{-\frac{2}{d}(z+\frac{5}{2}d)} + e^{-\frac{2}{d}(z+\frac{7}{2}d)} \right]$$

$$= \frac{e^{2z}}{d}\left[e^{-7} + e^{-5} + e^{-3} + e^{-1} \right] + \frac{e^{-2z}}{d}\left[e^{-1} + e^{-3} + e^{-5} + e^{-7} \right]$$

$$= 2\frac{e^{-2|z|}}{d}\left[e^{-1} + e^{-3} + e^{-5} + e^{-7} \right]$$

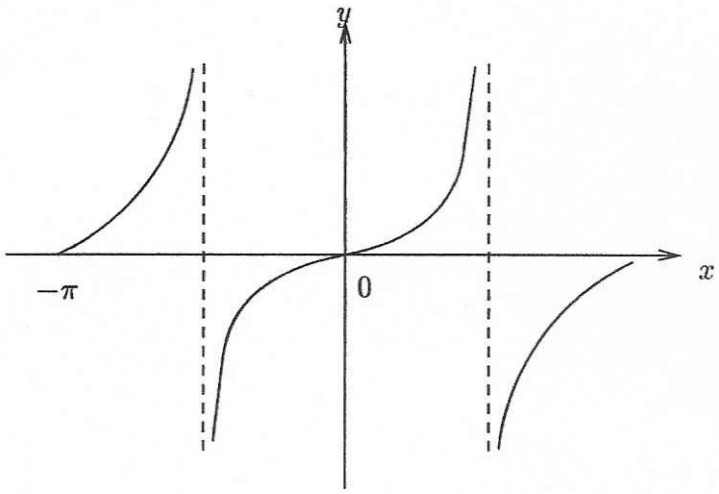
pdf for $z < -\frac{d}{2}$ and $z > \frac{d}{2}$ involve only one root.

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4.94 $Y = a \tan X$.

$$x = \tan^{-1}(y/a), \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\frac{dx}{dy} = \frac{1}{1 + (y/a)^2}\frac{1}{a} = \frac{a}{y^2 + a^2}$$



$$\begin{aligned} f_X(y) &= \sum_k f_X(x)\left|\frac{dx}{dy}\right|_{x=x_k} \\ &= 2 \cdot \frac{1}{2\pi}\frac{a}{y^2 + a^2} \\ &= \frac{a/\pi}{y^2 + a^2} \end{aligned}$$

Y is a Cauchy RV.

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4.95

$$Y = \left(\frac{X}{\lambda}\right)^{\beta}$$

$$X = \lambda \sqrt[\beta]{Y}$$

$$F_Y(y) = P\left[\left(\frac{X}{\lambda}\right)^{\beta} \le y\right] = P\left[X \le \lambda \sqrt[\beta]{y}\right]$$

$$= \begin{cases} 0 & y<0 \\ 1 - e^{-\left(\frac{\lambda\sqrt[\beta]{y}}{\lambda}\right)^{\beta}} & y \ge 0 \end{cases}$$

$$= \begin{cases} 0 & y<0 \\ 1 - e^{-y} & y \ge 0 \end{cases}$$

$$f_Y(y) = e^{-y} \qquad y \ge 0$$

4.96

$$X = -\ln(1-U)$$

$$e^{-X} = (1-U)$$

$$U = 1 - e^{-X}$$

$$f_X(x) = f_U(u)\left|\frac{du}{dx}\right|$$

$$= f_U(1-e^{-x})\left|e^{-x}\right|$$

$$= e^{-x}$$

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4.6 The Markov and Chebyshev Inequalities

4.97

a) $P[X>c] = 1 - \frac{c}{b}$ $0 \le c \le b$

$E[X] = \frac{b}{2}$

$1 - \frac{c}{b} \overset{?}{\le} \frac{b}{2c}$ $c=0$ $1 \le \infty$ ✓ $c=b$ $0 \le 1/2$ ✓

b) $P[X>c] = e^{-\lambda c}$ $c \ge 0$

$E[X] = 1/\lambda$

$e^{-\lambda c} \overset{?}{\le} \frac{1}{\lambda c}$ $c=0$ $1 \le \infty$ ✓ $c \to \infty$ $0 \le 0$ ✓

c) $P[X>c] = \frac{X_m^\alpha}{c^\alpha}$ $c \ge X_m$

$E[X] = \frac{\alpha X_m}{\alpha - 1}$ $\alpha > 1$

$\frac{X_m^\alpha}{c^\alpha} \overset{?}{\le} \frac{\alpha X_m}{\alpha - 1}\frac{1}{c}$ $c = X_m$ $1 \le \frac{\alpha}{\alpha-1}$ ✓ $\alpha > 1$ $c \to \infty$ $0 \le 0$ ✓

d) $P[X>c] = e^{-c^2/2\alpha^2}$ $c \ge 0$

$E[X] = \alpha\sqrt{\pi/2}$ $\alpha > 0$

$e^{-c^2/2\alpha^2} \overset{?}{\le} \frac{\alpha\sqrt{\pi/2}}{c}$ $c=0$ $1 \le \infty$ ✓ $c \to \infty$ $0 \le 0$ ✓

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4.98

a) $P[X>c] = 1 - \left(\frac{c-1+1}{L-1+1}\right)$ $\quad c \in \{1,2,\ldots,L\}$

$E[X] = \frac{L+1}{2}$

$1 - \frac{c}{L} \overset{?}{\leq} \frac{L+1}{2c} - \frac{1}{L}$

$c=1$
$1 - \frac{1}{L} \leq \frac{L}{2} + \frac{L}{2} - \frac{1}{L}$ ✓ $L \geq 1$

$c=L$
$1 - \frac{L}{L} \leq \frac{L+1}{2L} - \frac{1}{L}$
$0 \leq \frac{1}{2} - \frac{1}{2L}$ ✓

b) $P[X>c] = (1-p)^{c+1}$ $\quad c \in \{0,1,2,\ldots\}$

$E[X] = \frac{1-p}{p}$

$(1-p)^{c+1} \overset{?}{\leq} \frac{1-p}{pc} - (1-p)^c p$

$(1-p)^c \leq \frac{1-p}{pc}$

$c=0$ $\quad 1 \leq \infty$ ✓
$c \to \infty$ $\quad 0 \leq 0$ ✓
$c=1$ $\quad 1-p \leq \frac{1-p}{p}$ ✓

c) $p_X[k] = \frac{1}{C_L}\frac{1}{k}$ $\quad k=1,2,\ldots,L$ $\qquad P[X>k] = 1 - \frac{C_k}{C_L}$

$E[X] = \frac{L}{C_L}$

$L=10$ $\quad C_L = 2.929$
$L=100$ $\quad C_L = 5.1874$

$1 - \frac{C_k}{C_L} \overset{?}{\leq} \frac{1}{k}\frac{L}{C_L}$

$C_L - C_k \leq \frac{L}{k}$

$L=10,\ k=1$ $\quad 1.929 \leq 10$ ✓
$k=10$ $\quad 0 \leq 1$ ✓

$L=100$ $k=1$ $\quad 4.1874 \leq 100$ ✓
$k=100$ $\quad 0 \leq 1$ ✓

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... 4.98

d) $P[X \geq c] = 1 - \sum_{j=0}^{c} \frac{n!}{j!(n-j)!} p^n$

$E[X] = np$

$P[X \geq c] \leq \frac{np}{c} - P[X = c]$

$1 - \sum_{j=0}^{c-1} \binom{n}{j} p^n \leq \frac{np}{c} \qquad 0 \leq c \leq n$

$n=10$, $c=0$ $\quad 1-(0.5)^{10} \leq \infty$ ✓

$c=1$ $\quad 1-(0.5)^{10} \leq 5$ ✓

$c=10$ $\quad 1-(0.5)^{10}(1023) \leq 0.5$ ✓

$n=100$, $c=0$ $\quad 1-(0.5)^{100} \leq \infty$ ✓

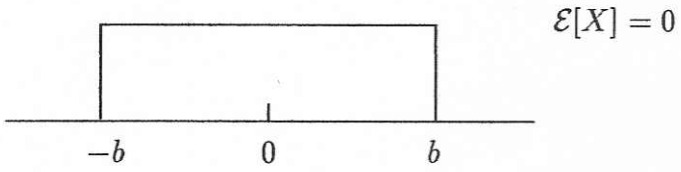
$c=1$ $\quad 1-(0.5)^{100} \leq 50$ ✓

$c=100$ $\quad 1-(0.5)^{100}(1.2677 \times 10^{30}) \leq 0.5$ ✓

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4.99

81 a) For a uniform random variable in $[-b, b]$ we have

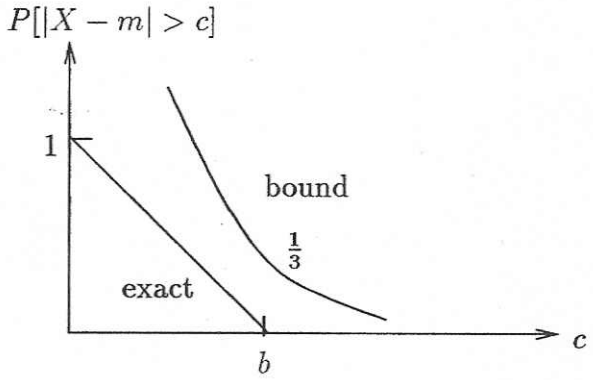


Exact:

$$P[|X - m| > c] = \begin{cases} 1 - \frac{c}{b} & 0 \le c \le b \\ 0 & c > b \end{cases}$$

Chebyshev Bound gives

$$P[|X - m| > c] \le \frac{\sigma_X^2}{c^2} = \frac{b^2}{3c^2}$$

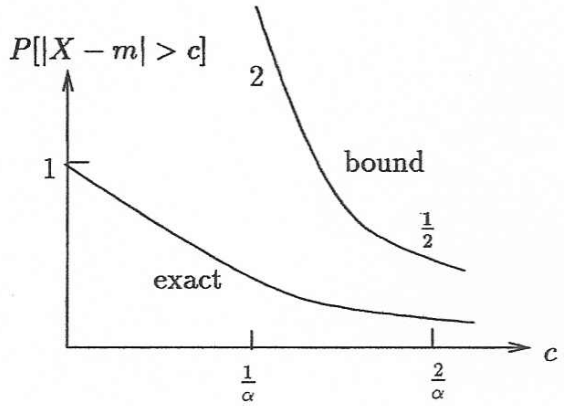


b) For the Laplacian random variable $\mathcal{E}[X] = 0$ and $VAR[X] = 2/\alpha^2$

Exact: $P[|X - m| > c] = P[|X| > c] = e^{-\alpha c}$

Bound: $P[|X - m| > c] \le \dfrac{2}{\alpha^2 c^2}$

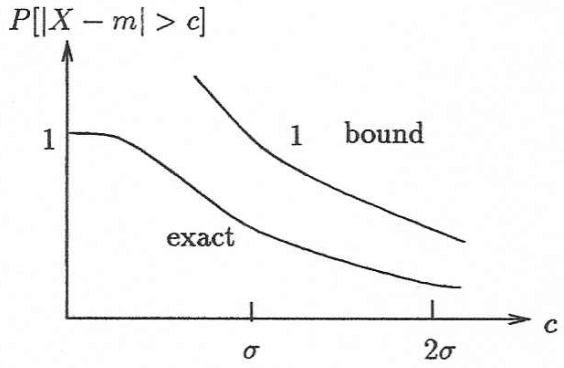
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c) For the Gaussian random variable $\mathcal{E}[X] = 0$ and $VAR[X] = \sigma^2$

Exact: $P[|X - m| > c] = 2Q\left(\frac{c}{\sigma}\right)$

Bound: $P[|X - m| > c] \leq \frac{\sigma^2}{c^2}$



(d) Binomial $n=10, p=\frac{1}{2}$ $n=50$ $p=\frac{1}{2}$

$m=np=5$ $\sigma^2=npq=2.5$ $m=np=25$ $\sigma^2=npq=12.5$

$P[|X-5| \geq c] \leq \frac{2.5}{c^2}$ $P[|X-25| > c] \leq \frac{12.5}{c^2}$

c	$P[\|X-5\| \geq c]$	$\frac{2.5}{c^2}$
1	0.754	2.5
2	0.344	0.625
3	0.109	0.277
4	0.0215	0.156
5	0.0019	0.1

c	$P[\|X-25\| > c]$	$12.5/c^2$
5	0.119	0.5
10	0.0026	0.125
15	5.61×10^{-6}	0.0556
20	4.46×10^{-10}	0.03125

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4.100

$Y = X/n$

$$E[Y] = E[X]/n = np/n = p$$

$$VAR[Y] = VAR[X]/n^2 = npq/n^2 = pq/n,\ q = 1 - p$$

$$P[\{|Y - p|\} > a] \leq \frac{\sigma^2}{a^2} = \frac{pq}{na^2}$$

as $n \to \infty$ $P[\{|Y - p|\} > a] \to 0$ for any fixed $a > 0$

4.101

$$Y = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$E[Y] = \frac{1}{n}\sum_{i} E[X_i] = E[X]$$

$$Var[Y] = \frac{1}{n^2} Var[\sum_{i} X_i] = \frac{1}{n^2} \cdot \frac{n}{\lambda^2} = \frac{1}{n\lambda^2}$$

$$P[\{|Y - E[X]| > a\}] = P[\{|Y - E[Y]| > a\}]$$

$$\leq \frac{1}{n\lambda^2 a^2}$$

as $n \to \infty$ $P[\{|Y - E[X]| > a\}] \to 0$

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4.7 Transform Methods

4.102

$$\begin{aligned}\phi_X(w) &= \int_{-\infty}^{\infty} f_X(x)e^{jwx}dx \\ &= \int_{-b}^{b} \frac{1}{b-a}e^{jwx}dx \\ &= \frac{e^{jwb}-e^{-jwb}}{jw(b-a)} \quad = \quad \frac{e^{jwb}-e^{-jwb}}{j2wb}\end{aligned}$$

$$\begin{aligned}E[X] &= \frac{1}{j}\frac{d\phi_X(w)}{dw}|_{w=0} \\ &= -\frac{1}{b+b}\left[-\frac{1}{2}b^2+\frac{1}{2}b^2\right] \\ &= \frac{1}{2}(b+b) = 0\end{aligned}$$

$$\begin{aligned}E[X^2] &= \frac{1}{j^2}\frac{d^2\phi_x(w)}{dw^2}|_{w=0} \\ &= -\frac{1}{j(b-a)}\left[-\frac{1}{3}jb^3+\frac{1}{3}ja^3\right]_{a=-b} \\ &= \frac{1}{3}(b^2+ab+a^2) = \frac{b^2}{3}\end{aligned}$$

$$\begin{aligned}VAR[X] &= E[X^2]-E^2[X] \\ &= \frac{1}{3}(b^2+ab+a^2)-\frac{1}{4}(b+a)^2 \\ &= \frac{1}{12}(b-a)^2 \\ &= \frac{4b^2}{12} = \frac{b^2}{3}\end{aligned}$$

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4.103

$$\begin{aligned}\phi_X(w) &= \int_{-\infty}^{\infty} f_X(x)e^{jwx}dx \\ &= \int_{-\infty}^{0} \frac{\alpha}{2}e^{\alpha x}e^{jwx}dx + \int_{0}^{\infty} \frac{\alpha}{2}e^{-\alpha x}e^{jwx}dx \\ &= \frac{\alpha}{2}\frac{1}{\alpha + jw} + \frac{\alpha}{2}\frac{1}{\alpha - jw} \\ &= \frac{\alpha^2}{\alpha^2 + w^2}\end{aligned}$$

$$\begin{aligned}E[X] &= \frac{1}{j}\frac{d\phi_X(w)}{dw}|_{w=0} \\ &= \frac{1}{j}\cdot\frac{\alpha^2 \cdot 2w}{-(\alpha^2+w^2)^2}|_{w=0} \\ &= 0\end{aligned}$$

$$\begin{aligned}E[X^2] &= \frac{1}{j^2}\frac{d\phi_X(w)}{dw^2}|_{w=0} \\ &= -\frac{\alpha^2}{j^2}\frac{2(\alpha^2+w^2)^2 - 2w\cdot 2(\alpha^2+w^2)\cdot 2w}{(\alpha^2+w^2)^4}|_{w=0} \\ &= \frac{2}{\alpha^2} \\ VAR[X] &= E[X^2] - E^2[X] = \frac{2}{\alpha^2}\end{aligned}$$

4.104

X exponential

$\Phi_X(\omega) = \frac{\lambda}{\lambda - j\omega}$

$\Phi_X^n(\omega) = \left(\frac{\lambda}{\lambda - j\omega}\right)^n = \frac{1}{\left(1 - \frac{j\omega}{\lambda}\right)^n}$

Corresponds to a Gamma RV, and specifically, an n-Erlang RV

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4.105

$$\begin{aligned}
E[X] &= \frac{1}{j}\frac{d}{d\omega}e^{jm\omega-\sigma^2\omega^2/2}\Big|_{\omega=0} \\
&= \frac{1}{j}(jm-\sigma^2\omega)e^{jm\omega-\sigma^2\omega^2/2}\Big|_{\omega=0} \\
&= m \\
E[X^2] &= \frac{1}{j^2}\frac{d^2}{d\omega^2}e^{jm\omega-\sigma^2\omega^2/2}\Big|_{\omega=0} \\
&= \frac{1}{j^2}\left[-\sigma^2 e^{jm\omega-\sigma^2\omega^2/2}+(jm-\sigma^2\omega)^2 e^{jm\omega-\sigma^2\omega^2/2}\right]_{\omega=0} \\
&= \frac{1}{j^2}[-\sigma^2+j^2m^2] = \sigma^2+m^2 \\
VAR[X] &= E[X^2]-\mathcal{E}[X]^2 = \sigma^2
\end{aligned}$$

4.106

$$\Phi_Y(\omega) = E[e^{j\omega Y}] = E[e^{j\omega(aX+b)}]$$

$$= E[e^{j a\omega X}]e^{j\omega b}$$

$$= e^{j\omega b}\Phi_X(a\omega)$$

$$= e^{j\omega b}\, e^{j\omega m-\frac{\sigma^2\omega^2}{2}}\Big|_{\omega=a\omega}$$

$$= e^{j\omega b}\, e^{j\omega am-\frac{a^2\sigma^2\omega^2}{2}}$$

characteristic fcn for Gaussian RV with mean $a\omega+b$ and variance $a^2\sigma^2$

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4.107

91 We take the inverse transform of $e^{-|\omega|}$ to show that it corresponds to a Cauchy pdf:

$$\begin{aligned}\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-|\omega|}e^{j-\omega x}d(x) &= \frac{1}{2}\int_{-\infty}^{0} e^{\omega}e^{-j\omega x}d\omega + \frac{1}{2\pi}\int_{0}^{\infty} e^{-\omega}e^{-j\omega}d\omega \\ &= \frac{1}{2\pi}\left[\frac{e^{\omega(1-jx)}}{1-jx}\right]_{-\infty}^{0} + \frac{1}{2\pi}\left[\frac{e^{-\omega(1+jx)}}{-(1+jx)}\right]_{0}^{\infty} \\ &= \frac{1}{2\pi}\left[\frac{1}{1-jx}+\frac{1}{1+jx}\right] = \frac{1}{\pi(1+x^2)} \quad \checkmark\end{aligned}$$

4.108

$$P[X \geq a] \leq e^{-sa}E[e^{sX}]$$

$$E[e^{sX}] = \int_0^{\infty} e^{sx}e^{-x}dx = \int_0^{\infty} e^{-(1-s)x}dx = \frac{1}{1-s}$$

$$P[X \geq a] \leq \min_{s\geq 0} \frac{e^{-sa}}{1-s}$$

$$0 = \frac{d}{ds}\frac{e^{-sa}}{1-s} = \frac{-ae^{-sa}}{1-s} + \frac{e^{-sa}}{(1-s)^2}$$

$$+a(1-s) = 1 \Rightarrow +a - as = 1$$

$$\Rightarrow s = \frac{a-1}{a} \qquad s \geq 0 \Rightarrow a \geq 1.$$

$$\Rightarrow P[X > a] \leq \frac{e^{-(a-1)}}{1-\frac{a-1}{a}} = ae^{-(a-1)} = ae^{-a}$$

Exact probability:

$$P[X \geq a] = e^{-a}$$

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4.109

$$G_X(z) = \frac{p}{1-qz}$$

$$E[X] = \frac{d}{dz}G_X(z)|_{z=1} = \frac{p}{(1-qz)^2}(q)\Big|_{z=1} = \frac{pq}{(1-q)^2} = \frac{q}{p}$$

$$E[X^2] - E[X] = \frac{d^2}{dz^2}G_X(z)\Big|_{z=1} = \frac{pq}{(1-qz)^3}2q\Big|_{z=1} = \frac{2pq^2}{(1-q)^3} = \frac{2q^2}{p^2}$$

4.110

$$G_N(z) = \sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}z^k$$

$$= \sum_{k=0}^{n}\binom{n}{k}(pz)^k(1-p)^{n-k}$$

$$= [pz+(1-p)]^n \quad \text{from Binomial Thm.}$$

$$E[N] = G_N'(z)\Big|_{z=1} = n(pz+q)^{n-1}p\Big|_{z=1} = np$$

$$E[N^2] - E[N] = G_N''(z)\Big|_{z=1} = n(n-1)[\cdot]p^2\Big|_{z=1}$$

$$= (n^2-n)p^2$$

$$VAR[N] = n^2p^2 - np^2 + np - (np)^2 = np(1-p) \checkmark$$

4.111

$G_X(z) = (pz+q)^n$

$G_Y(z) = (pz+q)^m$

$G_X(z)G_Y(z) = (pz+q)^n(pz+q)^m = (pz+q)^{n+m}$

corresponds to pgf of Binomial RV with parameters $(n+m)$ and p.

This is a legitimate pgf.

4.112

$G_N(z) = e^{\alpha(z-1)}$ $\quad G_M(z) = e^{\beta(z-1)}$

$G_N(z)G_M(z) = e^{(\alpha+\beta)(z-1)}$

corresponds to pgf of a Poisson RV with $(\alpha+\beta)$

this is a legitimate pgf.

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4.113

$$P[X=k] = \frac{\alpha^k}{k!}e^{-\alpha}$$

$$E[e^{sX}] = \sum_{k=0}^{\infty} \frac{e^{sk}\alpha^k}{k!}e^{-\alpha} = e^{-\alpha}\sum_{k=0}^{\infty}\frac{(\alpha e^s)^k}{k}$$

$$= e^{-\alpha}e^{(\alpha e^s)}$$

$$P[X \geq a] \leq \min_{s\geq 0} e^{-sa}e^{-\alpha(1-e^s)} =$$

$$= \min_{s\geq 0} e^{-(sa+\alpha(1-e^s))}$$

We minimize the exponential by maximizing the exponent

$$0 = \frac{d}{ds}(sa+\alpha(1-e^s)) = a - \alpha e^s \Rightarrow a = \alpha e^s$$

$$s = \ln\left(\frac{a}{\alpha}\right) \qquad s \geq 0 \text{ for } a \geq \alpha$$

$$P[X \geq a] \leq e^{-a\ln\frac{a}{\alpha}}e^{-\alpha(1-\frac{a}{\alpha})} = e^{(a-\alpha-a\ln\frac{a}{\alpha})}$$

For $\alpha = 1$, $a = 5$ $\quad P[X \geq 5] \leq e^{(5-1-5\ln 5)} = e^{4-5\ln 5}$

$= 0.01747$

Exact Probability

$$P[X \geq 5] = 1 - \sum_{k=0}^{4}\frac{(1)^k}{k!}e^{-1}$$

$$= 0.00366$$

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4.114

a) $G_U(z) = E[z^U] = \sum_{k=0}^{\infty} p_U(k) z^k = \sum_{k=a}^{b} \frac{1}{b-a+1} z^k$

$$= \frac{z^a}{b-a+1}\left[\sum_{k=0}^{b-a} z^k\right] = \frac{z^a}{b-a+1}\left[\frac{1-z^{b+1-a}}{1-z}\right]$$

$$= \frac{1}{b-a+1}\left[\frac{z^a - z^{b+1}}{1-z}\right]$$

b) $E[U] = G_U'(1) = \frac{1}{b-a+1}\frac{1}{(1-z)^2}\left[z^a(az^{-1}-a+1) - z^b(b+1-zb)\right]\Big|_{z=1}$

$$\lim_{z\to 1} G_U'(z) = \frac{1}{(b-a+1)(2)(1-z)(-1)}\cdot\left[az^{a-2}(-1+a)+az^{a-1}(1-a)+bz^b(1+b)-bz^{b-1}(b+1)\right]$$

$$\cdots = \frac{a(a-1)z^{a-2} - b(1+b)z^{b-1}}{(b-a+1)(-1)(2)}$$

$$= \frac{a(a-1)-b(1+b)}{-2(b-a+1)} = \frac{b+a}{2}$$

$$VAR[U] = G_U''(1) + G_U'(1) - (G_U'(1))^2$$

$$\lim_{z\to 1} G_U''(1) = \frac{a(a-1)(a-2) - b(b-1)(b+1)}{-3(b-a+1)}$$

$$VAR[U] = \frac{b(b-1)(b+1) - a(a-1)(a-2)}{3(b-a+1)} + \frac{b+a}{2} - \left(\frac{b+a}{2}\right)^2$$

$$= \frac{(b-a+1)^2 - 1}{12}$$

c) $G_U(z)^2$ corresponds to a pgf

$$G_V(z) = G_U(z)^2$$

$$E[V] = G_V'(1) = 2G_U'(1)G_U(1) = 2G_U'(1)$$

$$= b+a$$

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4.115

The negative Binomial random variable has

$$\begin{aligned} G_X(z) &= \left(\frac{pz}{1-qz}\right)^r \\ P[X=r] &= \frac{1}{r!}\frac{d^r}{dz^r}G_X(z)\Big|_{z=0} = \frac{1}{r!}\frac{d^r}{dz^r}\left(\frac{pz}{1-pz}\right)^r \\ &= \frac{1}{r!}\frac{d^{r-1}}{dz^{r-1}}r\left(\frac{pz}{1-qz}\right)^{r-1}\frac{p}{(1-qz)^2} \end{aligned}$$

First, consider the $r=2$ negative Binomial random variable:

$$\begin{aligned} P[X=2] &= \frac{1}{2!}\frac{d^2}{dz^2}G_X(z)\Big|_{z=0} = \frac{1}{2!}\frac{d^2}{dz^2}\frac{(pz)^2}{(1-qz)^2}\Big|_{z=0} \\ &= \frac{p^2}{2!}\frac{d}{dz}\left[\frac{2z}{(1-qz)^2}+\frac{z^2 2q}{(1-qz)^3}\right]_{z=0} \\ &= \frac{p^2}{2!}\left[f2(1-qz)^2+2\frac{4zq}{(1-qz)^3}+\frac{z^2(z)(3)q^2}{(1-qz)^4}\right]_{z=0} \\ &= p^2 \end{aligned}$$

In the general case, we have

$$\begin{aligned} P[X=r] &= \frac{1}{r!}\frac{d^r}{dz^r}\frac{(pz)^r}{(1-qz)^r}\Big|_{z=0} \\ &= \frac{p^r}{r!}\sum_{k=0}^{r}\binom{r}{k}\frac{d^k}{dz^k}(z^r)\frac{d^{r-k}}{dz^{r-k}}\frac{1}{(10qz)^r}\Big|_{z=0} \end{aligned}$$

$$\frac{d^k}{dz^k}(z^r)\Big|_{z=0} = \begin{cases} z^r|_{z=0}=0 & k=0 \\ (r-r1)...(r-k+1)z^{r-k}|_{z=0}=0 & 0<k<r \\ r(r-1)...(3)(2)(1)=r! & k=r \end{cases}$$

$\therefore$ only the $k=r$ term in the summation is nonzero

$$\begin{aligned} P[X=r] &= \frac{p^r}{r!}\left\{\binom{r}{r}f!\frac{1}{(1-qz)^r}\right\}_{z=0} = p^r \quad \surd \\ \mathcal{E}[X] &= \frac{d}{dz}G_N(z)\Big|_{z=1} = \frac{d}{dz}\left(\frac{pz}{1-qz}\right)_{z=1}^r \\ &= \frac{d}{dz}\frac{p^r}{(z^{-1}-q)^r}\Big|_{z=1} = p^r(-r)(z^{-1}-q)^{-r-1}(-z^{-2})\Big|_{z=1} \\ &= \frac{rp^r}{(1-q)^{r+1}} = \frac{r}{p} \end{aligned}$$

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4.116

$$\begin{aligned} X^*(s) &= \int_{-\infty}^{\infty} f_X(x)e^{-sx}dx \\ \frac{d^nX^*(s)}{ds^n} &= \int_{-\infty}^{\infty} f_X(x)\cdot(-x)^n e^{-sx}dx \\ &= (-1)^n\int_{-\infty}^{\infty} x^n f_X(x)e^{-sx}dx \\ \Rightarrow E[X^n] &= \int_{-\infty}^{\infty} x^n f_X(x)dx \\ &= (-1)^n \frac{d^nX^*(s)}{ds^n}|_{s=0} \end{aligned}$$

4.117

$$\begin{aligned} \Phi_X(\omega) &= \int_0^{\infty} \frac{\lambda(\lambda x)^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)} e^{j\omega x}dx = \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^{\infty} x^{\alpha-1}e^{-(\lambda-j\omega)x}dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)}(\lambda-j\omega)^{-\alpha}\underbrace{\int_0^{\infty}((\lambda-j\omega)x)^{\alpha-1}e^{-(\lambda-j\omega)x}\alpha(\lambda-j\omega)x}_{\Gamma(\alpha)} \\ &= \frac{(\lambda^\alpha)}{(\lambda-j\omega)^\alpha} = \frac{1}{(1-j\omega/\lambda)^\alpha} \\ \mathcal{E}[X^n] &= \frac{1}{j^n}\frac{d^n}{d\omega^n}\Phi_X(\omega)\Big|_{\omega=0} = \lambda^{-n}\frac{(\alpha+n-1)!}{(\alpha-1)!}\left(1-\frac{j\omega}{\lambda}\right)^{-(\alpha+n)}\Big|_{\omega=0} \\ &= \lambda^{-n}\frac{(\alpha+n-1)!}{(\alpha-1)!} \end{aligned}$$

4.118

$$X = \begin{cases} X_1 & \text{with prob } p \\ X_2 & \text{with prob } 1-p \end{cases}$$

$$\begin{aligned} X^*(s) &= \mathcal{E}[e^{-sX}] = \mathcal{E}[e^{-sX}|X=X_1]p + \mathcal{E}[e^{-sX}|X=X_2](1-p) \\ &= \mathcal{E}[e^{-sX_1}]p + \mathcal{E}[e^{-sX_2}](1-p) \\ &= p\frac{\lambda_1}{s+\lambda_1} + (1-p)\frac{\lambda^2}{s+\lambda_2} \end{aligned}$$

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4.119

$$\begin{aligned} X^*(s) &= \left(\frac{\alpha}{s+\alpha}\right)\left(\frac{\beta}{s+\beta}\right) \\ &= \frac{\alpha\beta}{\alpha+\beta}\left[\frac{1}{s+\beta}-\frac{1}{s+\alpha}\right] \\ f_X(t) &= \frac{\alpha\beta}{\alpha-\beta}\left[\mathcal{L}^{-1}\left[\frac{1}{s+\beta}\right]-\mathcal{L}^{-1}\left[\frac{1}{s+\alpha}\right]\right] \\ &= \frac{\alpha\beta}{\alpha-\beta}\left[\frac{1}{\beta}e^{-\beta t}-\frac{1}{\alpha}e^{-\alpha t}\right] \\ &= \frac{\alpha}{\alpha-\beta}e^{-\beta t}-\frac{\beta}{\alpha-\beta}e^{-\alpha t} \end{aligned}$$

4.120

- Gamma(λ, α)

$$X^*(s) = \frac{\lambda^\alpha}{(\lambda+s)^\alpha}$$

- Gamma$(\lambda, \alpha-1)$

$$\begin{aligned} Y^*(s) &= \int_0^\infty \frac{\lambda(\lambda y)^{\alpha-2}}{\Gamma(\alpha-1)} e^{-\lambda y} e^{-sy}\, dy \\ &= \frac{\lambda^{\alpha-1}}{\Gamma(\alpha-1)} \int_0^\infty y^{\alpha-2} e^{-(\lambda+s)y}\, dy \\ &= \frac{\lambda^{\alpha-1}}{\Gamma(\alpha-1)} \int_0^\infty \left(\frac{\omega}{\lambda+s}\right)^{\alpha-2} e^{-\omega} \frac{d\omega}{\lambda+s} \\ &= \frac{\lambda^{\alpha-1}}{(\lambda+s)^{\alpha-1}} \cdot \frac{1}{\Gamma(\alpha-1)} \int_0^\infty \omega^{(\alpha-1)-1} e^{-\omega} d\omega \\ &= \frac{\lambda^{\alpha-1}}{(\lambda+s)^{\alpha-1}} \end{aligned}$$

$$X_1(s) = \frac{\lambda}{\lambda+s} Y^*(s)$$

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4.121

$$P[X \geq a] \leq \min_{s \geq 0} \frac{e^{-sa}}{(1 - s/\lambda)^{\alpha}}$$

$$0 = \frac{d}{ds} \frac{e^{-sa}}{(1-\frac{s}{\lambda})^{\alpha}} = \frac{-ae^{-sa}}{(1-\frac{s}{\lambda})^{\alpha}} - \frac{\alpha e^{-sa}}{(1-\frac{s}{\lambda})^{\alpha+1}} \left(-\frac{1}{\lambda}\right)$$

$$a\left(1-\frac{s}{\lambda}\right) = \frac{\alpha}{\lambda} \qquad 1 + \frac{s}{\lambda} = \frac{\alpha}{a\lambda}$$

$$s = \lambda\left(1 - \frac{\alpha}{a\lambda}\right) = \lambda - \frac{\alpha}{a} \qquad s \geq 0 \Rightarrow \lambda \geq \frac{\alpha}{a}$$

$$a \geq \frac{\alpha}{\lambda}$$

$$P[X \geq a] \leq \frac{e^{-sa}}{(1-s/\lambda)^{\alpha}} = \frac{e^{-(a\lambda-\alpha)}}{\left(1-(1-\frac{\alpha}{a\lambda})\right)^{\alpha}}$$

$$= \frac{e^{-(a\lambda-\alpha)}}{(\alpha/a\lambda)^{\alpha}}$$

(b) $P[X \geq 9]$ $m=3, \lambda=1$ Erlang $\Rightarrow \alpha=3, \lambda=1$

$$P[X \geq 9] \leq \frac{e^{-(9-3)}}{\left(\frac{3}{9}\right)^{3}} = 27e^{-6} = 0.067$$

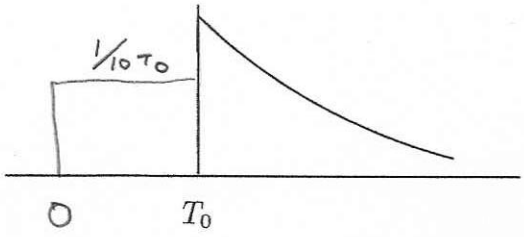
Exact for Gamma

$P[X \geq 9] = 0.00623$

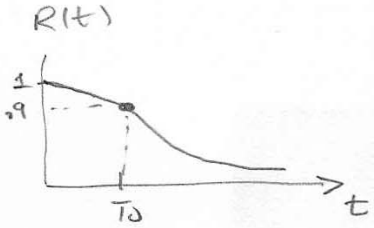
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4.8 Basic Reliability Calculations

4.122 a) $f_T(t) = \begin{cases} \frac{1}{10} & 0 \le t \le T_0 \\ \frac{1}{9}\lambda e^{-\lambda(t-T_0)} & t \ge T_0 \\ 0 & t < 0 \end{cases}$



$$R(t) = P[T > t] = \begin{cases} 1 - t/10T_0 & 0 < t < T_0 \\ e^{-\lambda(t-T_0)} & t \ge T_0 \end{cases}$$



where we used the fact that

$$P[T > t] = \int_t^\infty \lambda e^{-(\lambda t' - T_0)} dt' = e^{-\lambda(t-T_0)} \quad t > T_0$$

The $MTTF$ is given by the expected value of X:

$$MTTF = E[T] = \int_0^\infty R(t')dt'$$
$$= \int_0^{T_0} 1\, dt' + \int_{T_0}^\infty e^{-\lambda(t'-T_0)} dt'$$
$$= T_0 + \frac{1}{\lambda}$$

b) $r(t) = \frac{-R'(t)}{R(t)} = \begin{cases} 0 & 0 \le t \le T_0 \\ \lambda & t < T_0 \end{cases}$

c) $R(t) = e^{-\lambda(t-T_0)} = 0.99 \Rightarrow$

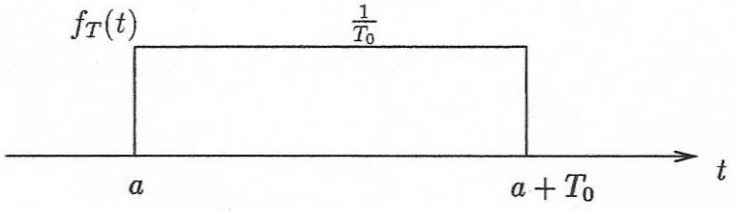
$$\Rightarrow \lambda(t - T_0) = \ln \frac{1}{0.99}$$
$$\Rightarrow t = T_0 + \frac{1}{\lambda} \ln \frac{1}{0.99} = T_0 + \frac{.01}{\lambda}$$

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4.123 a)

$$R(t) = P[T > t] = \int_t^\infty f_T(t')dt'$$

$$= \begin{cases} 1 & t < a \\ 1 - \dfrac{t-a}{T_0} & a < t < a + T_0 \\ 0 & t > T_0 \end{cases}$$



$$MTTF = \int_0^\infty R(t)dt = a + \frac{T_0}{2}$$

b) $$r(t) = \frac{-R'(t)}{R(t)} = \begin{cases} \dfrac{1}{a + T_0 - t} & a < t < a + T_0 \\ 0 & \text{elsewhere} \end{cases}$$

c) $$R(t) = 1 - \frac{t-a}{T_0} = 0.99 \Rightarrow t = \alpha + 0.01T_0$$

4.124

3 a) $$R(t) = \int_0^\infty \frac{x}{\alpha^2} e^{-s^2/2\alpha^2} dx = -e^{-x^2/2\alpha^2}\Big|_t^\infty = e^{-t^2/2\alpha^2}$$

b) $$r(t) = \frac{f_T(t)}{R(t)} = \frac{t}{\alpha^2} \qquad t \geq 0$$

(c) $R_{series} = e^{-t^2/2\alpha^2} \cdot e^{-t^2/2\alpha^2} = e^{-t^2/\alpha^2}$

(d) $R_{parallel} = 1 - (1 - e^{-t^2/2\alpha^2})(1 - e^{-t^2/2\alpha^2})$

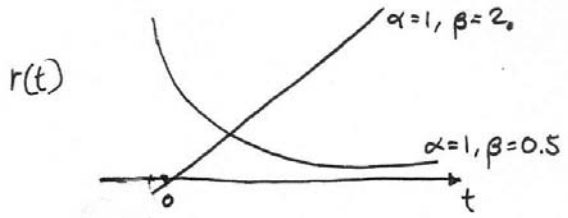
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4.125

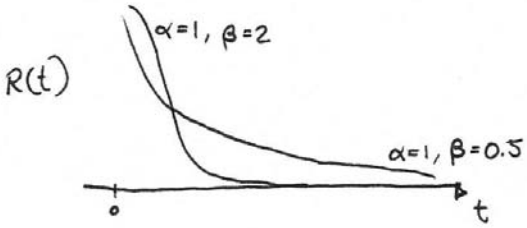
a) $R(t) = 1 - F_T(t) = e^{-(t/\lambda)^\beta}$

$$R'(t) = -\frac{\beta}{\lambda}\left(\frac{t}{\lambda}\right)^{\beta-1} e^{-(t/\lambda)^\beta}$$

$$r(t) = \frac{\beta}{\lambda}\left(\frac{t}{\lambda}\right)^{\beta-1} = \alpha\beta t^{\beta-1}$$



b)



c) $R(t) = R_1(t)R_2(t) = \alpha_1\beta_1 t^{\beta_1-1}\alpha_2\beta_2 t^{\beta_2-1} = e^{-(\alpha_1 t^{\beta_1} + \alpha_2 t^{\beta_2})}$

d) $R(t) = 1-(1-R_1(t))(1-R_2(t)) = 1-(1-e^{-\alpha_1 t^{\beta_1}})(1-e^{-\alpha_2 t^{\beta_2}})$

$$= \cancel{1} - \cancel{1} + e^{-\alpha_2 t^{\beta_2}} + e^{-\alpha_1 t^{\beta_1}} - e^{-\alpha_1 t^{\beta_1}} e^{-\alpha_2 t^{\beta_2}}$$

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4.126

$$R(t) = P[T>t]$$
$$= \sum_{n=0}^{m-1} e^{-\lambda t}\frac{(\lambda t)^n}{n!} \qquad \text{Erlang.}$$

b)
$$r(t) = \frac{-f_T(t)}{R(t)}$$
$$= \frac{-\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!} \cdot \frac{1}{\sum_{n=0}^{m-1} e^{-\lambda t}\frac{(\lambda t)^n}{n!}}$$
$$= \frac{-\lambda (\lambda t)^{m-1}/(m-1)!}{\sum_{n=0}^{m-1} (\lambda t)^n/n!}$$

4.127

3.105 The failure rate function of the memory chips is obtained as follows:

$$\begin{aligned}
F_T(x|t>t) &= P[T \le x | T > t] \\
&= \begin{cases} 0 & x < t \\ \frac{F_T(x)-F_T(t)}{1P[T>t]} & x \ge t \end{cases} \\
&= \begin{cases} 0 \\ \frac{[1-(1-p)e^{-\alpha x}-pe^{-1000\alpha x}]-[1-(1-p)e^{\alpha t}-pe^{-1000\alpha t}]}{(1-p)e^{-\alpha t}+pe^{-1000\alpha t}} \end{cases} \\
f_T(x|T>t) &= \frac{\alpha(1-p)e^{-\alpha x}+1000\alpha pe^{-1000\alpha x}}{(1-p)e^{-\alpha t}+pe^{-1000\alpha t}} \\
r(t) &= f_T(t|T>t) = \frac{\alpha(1-p)e^{-\alpha t}+1000\alpha pe^{-1000\alpha t}}{(1-p)e^{-\alpha t}+pe^{-1000\alpha t}}
\end{aligned}$$

For small t, $r(t)$ is dominated by the second term is the numerator. For large t, $r(t)$ is dominated by the first term.

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4.128

(a) $R(t) = P[T > t]$

$$= P[T > t \mid s=1]\,p + P[T > t \mid s=2](1-p)$$

$$= (1 - F_{exp}(t))\,p + (1 - F_{pareto}(t))(1-p)$$

$$= \begin{cases} p e^{-t/m} + (1-p)\left(\frac{x_m}{t}\right)^{\alpha} & t > x_m \\ p e^{-t/m} + (1-p)\cdot 1 & x_m \geq t > 0 \end{cases}$$

$$x_m = \frac{m(\alpha-1)}{\alpha}$$

(b)

$$r(t) = -\frac{R'(t)}{R(t)}$$

$$= \begin{cases} \dfrac{-\frac{p}{m}e^{-t/m} - \alpha x_m^{\alpha} t^{-\alpha-1}(1-p)}{p e^{-t/m} + (1-p) x_m^{\alpha} t^{-\alpha}} & t > x_m \\ \dfrac{-\frac{p}{m}e^{-t/m}}{p e^{-t/m} + (1-p)} & 0 < t < x_m \end{cases}$$

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4.129

$$R(t) = \exp\{-\int_0^t r(t')dt'\}$$

For $0 \leq t < 1$:

$$R(t) = \exp\{-\int_0^t [1 + 9(1-t')]dt'\}$$
$$= \exp\{-10t + \frac{9}{2}t^2\}$$

For $1 \leq t < 10$:

$$R(t) = \exp\{-\int_0^1 [1 + 9(1-t')]dt' - \int_1^t 1dt'\}$$
$$= \exp\{-10 + \frac{9}{2} - (t-1)\}$$
$$= \exp\{-4,5 - t\}$$

For $t \geq 10$:

$$R(t) = \exp\{-\int_0^1 [1 + 9(1-t')]dt' - \int_1^{10} 1dt' - \int_{10}^t [1 + 10(t'-10)]dt'\}$$
$$= \exp\{-4,5 - 10 - [-99(t-10) + 5(t^2 - 10^2)]\}$$
$$= \exp\{-5t^2 + 99t - 1504.5\}$$

$$f_T(t) = -r(t)R(t)$$
$$= \begin{cases} -[1 + 9(1-t)]\exp\{-10t + \frac{9}{2}t^2\} & 0 \leq t < 1 \\ 1\exp\{-4.5 - t\} & 1 \leq t < 10 \\ 1 + 10(t-10)\exp\{-5t^2 + 99t - 1504.5\} & t > 10 \end{cases}$$

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4.130

3.108 Each component has reliability: $R_i(t) = e^{-t}$

a) $R(t) = P[\text{system working at time}] = P[\text{2 or more working at time } t]$

$$= \binom{3}{2}(e^{-t})^2(1-e^{-t}) + \binom{3}{3}(e^{-t})^3$$
$$= 3e^{-2t} - 2e^{-3t}$$

$$MTTF = \int_0^\infty R(t')dt' = \int_0^\infty (3e^{-2t'} - 2e^{-3t'})dt'$$
$$= \frac{3}{2} - \frac{2}{3} = \frac{5}{6}$$

b) Now $R_1(t) = R_2(t) = e^{-t}$ and $R_3(t) = e^{-t/2}$. $R(t) = P[\text{2 or more working at time } t]$

$$= R_1(t)R_2(t)(1-R_3(t)) + R_1(t)(1-R_2(t))R_3(t)$$
$$+(1-R_1(t))R_2(t)R_3(t) + R_1(t)R_2(t)R_3(t)$$
$$= e^{-2t}(1-e^{-t/2}) + 2e^{-t}(1-e^{-t})e^{-t/2} + e^{-2t}e^{-t/2}$$
$$= e^{-2t} + 2e^{-3t/2} - 2e^{-st/2}$$

$$MTTF = \int_0^\infty R(t')dt' = \int_0^\infty (3e^{-2t'} + 2e^{-\frac{3t}{2}} - 2e^{-\frac{5t'}{2}})dt'$$
$$= \frac{1}{2} + 2\frac{2}{3} - 2\frac{2}{5} = \frac{31}{30}$$

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4.131 $R(t) = e^{-\alpha t^\beta}$ Weibull $E[X] = 1 = \frac{\Gamma(4/3)}{\alpha^{1/3}}$

(a) $P[2 \text{ or more working at time } t]$

$$= \binom{3}{2}(e^{-\alpha t^\beta})^2(1-e^{-\alpha t^\beta}) + \binom{3}{3}(e^{-\alpha t^\beta})^3$$

$$= 3e^{-2\alpha t^\beta} - 3e^{-3\alpha t^\beta} + 1e^{-3\alpha t^\beta}$$

$$= 3e^{-2\alpha t^\beta} - 2e^{-3\alpha t^\beta} = 3e^{-2\alpha t^3} - 2e^{-3\alpha t^3}$$

(b) $P[2 \text{ or more working at time } t]$

$$= (R_1(t))^2(1-e^{-\alpha' t^{\beta'}}) + 2e^{-\alpha t^\beta}(1-e^{-\alpha t^\beta})e^{-\alpha' t^{\beta'}} + (e^{-\alpha t^\beta})^2 e^{-\alpha' t^{\beta'}}$$

$$= e^{-2\alpha t^3}(1-e^{-\alpha' t^3}) + 2e^{-\alpha t^3}(1-e^{-\alpha t^3})e^{-\alpha' t^3} + e^{-2\alpha t^3}e^{-\alpha' t^3}$$

$$= e^{-2\alpha t^3} - e^{-(2\alpha+\alpha')t^3} + 2e^{-(\alpha+\alpha')t^3} - 2e^{-(2\alpha+\alpha')t^3} + e^{-(2\alpha+\alpha')t^3}$$

$$= e^{-2\alpha t^3} + 2e^{-(\alpha+\alpha')t^3} - 2e^{-(2\alpha+\alpha')t^3}$$

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$$MTTF = \int_0^\infty R(t)\,dt$$

for Weibull $\int_0^\infty e^{-\alpha t^\beta}\,dt = \frac{\Gamma(1+\frac{1}{\beta})}{\alpha^{1/\beta}} \overset{\beta=3}{=} \frac{\Gamma(\frac{4}{3})}{\alpha^{1/3}} = E[X]$

$$\int_0^\infty e^{-2\alpha t^\beta}\,dt = \frac{\Gamma(1+\frac{1}{\beta})}{(2\alpha)^{1/\beta}} = \frac{\Gamma(1+1/\beta)}{2^{1/\beta}(\alpha)^{1/\beta}} = \frac{E[X]}{2^{1/\beta}}$$

$$\int_0^\infty e^{-\alpha' t^\beta}\,dt = 2 = \frac{\Gamma(1+\frac{1}{\beta})}{(\alpha')^{1/\beta}} = \frac{\Gamma(\frac{4}{3})}{(\alpha')^{1/\beta}}$$

$$\Rightarrow \Gamma(\tfrac{4}{3}) = 2(\alpha')^{1/\beta} \qquad \Rightarrow 2(\alpha')^{1/\beta} = \alpha^{1/\beta}$$

$$\text{vs} \Rightarrow \Gamma(\tfrac{4}{3}) = 1(\alpha)^{1/\beta} \qquad \Rightarrow (\alpha')^{1/\beta} = \tfrac{1}{2}\alpha^{1/\beta}$$

$$\Rightarrow \alpha' = \alpha/2^\beta = \frac{\alpha}{8}$$

(a) $$MTTF = \int_0^\infty 3e^{-2\alpha t^3}\,dt - 2\int_0^\infty e^{-3\alpha t^3}\,dt = 3\frac{\Gamma(4/3)}{(2\alpha)^{1/3}} - 2\frac{\Gamma(4/3)}{(3\alpha)^{1/3}}$$

$$= \left(\frac{3}{2^{1/3}} - \frac{2}{3^{1/3}}\right)\underbrace{\frac{\Gamma(4/3)}{\alpha^{1/3}}}_{1} = 0.381$$

(b) $$MTTF = \int_0^\infty e^{-2\alpha t^3}\,dt + 2\int_0^\infty e^{-(\alpha+\alpha')t^3}\,dt - 2\int_0^\infty e^{-(2\alpha+\alpha')t^3}\,dt$$

$$= \frac{\Gamma(4/3)}{(2\alpha)^{1/3}} + \frac{2\Gamma(4/3)}{(\alpha+\alpha')^{1/3}} - 2\frac{\Gamma(4/3)}{(2\alpha+\alpha')^{1/3}}$$

$$= \frac{\Gamma(\frac{4}{3})}{\alpha^{1/3}}\left[\frac{1}{(2)^{1/3}} + \frac{2}{(9/8)^{1/3}} - \frac{2}{(17/8)^{1/3}}\right]$$

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4.132

.110 a) Reliability of processor $R_S(t) = e^{-t/5}$
Reliability of peripheral units $R_P(t) = e^{-t/10}$

$$\begin{aligned}
R(t) &= P[\text{1 or more processors functioning at time } t] \\
&\quad \times P[\text{2 or more peripherals functioning at time } t] \\
&= \left[\binom{2}{1} R_S(t)(1-R_S(t)) + R_S^2(t)\right]\left[\binom{3}{2} R_P^2(t)(1-R_P(t)) + \binom{3}{3} R_P^3(t)\right] \\
&= 2e^{-tt/5}(1-e^{-t/5}) + e^{-2t/5}][3e^{-2t/20}(1-e^{-t/10}+e^{-3t/10}] \\
&= e^{-2t/5}[(2-e^{-t/5})(3-2e^{-t/10})]
\end{aligned}$$

$$MTTF = \int_0^\infty R(t)dt = 6\left(\frac{5}{2}\right) - 3\left(\frac{5}{3}\right) - 4\left(\frac{10}{5}\right) + 2\left(\frac{10}{7}\right) = \frac{34}{7}$$

b) If $R_S(t) = e^{-t/10}$ $R_P(t) = e^{-t/5}$

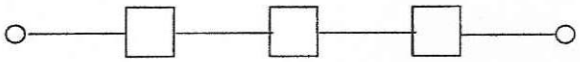
$$\begin{aligned}
R(t) &= [2e^{-t/10}(1-e^{-t/10}) + e^{-2t/10}][3e^{-2t/5}(1-e^{-t/5}) + e^{-3t/5}] \\
&= \underbrace{e^{-t/10}e^{-2t/5}}_{e^{-t/2}}[(2-e^{-t/10})(3-2e^{-t/5})]
\end{aligned}$$

$$MTTF = \int_0^\infty R(t)dt = 6(2) - 3\left(\frac{10}{6}\right) - 4\left(\frac{10}{7}\right) + 2\left(\frac{10}{8}\right) = \frac{53}{14}$$

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4.133

(a)



$$R_{subsystem}(t) = R^3(t) = e^{-3t}$$

$$R_{parallel}(T) = 1 - (1 - e^{-3T})^n = 0.99$$

$$n \ln(1 - e^{-3T}) = \ln \frac{1}{100}$$

$$n = \frac{\ln 100}{\ln \frac{1}{1-e^{-3T}}}$$

(b) $1 - (1 - e^{-T^2/2\alpha^2})^n = 0.99$

$$n \ln(1 - e^{-T^2/2\alpha^2}) = \ln \frac{1}{100}$$

$$n = \frac{\ln 100}{\ln(1 - e^{-T^2/2\alpha^2})}$$ Rayleigh

(c)

$$n = \frac{\ln 100}{\ln(1 - e^{-\alpha T^3})}$$ Weibull

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4.9 Computer Methods for Generating Random Variables

4.134 The following Octave commands generate the requested plots:

(a)

```
x = [-4:0.01:4];
y0 = normal_pdf(x, -2, 1);
y1 = normal_pdf(x, 2, 1);
figure;
hold on;
plot(x, y0, "1");
plot(x, y1, "3");
```

(b)

```
x = [-5:0.01:5];
y0 = 1-normal_cdf(x, -2, 1);
y1 = 1-normal_cdf(x, 2, 1);
ey0 = e.^(-(x+2).^2/2);
ey1 = e.^(-(x-2).^2/2);
figure;
hold on;
plot(x, y0, "1");
plot(x, y1, "3");
plot(x, ey0, "1");
plot(x, ey1, "3");
```

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4.135 The following Octave commands generate plots of the pdf and cdf of the gamma random variable:

(a)

```
x = [0:0.01:15];
figure;
hold on;
plot(x, gamma_pdf(x, 1, 1), "1");
plot(x, gamma_pdf(x, 2, 1), "2");
plot(x, gamma_pdf(x, 4, 1), "3");
figure;
hold on;
plot(x, gamma_cdf(x, 1, 1), "1");
plot(x, gamma_cdf(x, 2, 1), "2");
plot(x, gamma_cdf(x, 4, 1), "3");
```

(b)

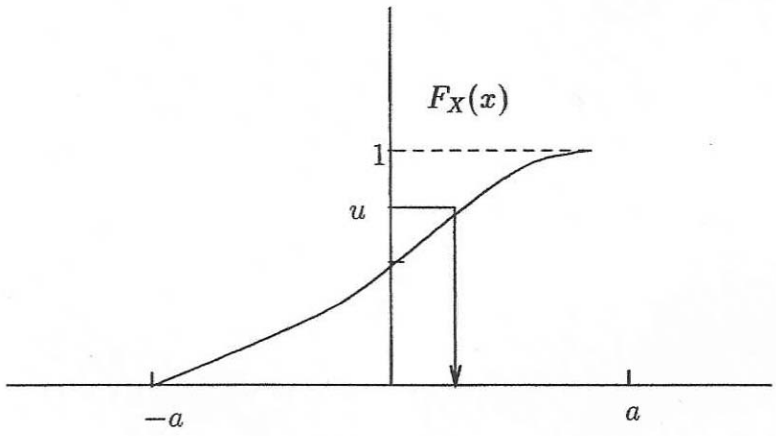
```
x = [0:0.01:15];
figure;
hold on;
plot(x, gamma_pdf(x, 1/2, 1/2), "1");
plot(x, gamma_pdf(x, 1, 1/2), "2");
plot(x, gamma_pdf(x, 3/2, 1/2), "3");
plot(x, gamma_pdf(x, 5/2, 1/2), "4");
figure;
hold on;
plot(x, gamma_cdf(x, 1/2, 1/2), "1");
plot(x, gamma_cdf(x, 1, 1/2), "2");
plot(x, gamma_cdf(x, 3/2, 1/2), "3");
plot(x, gamma_cdf(x, 5/2, 1/2), "4");
```

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4.136

$$f_X(x) = \begin{cases} (a+x)/a^2 & -a \le x \le 0 \\ (a-x)/a^2 & 0 \le x \le a \\ 0 & \text{elsewhere} \end{cases}$$

$$\Rightarrow F_X(x) = \begin{cases} 0 & x < -a \\ \frac{1}{2}\left[\left(\frac{x}{a}\right)^2 + 2\left(\frac{x}{a}\right) + 1\right] & -a \le x \le 0 \\ \frac{1}{2} + \frac{1}{2}\left[2\left(\frac{x}{a}\right) - \left(\frac{x}{a}\right)^2\right] & 0 \le x \le a \\ 1 & x > a \end{cases}$$



Solving the equation $u = F_X(x)$ for x we obtain

$$= F_X^{-1}(u) = \begin{cases} -a + a\sqrt{2u} & 0 \le u \le \frac{1}{2} \\ a - a\sqrt{2-2u} & \frac{1}{2} \le u \le 1 \end{cases}$$

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4.137 The following Octave commands generate the requested samples and plots:

(a)

```
x = [-6:0.01:6];
u = rand(1, 1000);
%Multiply all values by discretely generated -1 or 1
z = -log(u).*discrete_rnd(length(u), [-1 1], [0.5 0.5]);
figure;
hold on;
%Normalize to 2 because bar width is 0.5
hist(z, [-6:0.5:6], 2);
plot(x, laplace_pdf(x), "1");
```

(b)

```
x = [1:0.01:10];
u = rand(1, 1000);
k = 1.5;
z = u.^(-1/k);
figure;
hold on;
hist(z, [1.25:0.5:10], 2);
plot(x, k./x.^(k+1));

x = [1:0.01:10];
u = rand(1, 1000);
k = 2;
z = u.^(-1/k);
figure;
hold on;
hist(z, [1.25:0.5:10], 2);
plot(x, k./x.^(k+1));

x = [1:0.01:10];
u = rand(1, 1000);
k = 2.5;
z = u.^(-1/k);
figure;
hold on;
hist(z, [1.25:0.5:10], 2);
plot(x, k./x.^(k+1));
```

(c)

```
x = [0:0.01:5];
u = rand(1, 1000);
b = 0.5;
z = log(1./u).^(1./b);
figure;
hold on;
hist(z, [0:0.25:5], 4);
plot(x, b.*x.^(b-1).*e.^(-x.^b), "1");

x = [0:0.01:5];
u = rand(1, 1000);
```

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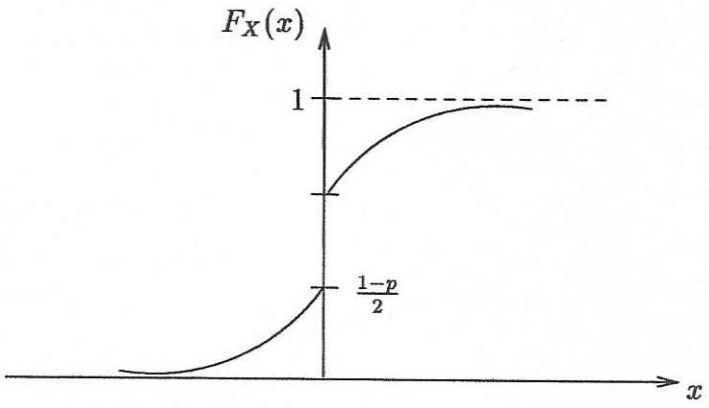
```
b = 2;
z = log(1./u).^(1./b);
figure;
hold on;
hist(z, [0:0.125:5], 8);
plot(x, b.*x.^(b-1).*e.^(-x.^b), "1");

x = [0:0.01:5];
u = rand(1, 1000);
b = 3;
z = log(1./u).^(1./b);
figure;
hold on;
hist(z, [0:0.125:5], 8);
plot(x, b.*x.^(b-1).*e.^(-x.^b), "1");
```

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4.138

$$F_X(x) = \begin{cases} \frac{1-p}{2}e^{\alpha x} & x < 0 \\ 1 - \frac{1-p}{2}e^{-\alpha x} & x \geq 0 \end{cases}$$



Note that $\mu = F_X(x)$ for

$$\frac{1-p}{2} < \mu < 1 - \frac{1-p}{2} \Rightarrow z = 0$$

$$Z = F_X^{-1}(\mu) = \begin{cases} \frac{1}{\alpha}\ln\frac{2U}{1-p} & 0 \leq \mu \leq \frac{1-p}{2} \\ -\frac{1}{\alpha}\ln\frac{2(1-U)}{1-p} & 1 - \frac{1-p}{2} \leq \mu \leq 1 \\ 0 & \frac{1-p}{2} \leq \mu \leq 1 - \frac{1-p}{2} \end{cases}$$

4.139

$$\begin{aligned} 0 < U < \tfrac{1}{2} &\Rightarrow X = 1 \\ \tfrac{1}{2} < U < \tfrac{3}{4} &\Rightarrow X = 2 \\ \tfrac{3}{4} < U < \tfrac{7}{8} &\Rightarrow X = 3 \\ \vdots & \quad \vdots \\ \mathcal{E}[N] = \mathcal{E}[X] &= \tfrac{1}{\frac{1}{2}} = 2 \end{aligned}$$

4.140

$$\begin{aligned} 0 < U < e^{-\alpha} &\Rightarrow X = 0 \\ e^{-\alpha} < U < e^{-\alpha}(1+\alpha) &\Rightarrow X = 1 \\ e^{-\alpha}(1+\alpha) < U < e^{-\alpha}(1+\alpha+\tfrac{\alpha^2}{2!}) &\Rightarrow X = 2 \\ \vdots & \quad \vdots \\ \text{Average number of comparisons} &= \textstyle\sum_{k=0}^{\infty}(k+1)\frac{\alpha^k}{k!}e^{-\alpha} = \alpha + 1 \end{aligned}$$

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4.141 The following Octave commands describe the function for performing the rejection method and the code to call the function:

```
function z = gaussian_rejection_method(N)
     z = zeros(1, N);
     k = 1;
     while k <= N
          while true
               u1 = rand;
               u2 = rand;
               x1 = -log(u1);
               if (u2 <= e.^(-((x1-1).^2))/2)
                    z(k) = x1.*discrete_rnd(1,[-1 1],[0.5 0.5]);
                    break;
               end
          end
          k = k + 1;
     end
end

x = [-4:0.01:4];
z = gaussian_rejection_method(10000);
figure;
hold on;
hist(z, [-4:0.125:4], 8);
plot(x, normal_pdf(x, 0, 1), "1");
```

The cdf of X_1 is

$$P[X_1 \leq x] = P[-\ln U_1 \leq x] = P[U_1 \geq e^{-x}] = 1 - e^{-x}$$

$\therefore X_1$ is exponential with parameter $\lambda = 1$, and $f_{X_1}(x) = e^{-x}$.
If X_1 is accepted, its pdf is given by:

$$P[x \leq X_1 < x + dx | X_1 \text{ accepted}] = \frac{P[\{X_1 \text{ accepted}\} \cap \{x \leq X_1 < x + dx\}}{P[X_1 \text{ accepted}]}$$

$$\begin{aligned} P[X_1 \text{ accepted}] &= \int_0^\infty P[X_1 \text{ accepted} | X_1 = x] e^{-x} dx \\ &= \int_0^\infty e^{-(x_1-1)^2/2} e^{-x} dx \\ &= e^{-\frac{1}{2}} \int_0^\infty e^{-x^2/2} dx \\ &= \sqrt{\frac{\pi}{2}} e^{-\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} P[x \leq X_1 < x + dx | X_1 \text{ accepted}] &= \frac{e^{-(x_1-1)^2/2} e^{-x} dx}{\sqrt{\frac{\pi}{2}} e^{-\frac{1}{2}}} = 2\frac{e^{-x^2/2}}{\sqrt{2\pi}} \\ &= f_Y(x) dx \end{aligned}$$

where $Y = |X|$ and X is a zero-mean, unit-variance random variable.

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4.142

$$F_Z(x) = \int_0^\infty \frac{\lambda\alpha^\lambda t^{\lambda-1}dt}{(\alpha^\lambda+\lambda)^2} = \int_0^{x^\lambda} \frac{\alpha^\lambda dy}{(\alpha^\lambda+y)^2}$$

where we let
$y = t^\lambda$
$dy = \lambda t^{\lambda-1}dt$

$$= \alpha^\lambda \left[\frac{-1}{(\alpha^\lambda+y)}\right]_0^{x^\lambda} = \alpha^\lambda\left[\frac{1}{\alpha^3} - \frac{1}{\alpha^\lambda+x^\lambda}\right]$$

$$= 1 - \frac{\alpha^\lambda}{\alpha^\lambda+x^\lambda} \qquad x > 0$$

To generate Z we need to solve

$$\mu = 1 - \frac{\alpha^\lambda}{\alpha^\lambda+x^\lambda}$$

$$1-\mu = \frac{\alpha^\lambda}{\alpha^\lambda+x^\lambda}$$

$$\Rightarrow x^\lambda = \alpha^\lambda\left[\frac{1}{1-\mu} - 1\right] = \alpha^\lambda\frac{\mu}{1-\mu}$$

$$\Rightarrow x = \alpha\left[\frac{\mu}{1-\mu}\right]^{1/\lambda}$$

$$\therefore \hat{Z} = \alpha\left[\frac{U}{1-U}\right]^{1/\lambda} \qquad \text{where } U \text{ is uniform in } [0,1]$$

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4.143

a) The key observation is that

$$P[X_1 \text{ is accepted} | X_1 = x] = \frac{f_X(x)}{K f_W(x)}$$

since Y is uniform in $[0, K f_W(x)]$. We then have that

$$\begin{aligned} P[X_1 \text{ is accepted}] &= \int_{-\infty}^{\infty} P[X_1 \text{ is accepted} | X_1 = x] f_W(x) dx \\ &= \int_{-\infty}^{\infty} \frac{f_X(x)}{K f_W(x)} f_W(x) dx \\ &= \frac{1}{K} \end{aligned}$$

b)

$$\begin{aligned} P[x < X_1 < x + dx | X_1 \text{ accepted}] &= \frac{P[\{X_1 \text{ accepted}\} \cap \{x < X_1 < x + dx\}]}{P[X_1 \text{ accepted}]} \\ &= \frac{\frac{f_X(x)}{K f_W(x)} f_W(x) dx}{1/K} \\ &= f_X(x) dx \end{aligned}$$

$\therefore X_1$ when accepted as pdf $f_X(x)$ as desired.

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4.144

~~3.123~~ The first approach involves performing n Bernoulli Trials, where each trial requires generating a uniform random number and a comparison to a threshold.

The second approach involves generating one uniform random number and comparing it to one or more thresholds. The maximum number of comparisons is n.

The following Octave commands generate the requested samples and plots:

```
function z = binomial_bernoulli_method(N, P)
        z = sum(discrete_rnd(N, [0, 1], [P, 1-P]));
end

function z = binomial_unit_interval_method(N, P)
     u = rand;
     z = 0;
     pos = 0;
     for j = 0:N
          pos = pos + bincoeff(N, j).*P.^j.*(1-P).^(N-j);
          if u < pos
               return;
          end
          z = z + 1;
     end
end

z1 = zeros(1,1000);
z2 = zeros(1,1000);
for i = 1:1000
     z1(i) = binomial_bernoulli_method(5, 0.5);
     z2(i) = binomial_unit_interval_method(5, 0.5);
end
figure;
hist(z1, [0:10], 1);
figure;
hist(z2, [0:10], 1);
```

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4.145

~~3.124~~ Let $T_1, T_2, ...$ be exponential interarrival times, then

$$S_n = T_1 + T_2 + ... + T_n$$

is the time of the nth arrival. Thus

$$N(t) = k \quad \text{iff} \quad S_k \leq t < S_{k+1} \quad (*)$$

Therefore to generate $N(t)$ we generate interarrival times $S_1, S_2, ...$ until the time t is exceeded as in (*). Then $N(t) = k$.

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4.146 The following Octave commands create the necessary functions for the requested program:

```
%This generates random numbers from the gamma distribution
%for alpha > 1.
function X = gamma_rejection_method_agtone(alpha, lambda)
     while(true),
          %Step 1: Generate X with pdf fx(x).
          X = cheng_inverse(alpha, lambda);
          %Step 2: Generate Y uniform in [0, Kfx(X)].
          B = cheng_pdf(X, alpha, lambda);
          Y = rand.*B;
          %Step 3: Accept or reject...
          if (Y <= fx_gamma_pdf(X, alpha, lambda)),
               break;
          end
     end
end

%This helper function generates RVs according to Kfz(x) that will bound
%our distribution.
%We will first generate random numbers according to the following pdf:
%fz(x) = (l.a^l).(x^(l-1))/(a^l + x^l)^2
%and with K = (2a-1)^(1/2)
%First we integrate to obtain the cdf:
%Fz(x) = x^l/(x^l + a^l)
%We have u = Kfz(x). Inverting the function by solving for x, we obtain:
%x = ((u.a^l)/(K-u))^(1/l)
function X = cheng_inverse(alpha, lambda)
     u = rand;
     X = ((u.*alpha.^lambda)./(1-u)).^(1./lambda);
end

%We also want to return B as we have to generate uniform variables in
%[0, Kfz(X)]
function B = cheng_pdf(X, alpha, lambda)
     K = (2.*alpha-1).^(1/2);
     B = (K.*lambda.*alpha.^lambda.*X.^(lambda-1))./((alpha.^lambda +
X.^lambda).^2);
end

%pdf of the gamma distribution.
%You could also use the Octave function gamma_pdf(X,A,B).
function y = fx_gamma_pdf(x, alpha, lambda)
     y = (x.^(alpha-1)).*(e.^(-
x./lambda))./(gamma(alpha).*lambda.^alpha);
end

function X = m_erlang_sum_of_m_exponentials(m, lambda)
     X = sum(exponential_rnd(lambda, 1, m));
end
```

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4.10 *Entropy

4.147

~~3.126~~ a) $H_X = \log 6$

b) $H_{X|A} = \log 3$

$H_X - X_{X|A} = \log 6 - \log 3 = \log 2$

4.148

~~3.127~~ a)

$$\begin{aligned} H_X &= \sum_k P_k \log \frac{1}{P_k} \\ &= -p^3 \log p^3 - p^2 q \log p^2 q - pqp \log pqp - pqq \log pqq \\ &\quad -qpp \log qpp - qpq \log qpq - q^2 p \log q^2 p - q^3 \log q^3 \\ &= -p^3 \log p^3 - 3p^2 \log p^2 q - 3pq^2 \log pq^2 - q^3 \log q^3 \end{aligned}$$

b)

$$\begin{aligned} H_X &= -\sum_k P_k \log P_k \\ &= -p^3 \log p^3 - 3p^2 q \log 3p^2 q - 3pq^2 \log 3pq^2 - q^3 \log q^3 \end{aligned}$$

c) The sample space and the number of outcomes are different in the experiments of parts a) and b).

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4.149

$P[X = n] = q^n p$, $N = 0, 1, 2, \ldots$, $q = 1 - p$

a)

$$\begin{aligned} P[X = n|X \geq k] &= P[X = n]/P[X \geq k] \text{ for } n \geq k \\ &= \frac{q^n p}{\sum_{i=k}^{\infty} q^i p} \\ &= \frac{q^n}{q^k \frac{1}{1-q}} \\ &= pq^{n-k} \\ H_{X|A} &= -\sum_{n=k}^{\infty} pq^{n-k} \log pq^{n-k} \\ &= -\sum_{n=0}^{\infty} pq^n \log pq^n \\ &= \frac{h(P)}{P} \end{aligned}$$

This is consistent with the memoryless property of the geometric random variable.

b)
$$\begin{aligned} P[X = n|X \leq k] &= P[X = n]/P[X \leq k] \text{ for } n \leq k \\ &= \frac{q^n p}{\sum_{i=0}^{k} q^i p} \\ &= \frac{q^n}{\frac{1-q^{k+1}}{1-q}} \\ &= \frac{pq^n}{1 - q^{k+1}} \end{aligned}$$

$$H_{X|A} = -\sum_{n=0}^{k} \frac{pq^n}{1 - q^{k+1}} = \log \frac{pq^n}{1 - q^{k+1}} \log(1 - q^{k+1}) - E[X|A] \log pq$$

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4.150

a)

$$\begin{aligned}
P[\text{Head}] &= P[\text{Head}|A]P(A) + P[\text{Head}|B]P(B) \\
&= \frac{1}{10}\cdot\frac{1}{2} + \frac{9}{10}\cdot\frac{1}{2} \\
&= \frac{1}{2} \\
P[\text{Tail}] &= \frac{1}{2} \\
H_X &= \log 2 = 1 \text{ bit}
\end{aligned}$$

b)

$$\begin{aligned}
P[HH] &= P[HH|A]P(A) + P(HH|B)P(B) \\
&= \frac{1}{10}\cdot\frac{1}{10}\cdot\frac{1}{2} + \frac{9}{10}\cdot\frac{9}{10}\cdot\frac{1}{2} \\
&= \frac{41}{100} \\
P[HT] &= P[HT|A]P(A) + P(HT|B)P(B) \\
&= \frac{1}{10}\cdot\frac{9}{10}\cdot\frac{1}{2} + \frac{9}{10}\cdot\frac{1}{10}\cdot\frac{1}{2} \\
&= \frac{9}{100} \\
P[TH] &= \frac{9}{100} \\
P[TT] &= 1 - \frac{41}{100} - \frac{9}{100} - \frac{9}{100} = \frac{41}{100} \\
H_X &= -\frac{41}{50}\log\frac{41}{100} - \frac{9}{50}\log\frac{9}{100} = 1.68 \text{ bits}
\end{aligned}$$

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4.151

$$\begin{aligned}
P(A|kth\,\text{toss}) &= \frac{P[A, kth\,\text{toss}]}{P[kth\,\text{toss}]} \\
&= \frac{P[A, kth\,\text{toss}]}{P[kth\,\text{toss}\,|A]P(A) + P[kth\,\text{toss}\,|B]P(B)} \\
&= \frac{\left(\frac{9}{10}\right)^{k-1} \cdot \frac{1}{10}}{\left(\frac{9}{10}\right)^{k-1} \cdot \frac{1}{10} + \left(\frac{1}{10}\right)^{k-1} \cdot \frac{9}{10}} \\
&= \frac{9^{k-1}}{9^{k-1} + 9} \\
&= \frac{9^{k-2}}{9^{k-2} + 1}
\end{aligned}$$

$$P(B|kth\,\text{toss}) = \frac{1}{9^{k-2} + 1}$$

$$H_X = -\frac{9^{k-2}}{9^{k-2}+1} \log \frac{9^{k-2}}{9^{k-2}+1} - \frac{1}{9^{k-2}+1} \log \frac{1}{9^{k-2}+1}$$

$$P[A|1] = \frac{1}{10} \;\; P[A|2] = \frac{1}{2} \;\; P[A|3] = 0.9 \;\; P[A|4] = .9878...$$

The entropy peaks at $k = 2$ and approaches 0 as $k \to \infty$ as we become certain that coin A was selected.

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4.152

~~3.131~~ a) $H_I = \log 7$

b) $X = 4$, $I = 3$ or 5, $H_I = \log 2$

4.153

~~3.132~~ The entropy is a function of probabilities and it does not depend on the values taken by the RV. Thus $H_Y = H_X$.

4.154

~~3.133~~ a) $H_X = \log 6$

b) $H_{X,Y} = \log 36 = 2 \log 6$

c) $P[\text{every outcome}] = \left(\frac{1}{6}\right)^n$
$H = \log 6^n = n \log 6$
The uncertainty in each toss is $\log 6$. In n independent tosses, the uncertainty increases linearly, $n \log 6$.

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4.155
3.134 a)

$$\begin{aligned} P[Y=1] &= \sum_{i=1}^{6} P[Y=1|X=i]P[X=i] \\ &= \left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{6}\right)\cdot\frac{1}{6} = \frac{147}{360} \\ P[Y=2] &= \left(\ldots\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{6}\right)\cdot\frac{1}{6} = \frac{87}{360} \\ P[Y=k] &= \left(\frac{1}{k}+\ldots+\frac{1}{6}\right)\cdot\frac{1}{6} \\ H_Y &= -\sum_k P_k \log P_k = 1.51 \end{aligned}$$

b)
$$\begin{aligned} H(X,Y) &= -\sum_j \sum_k P[X=j,Y=k]\log P[X=j,Y=k] \\ &= -\sum_{j=1}^{6}\sum_{k=1}^{j}\frac{1}{6}\frac{1}{j}\log\frac{1}{6}\frac{1}{j} \\ &= \sum_{j=1}^{6}\frac{1}{6}\log 6j \\ &= \log 6 + \frac{1}{6}\log 6! \end{aligned}$$

c) $H(Y|X=k) = \log k$, therefore

$$E[H(Y|X)] = \sum_{k=1}^{6} H(Y|X=k) = \sum_{k=1}^{6}\frac{1}{6}\log k = \frac{1}{6}\log 6!$$

d)
$$\begin{aligned} P(x,y) &= P(y|x)P(x) \\ \log P(x,y) &= \log P(y|x) + \log P(x) \end{aligned}$$

Take expectation on both sides.

$$H(X,Y) = H(Y|X) + H(X)$$

The joint uncertainty (X,Y) is equal to the sum of uncertainty in X and uncertainty of Y given X is observed.

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4.156

$$\begin{aligned} H_X &= -\sum_{k=1}^{K} P_k \log P_k \\ &= -\sum_{k=1}^{K-1} P_k \log P_k - P_K \log P_K \\ &\quad -(1-P_K)\log(1-P_K) + (1-P_K)\log(1-P_K) \end{aligned}$$

But $(1-P_K) = \sum_{k=1}^{K-1} P_k$. Therefore,

$$\begin{aligned} H_X &= -P_K \log P_K - (1-P_K)\log(1-P_K) - \sum_{K=1}^{K-1} P_k \log P_k + \sum_{k=1}^{K-1} P_k \log(1-P_K) \\ &= -P_K \log P_K - (1-P_K)\log(1-P_K) - \sum_{k=1}^{K-1} P_k \log \frac{P_k}{(1-P_K)} \end{aligned}$$

We finish the proof by noting that

$$H_Y = -\sum_{k=1}^{K-1} P_k \log \frac{P_k}{(1-P_K)} \text{ since } P[Y=k|X \neq K] = \frac{P_k}{1-P_K}$$

4.157

$$VAR[X-Q(X)] = \Delta^2/12 = \alpha^2$$

$$\Delta = \sqrt{12}\alpha$$

$$\begin{aligned} H_Q &= -\ln \Delta - \sum_{k=1}^{K} f_X(x_k) \log f_X(x_k)\Delta \\ &= -\ln \sqrt{12}\alpha - \sum_{k=1}^{K} \frac{1}{b-a} \cdot \sqrt{12}\alpha \log \frac{1}{b-a} \\ &= -\ln \sqrt{12}\alpha - \frac{K \cdot \sqrt{12}\alpha}{b-a} \log \frac{1}{b-a} \end{aligned}$$

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4.158

$$\begin{aligned} P[\text{Input 000}|\text{Output 000}] &= \frac{P[\text{Output 000, Input 000}]}{P[\text{Output 000}]} \\ &= \frac{(1-P)^3 \cdot \frac{1}{2}}{(1-P)^3 \cdot \frac{1}{2} + P^3 \cdot \frac{1}{2}} \\ &= \frac{(1-P)^3}{(1-P)^3 + P^3} \end{aligned}$$

$$P[\text{Input 111}|\text{ Output 000}] = \frac{P3}{(1-P)^3 + P^3}$$

$$H_{X|A} = -\frac{(1-P)^3}{(1-P)^3+P^3}\log\frac{(1-P)^3}{(1-p)^3+P^3} - \frac{P^3}{(1-P)^3+P^3}\log\frac{P^3}{(1-P)^3+P^3}$$

If the output is 010,

$$\begin{aligned} H_{X|A} &= -\frac{P(1-P)^2}{P(1-P)^2+P^2(1-P)}\log\frac{P(1-P)^2}{P(1-P)^2+P^2(1-P)} \\ &\quad -\frac{P^2(1-P)}{P(1-P)^2+P^2(1-P)}\log\frac{P(1-P)^2}{P(1-P)^2+P^2(1-P)} \end{aligned}$$

4.159

[38] X is uniform RV in $[-a, a]$, $f_X(x_k) = \frac{1}{2a}$,

$$\begin{aligned} H_Q &= -\log\Delta - \sum_{k=1}^{K} f_X(x_k)\Delta\log\left(f_X(x_k)\right) \\ &= -\log\Delta - \log\left(f_X(x)\right) \\ &= -\log\Delta - \log\frac{1}{2a} \\ H_{Q|A} &= -\log\Delta - \log\left(f_{X|A}(x)\right) \\ &= -\log\Delta - \log\frac{1}{a} \\ H_Q - H_{Q|A} &= \log\frac{1}{a} - \log\frac{1}{2a} = \log 2 \end{aligned}$$

The difference of the differential entropy $\log(a-(-a)) - \log(a-0) = \log 2$

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4.160

$$\begin{aligned} H_Y &= -\int f_Y(y) \log f_Y(y) dy \\ &= -\int \frac{1}{2} f_X(x) \log \frac{1}{2} f_X(x) \cdot 2dx \\ &= -\log \frac{1}{2} + H_X \\ &= \log 2 + H_X \end{aligned}$$

Note that $f_Y(y)$ is different from $f_X(x)$.

4.161

X	P	l(X)
1	1/2	1
2	1/4	2
3	1/8	3
4	1/16	4
5	1/32	5
6	1/64	6
7	1/128	7
8	1/128	7

For this pmf Equation 3.114 gives $E[L] - H_X = 0$ implying that the code is optimum. An intutive way of seeing that this is optimum is to note that the alternatives in each question are always equiprobable for this pmf.

4.162

$$\begin{aligned} H_X &= -2 \cdot \frac{3}{8} \log \frac{3}{8} - \frac{1}{8} \log \frac{1}{8} - \frac{1}{16} \log \frac{1}{16} - 2 \cdot \frac{1}{32} \log \frac{1}{32} \\ &= 1.06 + \frac{3}{8} + \frac{4}{16} + \frac{5}{16} \\ &= 2.0 \text{ bits} \end{aligned}$$

X	P	Codeword
1	3/8	0
2	3/8	10
3	1/8	110
4	1/16	1110
5	1/32	11110
6	1/32	11111

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4.163

~~3.142~~ $\log_2 \binom{52}{7} = 27$ bits

4.164

~~3.143~~ $P[X = k] = \left(\frac{1}{2}\right)^k$

$x = 1$, the codeword is 0 .

$x \geq 2$ the codeword begins with $(x-1)$ ones, and ends with 0 .

The outcomes at each toss is equiprobable implying that we use 1 bit to encode the result.

4.165

~~3.144~~ a)

$$E[L] = \frac{6}{10} \cdot 3 + \frac{4}{10} \cdot 4 = 3 \cdot 4 \text{ bits}$$
$$H_X = \log_2 10 = 3.3 \text{ bits}$$

b) Choose the smallest k, s.t. $2^k > 10^n$

In the full binary tree with depth $(k-1)$, $(10^n - 2^{k-1})$ nodes have to be expanded.

$$E[C] = \frac{2^{k-1} - (10^n - 2^{k-1})}{10^n} \cdot (k-1) + \frac{2(10^n - 2^{k-1})}{10^n} \cdot k$$

$$H_X = \log_2 10^n = n \log_2 10.$$

The performance will be better if $2^k - 10^n$ is small.

4.166

~~3.145~~ a) There are $\binom{n}{k}$ equiprobable patterns, so a code with codewords of lengths $\lceil \log \binom{n}{k} \rceil$ and $\lfloor \log_2 \binom{n}{k} \rfloor$ will be optimum.

b) $\log_2(n+1)$ bits are sufficient.

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4.167

~~3.146~~ Set

$$
\begin{aligned}
P_k &= Ce^{-\lambda k} = C\alpha^k \\
&\begin{cases} 1 = \sum P_k = C\alpha + C\alpha^2 + C\alpha^3 + C\alpha^4 \\ 2 = E[X] = 1 \cdot C\alpha + 2 \cdot C\alpha^2 + 3 \cdot C\alpha^3 + 4 \cdot C\alpha^4 \end{cases} \\
&C = 0.64,\ a = 0.66 \\
&P_1 = 0.42,\ P_2 = 0.28,\ P_3 = 0.18,\ P_4 = 0.12
\end{aligned}
$$

4.168

~~.147~~ Set

$$
\begin{aligned}
f_X(x) &= Ce^{-\lambda x} \quad x \geq 0 \\
1 &= \int_0^\infty f_X(x)dx = \int_0^\infty Ce^{-\lambda x}dx = C/\lambda \\
C &= \lambda \\
10 &= E[X] = \int_0^\infty xCe^{-\lambda x}dx \\
&= -\int_0^\infty x de^{-\lambda x} \\
&= -xe^{-\lambda x}|_0^\infty + \int_0^\infty e^{-\lambda x}dx \\
&= \frac{1}{\lambda} \\
\lambda &= C = \frac{1}{10}
\end{aligned}
$$

$$f_X(x) = \frac{1}{10}e^{-\frac{1}{10}x} \text{ for } x \geq 0$$

X is an exponential RV.

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4.169

Set

$$\begin{aligned}
f_x(x) &= C_1 e^{-\lambda x^2} \\
1 &= \int_{-\infty}^{\infty} f_x(x)dx = 2C_1 \int_0^{\infty} e^{-\lambda x^2} dx = C_1 \sqrt{\frac{\pi}{\lambda}} \\
\lambda &= \pi C_1^2 \\
C &= E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)dx \\
2C_1 \int_0^{\infty} x^2 e^{-\lambda x^2} dx &= -\frac{1}{\pi C_1} \int_0^{\infty} x de^{-\pi C_1^2 x^2} \\
&= +\frac{1}{\pi C_1} \int_0^{\infty} e^{-\pi C_1^2 x^2} dx \\
&= +\frac{1}{\pi C_1} \frac{\sqrt{\pi}}{2\sqrt{\pi C_1^2}} \\
&= \frac{1}{2\pi C_1^2}
\end{aligned}$$

Therefore,

$$C_1 = \sqrt{\frac{1}{2\pi C}}, \lambda = \pi C_1^2 = \frac{1}{2c}$$

$$f_X(x) = \sqrt{\frac{1}{2\pi c}} e^{-\frac{x^2}{2c}}$$

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4.170

~~3.149~~ a)

$$\int f_X(x)dx = 1$$
$$\int g_1(x)f_X(x)dx = C_1$$
$$\int g_2(x)f_X(x)dx = C_2$$

Using Lagrange Multipliers,

$$-\int f_X(x)\ln f_X(x)dx + \lambda_1[\int g_1(x)f_X dx - C_1] + \lambda_2[\int g_2(x)f_X(x)dx - C_2]$$
$$= -\int f_X(x)\ln\frac{f_X(x)}{Ce^{-\lambda_1 g_1(x)-\lambda_2 g_2(x)}}dx$$

where $C = e^{-\lambda_1 C_2 - \lambda_2 C_2}$ so $f_X(x)$ has the form of $e^{-\lambda_1 g_1(x)-\lambda_2 g_2(x)}$

b)

$$\begin{aligned} h_X(x) &= -\int f_X(x)\ln f_X(x)dx \\ &= -\int f_X(x)[\ln C - \lambda_1 g_1(x) - \lambda_2 g_2(x)]dx \\ &= -\ln C + \lambda_1 C_1 + \lambda_2 C_2 \end{aligned}$$

4.171

~~3.150~~ From the results of Problem 149, $f_X(x) = Ce^{-\lambda_1 x - \lambda_2 x^2}$. It should be in the form of Gaussian RV.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\alpha}exp\left[-\frac{(x-m)^2}{2\alpha^2}\right]$$

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Problems Requiring Cumulative Knowledge

4.172

$$x = \begin{cases} X_1, & \text{exponential RV, } \frac{1}{\lambda_1} = 1, & \text{with} P_1 = 1/2 \\ X_2, & \text{exponential RV, } \frac{1}{\lambda_1} = 10, & \text{with} P_2 = 1/8 \\ X_3, & \text{constant 2,} & \text{with} P_3 = 3/8 \end{cases}$$

$$\begin{aligned} P[X > 15] &= P[X > 15|X = X_1]P_1 + P[X > 15|X = X_2]P_2 + P[X > 15|X = X_3]P_3 \\ &= e^{-\lambda_1 \cdot 15} \cdot \frac{1}{2} + e^{-\lambda_2 \cdot 15} \cdot \frac{1}{8} + 0 \\ &= 0.028 \\ E[X] = E[E[X|\text{type}]] &= P_1 E[X|X = X_1] + P_2 E[X|X = X_2] + P_3 E[X|X = X_3] \\ &= \frac{1}{2} \cdot 1 + \frac{1}{8} \cdot 10 + \frac{3}{8} \cdot 2 \\ &= 2.5 \end{aligned}$$

Markov's inequality $P[X \geq 15] \leq E[X]/15 = \frac{1}{6}$. The bound is loose.

4.173

$$P[X \geq 9|X \geq 1] = \frac{P[X \geq 9, X \geq 1]}{P[X \geq 1]} = \frac{\frac{1}{1+9}}{\frac{1}{1+1}} = \frac{1}{5}$$

$$P[X < 9|X \geq 1] = \frac{4}{5}$$

$$P[\text{At least one bulb is working}] = 1 - (\frac{4}{5})^3$$

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4.174

3.154 $Y = \max\{X_1, X_2, ..., X_n\}$.

$$\begin{aligned}
P[Y \le y] &= P[X_1 \le y, X_2 \le y, ..., X_n \le y] \qquad 0 \le y \le a \\
&= P[X \le y]^n \\
&= \left(\frac{y}{a}\right)^n \\
E[Y] &= \int_0^a y f_Y(y) dy = \int_0^1 y \frac{n y^{n-1}}{a^n} dy \\
&= \frac{n}{a^n} \frac{y^{n+1}}{n+1}\bigg|_0^1 = \frac{n}{n+1} a \\
E[Y^2] &= \frac{n}{a^n} \int_0^1 y^2 y^{n-1} dy = \frac{n}{a^n} \frac{y^{n+2}}{n+2}\bigg|_0^1 = \frac{n}{n+2} a^2 \\
VAR(Y) &= E[Y^2] - E[Y]^2 = \frac{n}{n+2} a^2 - \left(\frac{n}{n+1}\right)^2 a^2 \\
&= \left[\frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2\right] a^2
\end{aligned}$$

The value of "a" is by definition larger than any value Y can assume. In addition, when n is large the above results show that Y tends to be close to "a".

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4.175

a) $P[X \le -a] = \Phi_X(-a)$

$P[-a < X \le 0] = \Phi_X(0) - \Phi_X(-a)$

$P[0 < X \le a] = \Phi_X(a) - \Phi_X(0)$

$P[X > a] = 1 - \Phi_X(a)$

Since $P[X \le -a] = P[-a < X \le 0] = P[0 < X \le a] = P[X > a]$

$\Phi(-a) = 1/4$

$\Phi(0) = 1/2$

$\Phi(a) = 3/4$

$\Phi(-a) = Q(a) = 1/4 \longrightarrow a = 0.7$

b) $\int_0^a 2(x - x_1)(-1) f_X(x)\,dx = 0$

$2x_1 \int_0^a f_X(x)\,dx = 2\int_0^a x f_X(x)\,dx$

$2x_1 P[0 \le X \le a] = 2\int_0^a \frac{x}{\sqrt{2\pi}} e^{-x^2/2}\,dx$

$2x_1 \Phi(-a) = \frac{2}{\sqrt{2\pi}}(1 - e^{-a^2/2})$

$x_1 = \frac{(1 - e^{a^2/2})}{\sqrt{2\pi}\,\Phi(-a)} = \frac{1 - e^{-0.245}}{\sqrt{2\pi}}(4) = 0.3468$

Similarly

$x_2 = \frac{4}{\sqrt{2\pi}} e^{-0.245} = 1.249$

(x)

$\longrightarrow x_{-1} = -x_1$

$x_{-2} = -x_2$

c) $E[(X - q(X))^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx + (x_{-2}^2 + x_{-1}^2 + x_1^2 + x_2^2)\int_{-\infty}^{\infty} f_X(x)\,dx - 2(x_{-2} + x_{-1} + x_1 + x_2)\int_{-\infty}^{\infty} x f_X$

$= x_{-2}^2 + x_{-1}^2 + x_1^2 + x_2^2$

$= 2\left(\frac{4}{\sqrt{2\pi}} e^{-0.245}\right)^2 + 2\left(\frac{4}{\sqrt{2\pi}}(1 - e^{-0.245})\right)^2$

$= \frac{16}{\pi}(1 + e^{-0.49} - 2e^{-0.245}) = 0.2405$

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4.176

~~3.159~~ a)
$$P[\text{input is } 1 | y < Y < y+h] = \frac{P[\text{input is } 1, y < Y < y+h]}{P[y < Y < y+h]}$$
$$= \frac{\int_y^{y+h} f_1(t)dt \cdot p}{\int_y^{y+h} f_1(t)dt \cdot p + \int_y^{y+h} f_0(t)dt \cdot (1-p)}$$
$$\approx \frac{f_1(t)hp}{f_1(t)ph + f_0(t)h(1-p)}$$

where we assume $h \ll 1$.

b)
$$\begin{array}{lll} & P[\text{input is } 1 | y < Y < y+h] > P[\text{input is } 0 | y < Y < y+h] \\ \text{iff} & f_1(t)p > f_0(t)(1-p) \\ \text{iff} & \frac{p}{\sqrt{2\pi}} e^{-(y-1)^2/2} > \frac{(1-p)}{\sqrt{2\pi}} e^{-y^2/2} \\ \text{iff} & e^{-\frac{1}{2}(y^2-2y+1-y^2)} > \frac{1-p}{p} \\ \text{iff} & -\frac{1}{2}(-2y+1) > \ln \frac{1-p}{p} \\ \text{iff} & y > \frac{1}{2} + \ln \frac{1-p}{p} = T. \end{array}$$

c)
$$\begin{aligned} P_e &= P[Y > T, \text{input } 0](1-p) + P[Y < T, \text{input } 1]p \\ &= (1-p)\int_T^\infty \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy + p\int_{-\infty}^T \frac{1}{\sqrt{2\pi}} e^{-(y-1)^2/2} dy \\ &= (1-p)Q(T) + pQ(1-T) \end{aligned}$$

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